

# Targeting Higher Credit Card Payments

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## Abstract

Online experiments find that shrouding the credit card minimum payment increases hypothetical payments, leading a literature to call the minimum an anchor. In a field experiment, I test shrouding the minimum in a credit card payment app for six statements. Actual payments did not significantly increase. Clickstream data show why. Consumers first entered amounts below the minimum to reveal it, then paid it, hacking the nudge. I model the minimum as a target at a payoff notch; anchoring appears only when stakes are low, as in hypotheticals. In the field, what changes payments is the target's level, not its display.

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# 1 Introduction

One of the foundational biases in Tversky and Kahneman (1974) is the “anchoring and adjustment heuristic,” whereby consumers faced with an uncertain choice rely on salient but irrelevant numerical information and adjust too little away from it. In the fifty years since, there is broad evidence of anchoring across domains (Schley and Weingarten, 2025). If choices depend on irrelevant numbers, preferences are not stable, and removing the number should change the choice.

An economically important domain in which anchoring has been influential is credit cards. A large body of research—with publications across the fields of economics, finance, management, marketing, and psychology—attributes consumer credit card payment choices in part to anchoring.<sup>1</sup> Online experiments, the results of which have been repeatedly replicated, find that hiding (“shrouding”) the credit card minimum payment causes a large increase in hypothetical credit card payments, leading to the conclusion that the minimum payment acts as a psychological anchor.<sup>2</sup> Further evidence in support of credit card anchoring has been provided in credit card administrative data by Keys and Wang (2019) and Medina and Negrin (2022), who both examine consumer responses to changes in minimum payments and find that these responses are inconsistent with mere liquidity constraints. A key implication of this research is that consumers would pay more on their credit cards without anchoring.<sup>3</sup>

In this paper, I show that the results of online experiments do not extrapolate to the field. My evidence is inconsistent with the minimum payment acting as an anchor that lowers payments. I study a pre-registered field experiment on 6,714 consumers with a FinTech credit card lender in the United States whose customers primarily have below-prime credit scores: the consumers most likely to pay at or near the minimum (Keys and Wang, 2019), who face the highest interest rates (Consumer Financial Protection Bureau, 2025), who appear to rely most on heuristics that do not minimize interest costs (Keys and Wang, 2019; Medina and Negrin, 2022; Agarwal et al., 2026), and who therefore have the most to gain from paying

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<sup>1</sup>Thaler and Sunstein (2008) write that a credit card minimum payment “can serve as an anchor, and as a nudge that this minimum payment is an appropriate amount.” A series of papers attribute credit card payment choices to anchoring: Stewart (2009); Navarro-Martinez et al. (2011); Salisbury (2014); McHugh and Ranyard (2016); Guttman-Kenney et al. (2018); Keys and Wang (2019); Salisbury and Zhao (2020); Hendy et al. (2021); Medina and Negrin (2022); Sakaguchi et al. (2022); Ortiz et al. (2024); Vihriälä (2025); Durán-Micco et al. (2026); Incekara-Hafalir et al. (2026); Katz et al. (2026).

<sup>2</sup>Navarro-Martinez et al. (2011); Salisbury (2014); McHugh and Ranyard (2016); Guttman-Kenney et al. (2018); Salisbury and Zhao (2020); Hendy et al. (2021); Sakaguchi et al. (2022); Ortiz et al. (2024); Durán-Micco et al. (2026), and Incekara-Hafalir et al. (2026). Appendix Figure A1 is an example of such online experimental results. Online experiments are sometimes referred to as lab or survey experiments.

<sup>3</sup>Keys and Wang (2019) write “Anchoring to minimum payments on debt contracts can lead to lower payments, higher interest costs, and higher default rates, so our results highlight minimum payments as a potential target for regulators and innovators aiming to improve financial health.”

more. For six statement cycles, the lender’s app shrouded the minimum payment amount for one treatment group and both the minimum and a 50% payment option for another. I merge the lender’s administrative data with clickstream data on payment choices, daily liquid cash balances from linked bank accounts, and credit reporting data.

Neither nudge significantly increases credit card payments. After six statements, the effects on payments as a percentage of the statement balance are  $-1.26$  (95% C.I.  $-3.67, 1.15$ ) and  $-0.54$  (95% C.I.  $-2.87, 1.79$ ) percentage points, and pooling across cycles narrows the confidence intervals without changing the conclusion. The contrast with online experiments is sharp: shrouding the minimum increases hypothetical payments by  $12.4$  (95% C.I.  $10.8, 14.0$ ) percentage points in Guttman-Kenney et al. (2023). The field results are robust to other outcomes and regression specifications; if anything, after six statements payments in dollars fall and revolving debt rises. Adding a 50% payment option has the unintended effect of temporarily making consumers less likely to pay their balance in full.

Clickstream data on consumers’ initial in-app payment choices reveal an important new result: when the minimum is shrouded, consumers are significantly more likely to initially select lower payment amounts, in particular amounts less than the minimum. This is the opposite of what the anchoring explanation in the earlier literature predicts, which is that shrouding the minimum should raise payments. Forcing consumers to make an active choice (e.g., Carroll et al., 2009; Keller et al., 2011) does not change economic outcomes. Instead, consumers immediately exert effort to reveal the minimum payment and then pay it, hacking the active choice design.

My results are what one expects if the minimum is a target rather than an anchor. An anchor is an irrelevant number that pulls choices only while it is displayed (Tversky and Kahneman, 1974; Ariely et al., 2003); a target is a relevant number that consumers aim for and exert effort to reach whether or not it is displayed (Heath et al., 1999; Bartels et al., 2024).<sup>4</sup> Formalizing and extending ideas in prior work (Bartels et al., 2024; Herzog et al., 2025; Schwartz, 2026), in Section 3 I develop a conceptual framework that puts this distinction into a consumer’s payment problem, so that anchoring and targeting make opposite predictions for the experiment.

The model has three results. Consumers have reference points at the two amounts where economic payoffs jump (Kleven, 2016; Dean et al., 2026), the minimum and the statement balance; payments bunch at these amounts from above, with consumers who have cash to spare paying exactly the minimum, and amounts between them, such as 50% of the balance

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<sup>4</sup>In Tversky and Kahneman (1974), one anchor is randomly determined by spinning a wheel of fortune. In Ariely et al. (2003), one anchor is the last two digits of a consumer’s Social Security Number, digits that are randomly assigned. In both examples, the anchors are irrelevant to the decision, yet they affect choices.

or the CARD Act’s three-year payment scenario, cannot compete with them. Anchoring is nested in the model with a weight that falls as more rides on the exact amount paid, so the same preferences produce large effects in a hypothetical scenario, where the amount does not matter, and none at a payoff notch in the field. Payments respond to the level of the minimum but not to its display. A higher minimum moves consumers at the target one for one and pulls above-minimum payers whose own targets are indexed to it. This is consistent with the responses of unconstrained consumers to minimum increases in Keys and Wang (2019), Medina and Negrin (2022), Katz et al. (2026), and Allen et al. (2026).

The consumers in the experiment are cash-poor rather than credit-constrained. As in Keys and Wang (2019) and Medina and Negrin (2022), most are not at their borrowing limits: median credit card portfolio utilization is 25%, and near-minimum payers, who pay above the minimum and so are not constrained by it, do not respond to the nudges either. But 93% have a minimum liquid cash balance below \$501 over the ninety days before the experiment, consistent with Guttman-Kenney et al. (2025).

The results also speak to the gap between hypothetical and actual choices. List (2001) shows how hypothetical responses can be biased. Gandhi et al. (2025) conclude that hypothetical studies usually agree with field experiments in direction but vary widely in magnitude, and scaling interventions from one setting to another is a recognized challenge (List, 2022). My study provides a case where a highly replicated online result does not hold up in a field setting with the intertemporal trade-offs consumers actually face.

More broadly, anchoring effects appear robust in no- or low-stakes online experiments (Schley and Weingarten, 2025), whereas in markets with real incentives, eye-catching anchoring results such as Ariely et al. (2003) diminish on closer inspection (Fudenberg et al., 2012; Maniadis et al., 2014) or are explained by alternative theories (Guo, 2024; Bartels et al., 2024). My results speak to a long-standing concern that behavioral economics has accumulated a separate bias for each anomaly (Fudenberg, 2006; Stango and Zinman, 2023). The credit card facts attributed to anchoring, payments bunching at the minimum and unconstrained consumers responding to changes in it follow from a notch in economic payoffs once consumers treat it as a reference point. My framework also works through two questions in the literature (O’Donoghue and Sprenger, 2018): how consumers act when two reference points compete, and how reference dependence combines with present bias.

My results have direct policy implications. Persistently revolving credit card debt is costly and common (Grodzicki and Koulayev, 2021). To reduce it, regulators in the United Kingdom have considered requiring lenders to shroud minimum payments (Financial Conduct Authority, 2021), and similar proposals have been discussed in the United States citing the academic evidence on credit card anchoring (Tescher and Stone, 2022; Gdalmann et al.,

2023; Schneider, 2025). My study provides field evidence that shrouding the minimum on the payment screen does not raise payments, so such a policy is unlikely to reduce credit card debt, whereas raising the minimum itself does (Allen et al., 2026). Given this policy interest, documenting null results is important, especially given publication bias and the lack of self-correction in the academic literature (Abadie, 2020; Brodeur et al., 2023; Ankel-Peters et al., 2025).

The paper’s contribution is to show in the field that the credit card minimum payment is a target rather than an anchor, and to provide a framework in which payments resist nudges that change what is displayed but respond to changes in the minimum itself. This speaks to the household finance literature on consumer financial protection and credit cards (Agarwal et al., 2015, 2023; Seira et al., 2017; Nelson, 2025; Hirshman and Sussman, 2022; Matcham, 2025; Bakker et al., 2026; Bord et al., 2026; Brown et al., 2026; Katz et al., 2026; Schwartz, 2026), in which information and nudges have been largely ineffective at reducing debt (Agarwal et al., 2015; Keys and Wang, 2016; Seira et al., 2017; Adams et al., 2022; Guttman-Kenney et al., 2025; Batista et al., 2026), with any effects attenuating over time (e.g., Adams et al., 2022; Han and Yin, 2026; Schwartz, 2026) or offset elsewhere in the portfolio (e.g., Medina, 2021), and to the broader finding that nudges are less effective than they were originally thought to be (DellaVigna and Linos, 2022; Choukhmane, 2025; Choi et al., 2024). It also corroborates in administrative data the hand-to-mouth-but-not-constrained pattern that Lee and Maxted (2026) document in survey data and show requires present bias rather than borrowing limits.

Relative to the closest prior work, Guttman-Kenney et al. (2025), this paper differs in three respects. I test shrouding on the manual payments consumers make each month rather than on autopay enrollment at card opening.<sup>5</sup> I test a new intermediate payment option. And, with the framework and the merged data, I explain why consumers pay at or just above the minimum and why that behavior moves with the minimum but not with nudges.

The paper proceeds as follows. Section 2 explains the design of the experiment, the data, and the empirical methodology. Section 3 sets out a conceptual framework in which the minimum payment is a target, and states what targeting and anchoring each predict for the experiment. Section 4 tests those predictions, and Section 5 examines heterogeneity in the results. Section 6 explains why consumers target the minimum and what follows from it. Section 7 concludes.

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<sup>5</sup>The majority of the value of credit card payments is made manually, even among autopay enrollees (Consumer Financial Protection Bureau, 2025; Guttman-Kenney et al., 2025), so the manual channel is where anchoring to the minimum should matter most. In Guttman-Kenney et al. (2025), the nudge’s effect on autopay payments was offset by lower autopay enrollment and lower manual payments; neither applies here.

## 2 Experimental Design, Data, and Empirical Methodology

### 2.1 Experimental Design

I study a field experiment that varies the credit card payment options on one lender’s mobile app. Nationally, 76% of credit card accounts are enrolled in mobile apps, with this method growing over time and becoming more widely used than online portals in 2024, and 57% of payments are made manually—i.e., non-autopay channels such as via app or online (Consumer Financial Protection Bureau, 2025).

The lender in the experiment is Arro, a FinTech that provides credit cards in the United States. Its customer base is primarily consumers with below-prime credit scores. The experiment is a randomized controlled trial (RCT) on consumers who hold credit cards with this lender and who meet the selection criteria of having had at least one statement issued, a credit limit of over \$40, a non-zero statement balance, and who have never been over 35 days past due at the time of randomization, and excludes 2,337 consumers who were incorrectly randomized to treatment groups in December 2024. For this experiment, 6,905 consumers are randomly allocated across three groups:

1. 68% of consumers are in the **Control (C)** group. The payment options displayed to consumers in this group are: current balance, statement balance, 50% of the statement balance, minimum payment, and a free text box to choose any payment amount. These are the status quo options that consumers already experience with this lender in the absence of the experiment. This is displayed in Panel A of Appendix Figure A2.
2. 16% of consumers are in the **Treatment 1 (T1)** group. Consumers in this group are not shown the minimum payment option, which, in the control group, also displays the minimum payment amount. Consumers in T1 are shown the following payment options: current balance, statement balance, 50% of the statement balance, and a free text box to choose any payment amount. This is displayed in Panel B of Appendix Figure A2.
3. 16% of consumers are in the **Treatment 2 (T2)** group. As with T1, this does not show consumers in this group the minimum payment option. But T2 also does not show consumers the option for 50% of the statement balance. Consumers in T2 have the following payment options displayed to them: current balance, statement balance, and a free text box to choose any payment amount. This is displayed in Panel C of Appendix Figure A2.

The experiment went live in the field on March 24, 2025, with 6,714 consumers—of which 4,542 (68%) were in group C, 1,096 (16%) in group T1, and 1,076 (16%) in group T2. This total is smaller than the original 6,905 consumers because when the experiment went live, 191 (3%) consumers had closed their accounts or no longer met the selection criteria.

Importantly, in all three groups (C, T1, and T2), if a consumer submits a choice to pay an amount less than the minimum payment, a warning prompt appears with the minimum payment amount (for a visual, see Panel D of Appendix Figure A2). Such prompts appear as standard across all major credit card lenders in the United States and other countries to prevent accidentally paying less than the minimum, enabling consumers to revise their initial payment choice. The experiment therefore estimates the effect of shrouding the minimum on a screen that still reveals it to anyone who enters less. Sakaguchi et al. (2022) show in an online experiment how such prompts appear effective at preventing such choices, and Mrkva et al. (2026) documents how prompts are more broadly useful for consumers.

T1 and T2 are nudges because they do not restrict a consumer’s choice of payment amount. Instead, they only change the choice architecture (Thaler and Sunstein, 2008) regarding how salient particular choices are presented. The motivation behind T1 is that the minimum payment amount acts as a psychological default (e.g., Stewart, 2009; Medina and Negrin, 2022; Keys and Wang, 2019; Sakaguchi et al., 2022), potentially reducing some consumer payment choices to exactly this amount or to amounts just above it. By shrouding this option, it may increase payments by increasing the salience of the statement balance for consumers to use as a target and enabling consumers to make a more active choice (Carroll et al., 2009; Keller et al., 2011) without their choice potentially being distorted by the presence of the minimum. Shrouding the minimum payment without an alternative option may be socially optimal if there is a high degree of heterogeneity in consumers’ socio-economic circumstances and preferences (Carroll et al., 2009).

The motivation behind T2 is that the 50% statement balance option can also act as a target, potentially reducing some consumer payment choices to exactly 50%, or amounts close to it, which would otherwise be higher. Prior research on the CARD Act statement disclosures has indicated that more payment options may unintentionally reduce payments for some consumers (Agarwal et al., 2015; Hershfield and Roese, 2015; Keys and Wang, 2016). There are two reasons why there may be different results in my experiment. First, statements are easily ignored by consumers (Adams et al., 2022), whereas consumers are required to select a payment amount in the app and so this environment may affect consumers differently. Second, these earlier studies examined three-year payment scenarios that involve far lower payment amounts than a 50% statement balance amount, which may act as a high anchor or target. By shrouding the 50% option, along with the minimum payment option,

it may increase payments by enhancing the salience of the statement balance for consumers to use as a target.

Comparing outcomes in group T1 to those in group C shows the effect of shrouding the minimum payment option, to target the higher payment options. Comparing outcomes in group T2 to those in group C shows the effect of shrouding both the minimum payment and the 50% statement balance options, to target the statement balance as a higher payment option. Comparing outcomes from group T1 to those in group T2 shows the effect of adding a 50% statement balance option, while holding constant the shrouding of the minimum payment option.

## 2.2 Data

All of the data I access is anonymous. I primarily use administrative data from the credit card lender for every card in the experiment. This data contains statement-level information (e.g., statement balances, credit card limits, minimum payment amounts due, statement dates), and disaggregated information on each payment a consumer makes on their credit card, including the timing, amount, and payment method (e.g., manual or autopay).

I also use three other linked datasets for my analysis. First, I observe linked clickstream data that consists of each payment choice made by a consumer on the lender’s app. Second, I observe linked credit reporting data (Gibbs et al., 2025) from Equifax, which contains consumer-level aggregated information on the portfolio of credit cards held by consumers each month. Third, the lender requires all of its cardholders to link their household financial transaction data (Baker and Kueng, 2022) from Plaid. This linked Plaid data contains balances on each checking and cash savings account held by consumers, which I use to calculate the liquid cash balances of a consumer each day.

Table 1 shows summary statistics on the cards in the experiment using credit card administrative data before the experiment went into the field. This shows that the covariates (credit score, APR, card tenure, and credit limits) are balanced across the control and the two treatment groups. The average credit card limit is approximately \$290, with a typical statement balance just greater than \$200, a typical VantageScore credit score near 560 with consumers generally in the subprime or near-prime categories, and 41% of consumers being enrolled in autopay. In my data, 92% of the total value of payments is made via manual payments (i.e., via the app) rather than through autopay.

A natural concern of this sample is its external validity to the broader credit card market. The data are not nationally representative, but rich and selected data sources are still informative about consumer behavior; Indarte et al. (2026) is one example. Nationally, ap-

proximately a quarter of consumers with credit cards have below-prime credit scores. Among these consumers, their total cost of credit—interest plus fees, as a percentage of statement balance—is typically 25% or higher. This is because consumers with lower credit scores have higher interest rates, lower repayment rates, and are more likely to incur late fees than those with higher credit scores (Consumer Financial Protection Bureau, 2025). Consumers with lower credit scores are those most likely to pay at or near the minimum payment in Keys and Wang (2019), who estimate no anchoring effects for consumers with superprime credit scores whereas for all credit score groups lower than superprime, they find similarly-sized estimates of anchoring. This suggests that the lack of superprime consumers in my study is not important for evaluating anchoring, and that one might expect similar results for prime consumers as for near-prime consumers. On-time payments with this lender are reported to credit reporting agencies and can unlock higher credit limits, as with credit cards generally.

The main limitation of this data source is that the credit cards used in the experiment have lower credit limits relative to the credit card market in the United States. Despite this, I find that the distribution of payments for the control group, with approximately a quarter at or just above the minimum payment amount and a quarter that is paying the full statement balance amount (or more), follows a very similar pattern to market-wide data across lenders in the United States (Keys and Wang, 2016; Guttman-Kenney and Shahidinejad, 2025; Consumer Financial Protection Bureau, 2025). In addition, Guttman-Kenney et al. (2018) find large and statistically significant effects of shrouding the minimum payment in a hypothetical payment study that were similar across both a low statement balance (£532) and a high statement balance (£3,217) scenario. This indicates that, given the prior literature, one might still expect to find significant increases in payments from shrouding the minimum payment in my sample if the driver of payment choices is anchoring to the minimum payment.

## 2.3 Empirical Methodology

The analysis for this experiment was pre-registered (Guttman-Kenney, 2025) on the AEA RCT registry before the experiment was live in the field. This pre-registration included a Pre-Analysis Plan (PAP), which is also included in Appendix A of this paper.

I first use t-tests to evaluate statistical significance. To understand the dynamics of the treatment effects, I use the dynamic regression specified in Equation (1), using data from all completed statement cycles during the experiment:

$$Y_{i,t} = \sum_{\tau=0}^6 \left[ \delta_{\tau}^{T1(vsC)} (D_{\tau} \times T1_i) + \delta_{\tau}^{T2(vsC)} (D_{\tau} \times T2_i) \right] + X_i' \beta + \mu_t + \varepsilon_{i,t} \quad (1)$$

In Equation (1),  $Y_{i,t}$  denotes the outcome  $Y$  for consumer  $i$  at time  $t$ . Because consumers are restricted to holding one credit card with the lender, card account-level data correspond directly to the consumer level. The term  $\varepsilon_{i,t}$  is the error term. My coefficients of interest are  $\delta_\tau^k$ , for  $k \in \{T1(vsC), T2(vsC)\}$ . These are the coefficients on the interactions between the treatment groups and event time indicators. The term  $T1_i$  is an indicator for the consumers in treatment 1 group ( $T1$ ), and the term  $T2_i$  is an indicator for the consumers in treatment 2 group ( $T2$ ). The term  $D_\tau$  is an indicator for event time  $\tau$ , where  $\tau = 1$  is the first full statement cycle after the launch of the experiment. Given random assignment to treatment,  $\delta_\tau^k$  produces unbiased estimates of the dynamic average effects of treatment  $k$  after  $\tau$  statements. I calculate the effects of adding the 50% payment option by  $\delta_\tau^{T1(vsT2)} = \delta_\tau^{T1(vsC)} - \delta_\tau^{T2(vsC)}$ . I cluster standard errors at the consumer level in all regressions.

I also examine a static version of my earlier regression specification, showing the average treatment effects pooled across cycles,  $\delta^k$ , estimated by Equation (2):

$$Y_{i,t} = \delta^{T1(vsC)} T1_i + \delta^{T2(vsC)} T2_i + X_i' \beta + \mu_t + \varepsilon_{i,t} \quad (2)$$

After examining the data, I made two departures from my pre-registered regression specification. I still also report results for the original pre-registered specification. First, there are time trends in the data, and therefore I include granular time fixed effects ( $\mu_t$ ), which are fixed effects for each statement end date (unreported results were similar when including less granular event time fixed effects). Second, although covariates are balanced, there is a significant difference in payments as a percent of the statement balance and also in whether they have missed a payment between the control group and T1, but not with T2, in the event month preceding the start of the experiment as shown in Table 1. This difference could occur by chance, as Table 1 contains 34 tests and therefore one or two false positives would be expected at the 5% statistical significance level. Given this, I include a vector,  $X_i'$ , of pre-experiment controls for each consumer, which also helps to increase the precision of my estimates.<sup>6</sup> This vector contains linear controls for age, credit score, card tenure, the pre-experiment values of all of the primary and secondary outcomes, the number of open credit cards held in their portfolio, total credit card portfolio balances, and total credit card portfolio credit limits, and fixed effects for each of the following: the card's interest rate, credit limit, and the type of autopay the consumer was enrolled in (autopay to the minimum, autopay to the full amount, autopay to a fixed amount, or no autopay).<sup>7</sup>

<sup>6</sup>Goldsmith-Pinkham et al. (2024) highlight how multiple treatments with controls may, in theory, produce biased regression estimates, however, they show that in practice there is only limited evidence of bias in experimental studies. In unreported results, I obtain similar estimates from estimating the effects of each of the treatments in separate regressions. This shows that any contamination bias is small in my setting.

<sup>7</sup>To ensure that no observations are dropped from these regressions with consumer controls, I winsorize

As a robustness check, I also report results using consumer-level fixed effects ( $\gamma_i$ ) instead of this vector of consumer controls. and to enable me to estimate this, I include an additional observation for each consumer for the event time preceding the start of the experiment. These absorb the assignment indicators, so I add observations for the cycle before the experiment ( $\tau = -1$ ) and estimate

$$Y_{i,t} = \delta^{T1(vsC)} (Post_t \times T1_i) + \delta^{T2(vsC)} (Post_t \times T2_i) + \gamma_i + \mu_t + \varepsilon_{i,t}, \quad (3)$$

where  $Post_t$  equals one from cycle zero onward. Each  $\delta^k$  is the post-experiment change for its treatment group relative to the control group, which nets out any pre-experiment difference. The dynamic version replaces  $Post_t$  with the event-time indicators in Equation (1), with  $\tau = -1$  omitted.

My primary outcome in my PAP is payments (%), defined as the sum of payments (\$) divided by the statement balance (\$). If payments (%) is greater than one, it is winsorized at a value of one, and it is assigned a value of one if the statement balance is zero. After the third statement cycle, the minimum detectable effect sizes for this primary outcome, assuming 80% power, are 3.8 percentage points at a 5% significance threshold, 4.7 percentage points at a 1% threshold, and 5.0 percentage points at a 0.5% threshold. My use of regression controls and pooling data across cycles helps to detect smaller differences. This appears sufficiently powerful, given the effect sizes of nudges shrouding minimum payments previously tested in online experiments on hypothetical credit card payment scenarios are substantially larger. For example, in Guttman-Kenney et al. (2023), a nudge shrouding minimum payments causes an increase in hypothetical payments, as a percentage of the statement balance, of 12.4 percentage points.

I also present results for eleven secondary outcomes specified in my PAP. These consist of measuring payments in dollars, aspects of credit card behavior—spending in dollars and as a percentage of the statement balance, revolving debt, and credit limits—, and, because of the non-normal distribution of credit card payments, binary indicators of payment amounts: paying less than the minimum, paying the minimum, paying 50% of the statement balance, paying the full statement balance, and paying the current balance.

My PAP specified evaluating the experiment after three completed statement cycles and ending it then if there was no effect. The lender instead ran it for six in case the effects took

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credit limits at \$1,300, approximately the 99.5<sup>th</sup> percentile. There are five observations with unique credit limit values which I group with other observations that have a neighboring credit limit (randomly assigned above or below). There is one observation with a unique APR that I assign to the next APR value. The number of open credit cards held in their portfolio, total credit card portfolio balances, and total credit card portfolio credit limits variables added as controls are each winsorized at their 99<sup>th</sup> percentiles. None of these choices affect my results.

longer to emerge, a departure from the PAP, so the three-cycle results remain the primary evaluation and I report the six-cycle results alongside them. I report the dynamics over time and discuss my results over this longer horizon. Cycle zero is the month when the experiment started in the field. At this point-in-time, some consumers may have already made payments against their latest statement. Therefore, the effects of the RCT can only be clearly evaluated from cycle one, the first statement with an ending date after the start of the experiment. I follow my PAP in analyzing the card-cycles observed, and given that 99.7% and 95.2% of cards remain open by the end of the third and sixth cycles, and there are no significant differences across treatments (Appendix Table A1), sample attrition is not an issue.

### 3 Conceptual Framework

#### 3.1 The Consumer’s Problem

In this section I set out targeting, the alternative to my pre-registered anchoring hypotheses, and state what each theory predicts for the treatments. Formal statements and proofs are in Appendix C and I return to its implications in Section 6. I model a consumer who chooses a credit card payment  $p$  against her statement balance  $B$ , given spare cash  $a$ , to maximize her utility:

$$U(p) = \underbrace{-C(p; a)}_{\text{cash cost}} + \underbrace{\rho p}_{\text{future benefit}} - \underbrace{F \cdot \mathbf{1}\{p < m\}}_{\text{fee}} + \underbrace{\eta_m \mu(p - m) + \eta_B \mu(p - B)}_{\text{reference dependence}}. \quad (4)$$

The term  $C(p; a)$  is the cost of the cash used to pay  $p$ . Paying uses cash, and the value of a dollar of cash rises steeply as cash runs out, so the cost of paying is convex.

A dollar paid today is worth  $\rho = \delta(1 + r)$  today in reduced future debt, where  $r$  is the monthly interest rate and  $\delta$  the discount factor, so  $\rho$  exceeds one whenever the card’s interest rate exceeds the consumer’s impatience. The problem is static: it takes the statement balance and cash as given and values future debt reduction through  $\rho$ . Appendix C.2 shows it is the reduced form of a dynamic problem. Present bias lowers  $\rho$ , with its implications in Section 6.2.

The payoffs have a notch at the minimum  $m$ : paying less triggers a fee, a delinquency mark, and the risk of losing the card and accessing future credit limit increases, which I treat together as a lump-sum cost  $F$ . They have a corner at the full balance  $B$ , since paying it ends interest.

I take  $m$  and  $B$ , the two economically consequential amounts (Kleven, 2016; O’Donoghue

and Sprenger, 2018; Dean et al., 2026), as the reference points: relative to  $m$ , paying less is a loss and paying more a gain; relative to  $B$ , not paying in full is a loss, for which debt aversion is one foundation (Martínez-Marquina and Shi, 2024), and payments bunch at the balance (Schwartz, 2026).<sup>8</sup> Gains and losses enter the objective through  $\mu(p - m)$  and  $\mu(p - B)$ , where  $\mu(z) = z$  for  $z \geq 0$  and  $\mu(z) = \lambda z$  for  $z < 0$ , with  $\lambda \geq 1$  so that losses weigh more than gains. The weights  $\eta_m, \eta_B \geq 0$  scale how much each reference point matters. When  $\eta_m = \eta_B = 0$  this is the standard model, so bunching at  $m$  and  $B$  does not require loss aversion; loss aversion adds to both. Dean et al. (2026) model reference points as information: they matter most when a decision-maker is uncertain of her own valuation and the reference point is well known, and there can be more than one at a time. A monthly payment decision made against a minimum and a balance printed on every statement fits both conditions. Two features distinguish these reference points from those in the reference-dependence literature: they are set externally at payoff notches rather than chosen (Heath et al., 1999), and there are two of them, whereas that literature has focused on a single reference point and has little to say about how consumers act when two compete (O’Donoghue and Sprenger, 2018).

Maximizing Equation (4) gives a payment that is a step function of cash (Appendix C, Proposition C1):

$$p^*(a) = \begin{cases} B & a \geq B + x_H \\ a - x_H & m + x_H < a < B + x_H \\ m & m + x_F \leq a \leq m + x_H \\ \max\{a - x_L, 0\} & a < m + x_F, \end{cases} \quad (5)$$

where  $x_H$  is the cash buffer a consumer keeps once the minimum is met,  $x_L \leq x_H$  the smaller buffer she keeps when the fee is already unavoidable, and  $x_F < x_L$  the smallest buffer she will accept to avoid the fee. The buffers are where a dollar in hand is worth a dollar paid: with  $v(\cdot)$  the marginal value of cash,  $x_H$  solves  $v(x_H) = \rho + \eta_m + \lambda\eta_B$ , the benefit of paying above the minimum, and  $x_L$  the same with the larger benefit of paying below it, so buffers shrink as the incentive to pay grows. The third line is the *bunching interval*: for cash between  $m + x_F$  and  $m + x_H$  the consumer pays exactly  $m$ , because the incentive to pay drops discontinuously once the minimum is met and the marginal dollar is worth more in the pocket. In the model, then, a consumer who pays the minimum while holding cash to spare is not constrained; she is in the bunching interval, at the target. The buffers vary across consumers and, with cash needs, within a consumer over time, so the thresholds in Equation (5) are specific to a

<sup>8</sup>As Kleven (2016) notes, “an issue that has received relatively little attention in the literature is that kinks and notches may represent reference points.”

consumer statement cycle.

### 3.2 Anchoring Versus Targeting

Following Bartels et al. (2024), I define an anchor as an irrelevant number that pulls choices when it is displayed and stops pulling when it is not (Tversky and Kahneman, 1974), and a target as a relevant number that the consumer aims for and exerts effort to reach whether or not it is displayed (Heath et al., 1999). In my model, the reference points  $m$  and  $B$  are the targets. The treatments change what is displayed and nothing about payoffs, so shrouding the minimum leaves a target payer’s optimum unchanged and raises only the cost of paying exactly  $m$ : a consumer who does not know the amount to the cent either rounds slightly up or probes by entering a low amount, triggering the warning prompt that reveals  $m$ , and then paying it. Targeting predicts more initial payment choices below the minimum that are then revised up to it, no change in payments, and a mass at  $m$  that changes only through rounding. Anchoring, under which the display itself pulls payments down, predicts the reverse: initial payment choices increase, payments increase, and the mass at  $m$  spreads upward across the distribution of payments.

The 50% option lies between the two reference points and has no notch. Under targeting, its only pull is that it is a one-click option, which attracts consumers whose ideal payment is near it from both sides: some who would otherwise type a smaller amount and some who would round up to the full balance. Targeting therefore predicts that shrouding it removes a local spike at 50%, returning that mass to just below 50% and to full payment, with no change in payments elsewhere. Under anchoring the displayed 50% pulls the whole distribution toward it, so shrouding it moves payments throughout the distribution rather than only near 50%.

## 4 Experimental Results

### 4.1 Descriptive Analysis

I start by describing the CDFs of payments for the three experimental groups in Figure 1, after three and six completed statement cycles. Panels A and B show payments as a percentage of the statement balance. Panels C and D show excess payments among consumers who make any payment, where excess payments are the amounts paid above the minimum payment, normalized by the statement balance less the minimum; on the x-axis, “MIN” denotes a payment of exactly the minimum and “FULL” a payment of the full statement balance or more.

There are two takeaways from Figure 1. The first is that there is no substantial difference in the overall distribution of payments between the control group and the two treatments, T1 and T2. The second takeaway is that there are two small differences. Panels A and B show a clear discontinuous increase in the likelihood of payments being at 50% of the statement balance, compared to just below this amount, for the control group and T1, both of which display the 50% payment option, and no discontinuity at 50% for T2, which does not. Panels C and D show that payments at or just above the minimum become less likely in T1 and T2 than in the control group.

To test these differences formally, Table 2 reports t-tests of the differences in means between each treatment and the control group after three completed statement cycles, for the primary outcome and the eleven secondary outcomes. There are no significant differences in payments as a percentage of the statement balance or in nine of the eleven secondary outcomes. The two exceptions are the probability of paying exactly the minimum, which T1 and T2 both reduce by 4.6 percentage points (28%) from a control mean of 16.2%, and the probability of paying exactly 50% of the statement balance, which no consumer in T2 chooses, against 2.7% in the control group, and which adding the 50% option (T1 versus T2) raises by 3.4 percentage points. Results are robust after six cycles (Appendix Table A2).

## 4.2 Regression Analysis of Primary Outcome

I use regressions to estimate the effects of the treatments more precisely than the unconditional t-tests. Table 3 shows the estimates on the primary outcome. Panel A reports the dynamic estimates after three completed statement cycles in columns (1) to (4) and after six in columns (5) to (8). Within each set, the first column includes a constant and no controls, which follows my PAP for completeness; the second adds time fixed effects; the third is the regression specified in Equation (1), with time fixed effects and pre-experiment consumer controls; and the fourth replaces the consumer controls with consumer fixed effects. The results in the no-controls columns are not robust to the inclusion of time fixed effects.

I find no significant effects of either treatment on payments as a percentage of the statement balance, after three or six cycles. After three cycles, column (3) shows that, relative to the control group, shrouding the minimum payment in T1 changes payments by an insignificant 0.2 (s.e. 1.2) percentage points, which is small relative to the control mean of 32.9%, and shrouding both the minimum and the 50% option in T2 changes them by an insignificant 0.9 (s.e. 1.2) percentage points. These estimates are positive, which may suggest a small short-term effect that I do not have the power to detect, but after six cycles, in column (7), the point estimates have declined and turned negative:  $-1.3$  (s.e. 1.2) percentage points for

T1 and  $-0.5$  (s.e. 1.2) for T2. Figure 2 shows the estimates from Equation (1) for every cycle, with T1 in orange and T2 in blue: there are no significant positive effects in any cycle, T2’s estimates are slightly positive up to cycle four and close to zero by cycles five and six, and T1’s start insignificantly negative and end close to zero. Any small short-term effect of the treatments is therefore not sustained. The results are robust to the specifications without consumer controls, columns (2) and (6), and with consumer fixed effects, columns (4) and (8), where the estimates are also insignificant and decline over time.<sup>9</sup>

Panel B of Table 3 increases statistical power by pooling observations across cycles (Equation (2)), with the same ordering of specifications as columns (1) to (4) of Panel A. Pooled estimates average over the dynamics, so they understate declines over time. Panel B shows that T1 and T2 cause no significant changes to payments, with estimates of  $-0.7$  (s.e. 0.8) and  $0.3$  (s.e. 0.8) percentage points, respectively, relative to a control mean of 34.7%.

Comparing T1 to T2 shows the effect of adding a 50% payment option. With time fixed effects and consumer controls, Table 3 shows no significant effect on payments after three or six cycles or when pooling across cycles, and the dynamic estimates (green in Figure 2) are insignificant in every cycle. With consumer fixed effects (column (4) of Panel B), adding the option decreases payments by 3.4 (s.e. 1.5) percentage points, 7.8% of the T2 mean in the cycle before the experiment; the decline is concentrated in cycles zero to two and insignificant from cycle three (Panel A of Appendix Figure A3). Column (4) differences out the pre-experiment gap in payments between T1 and the control group in Table 1, which the other columns do not, and this accounts for the difference.

### 4.3 Regression Analysis of Secondary Outcomes

I now consider my secondary outcomes to better understand the effects of these treatments. Table 4 shows results for each secondary outcome after three cycles using the regression specification with time fixed effects and consumer controls (Equation (1)), and Figure 3 displays the estimates across cycles. Columns (1) and (2) show the effects of T1 and T2 relative to the control group, and column (3) the effect of adding the 50% payment option, comparing T1 to T2.<sup>10</sup>

The primary results are robust to measuring payments in dollars rather than as a percentage of the statement balance, and to examining the statement balance or revolving debt

<sup>9</sup>The only notable difference with the consumer fixed effects specification (Appendix Figure A3) is that T1 temporarily and significantly decreases payments by 2.9 and 3.2 (s.e. 1.4 and 1.5) percentage points in cycles zero and one, an effect that becomes insignificant in later cycles.

<sup>10</sup>Appendix Table A3 shows the secondary outcomes across specifications pooled across cycles, and Appendix Tables A4 and A5 the dynamic estimates after three and six cycles. Appendix Figure A3 presents the dynamic estimates for the models with both consumer and time fixed effects.

(the statement balance less payments against it, bounded at zero), where I find no significant reductions. If anything the effects go the other way: after six statements, shrouding the minimum payment significantly decreases payments in dollars by \$9.3 (s.e. \$3.3), a 12% decline from the baseline mean, and increases revolving debt by \$10.8 (s.e. \$5.3), a 7% increase (Panels A and E of Figure 3; Appendix Table A5). Adding the 50% option raises the statement balance by \$9.6 (s.e. \$4.5) after three cycles, a 4.1% increase over the control mean of \$231, which does not persist to cycle six (\$2.1, s.e. \$6.3). Two potential offsetting responses are absent: the treatments do not change credit limits, and they do not change spending, apart from a temporary increase in cycle zero when the 50% option is added (Panels C and D of Figure 3). Nor do they increase missed payments, except for T1 in cycle six.

The treatments do change the distribution of payments, as the descriptive analysis suggested. Both significantly reduce the probability of paying exactly the minimum, and the effects persist across cycles (Panel H of Figure 3): after three cycles, T1 and T2 decrease this probability by 4.2 (s.e. 1.1) and 3.9 (s.e. 1.1) percentage points, 25% and 24% of the control mean of 16.2%. This pattern, a credit card nudge that moves a proximate outcome such as paying exactly the minimum without moving distal outcomes such as payments or revolving debt, matches the autopay nudge in Guttman-Kenney et al. (2025).

The 50% option moves payments at exactly 50% and at the full balance. T2, which shrouds the option, effectively eliminates payments of exactly 50% of the statement balance, by  $-2.9$  (s.e. 0.3) percentage points after three cycles and persistently across cycles (Panel I of Figure 3), while T1, which retains it, has little effect on this outcome. Adding the option (T1 versus T2) makes consumers 3.6 (s.e. 0.6) percentage points more likely to pay exactly 50%, and temporarily less likely to pay the statement balance in full, an effect that does not persist after cycle two (Panel J). Neither treatment makes consumers more likely to pay in full relative to the control group.

## 4.4 Alternative Payment Choices

What payment amounts are consumers choosing instead of the minimum? I test this by using my pooled regression specification (Equation (2)) and changing the outcome to be indicators for one-dollar intervals of payments relative to the minimum payment amount in Panel A of Figure 4, relative to the 50% statement balance amount in Panel B, and relative to the statement balance amount in Panel C. Panel A shows no effects of either treatment on payments slightly below the minimum, with estimates close to zero, and a concentrated effect just above it: T1 and T2 make consumers significantly more likely to pay up to one dollar more than the minimum, by 57% and 45% respectively, relative to the 1.2% of consumers in

the control group who choose this amount. This is consistent with consumers rounding up to the next whole dollar rather than entering the exact minimum to the cent.

I find no consistent evidence that the treatments lead consumers to make payments much larger than the minimum. Panel A of Figure 5 repeats the exercise across the full distribution of payments as a percentage of the statement balance, with an indicator for each five-percentage-point interval as the outcome, and T1 has no significant effect at any point in the distribution. The mass of payments at the minimum therefore appears to move one dollar up and nowhere else, which is what targeting predicts and anchoring does not.

The 50% option acts locally, drawing consumers from both sides. Panel B of Figure 4 shows that when the option is displayed, consumers are more likely to pay exactly 50% of the statement balance or up to one dollar more, but not more than that, and that T2 removes these payments. Adding the option (T1 versus T2) makes consumers significantly more likely to pay between 45% and 55% of the statement balance and less likely to pay just below that range (30% to 45%) or to pay in full (Panel A of Figure 5), consistent with consumers moving to 50% from just below it and from full payment. There are no significant effects around the statement balance amount (Panel C of Figure 4). As Figure 1 shows, there is a limited mass of payments near 50%, and because the increases for some consumers are offset by decreases for others, these effects are not large enough to change overall payments. The pull away from full payment is concentrated in cycles zero to two, while the spike at exactly 50% persists, consistent with Remark C1 in Appendix C.

Shrouding the minimum payment also changes the heuristics that consumers use. Appendix Table A6 shows that T1 and T2 make consumers 0.76 (s.e. 0.25) and 0.94 (s.e. 0.23) percentage points significantly less likely to pay multiples of the minimum payment amount—the “multiple heuristic” documented by Medina and Negrin (2022). This is visible in Panel A of Figure 4 as the significant negative coefficient at \$11 more than the minimum, often a payment of \$22 or two times the common minimum of \$11. Instead, consumers are 2.51 (s.e. 0.62) and 3.61 (s.e. 0.62) percentage points significantly more likely to pay small, round-numbered amounts, which are also common choices in other credit card data (Sakaguchi et al., 2026). Consumers therefore appear to express their payment targets in terms of the minimum when it is salient, and in round dollar amounts when it is not.

## 4.5 Initial Payment Choices

I observe clickstream data that records each payment choice a consumer makes in the app, including the initial payment choice in each statement cycle and any revisions to it. Initial choices are especially informative for evaluating anchoring, which relies on consumers placing

too much weight on an initial piece of irrelevant information and adjusting insufficiently from it. In my treatments the minimum payment is not shown before the consumer’s first choice, so the two accounts in Section 3 predict opposite movements. If anchoring lowers payments, as proposed in Stewart (2009) and Keys and Wang (2016), initial choices should shift up when the minimum is shrouded, with less bunching at or near the minimum, as observed in online experiments (e.g., Guttman-Kenney et al., 2023). If consumers target the minimum, initial choices should shift down, as consumers who do not know the amount to the cent enter a low amount to reveal it. Prior work with administrative data, such as Keys and Wang (2016), could not observe this margin because the minimum was always displayed.

I take the first payment choice of each consumer in each statement cycle and apply my pooled regression specification to increase power. Panel A of Table 5 shows that T2 makes consumers significantly more likely to initially choose a lower payment amount, measured by adapting my primary outcome to the *initial* payment choice as a percentage of the statement balance: the effect is  $-4.33$  (s.e. 0.81) percentage points, relative to a control mean of 26.83%. Adding the 50% payment option raises initial choices by 3.43 (s.e. 1.07) percentage points.

The main margin is initial choices below the minimum. T1 and T2 make consumers 19.44 (s.e. 1.16) and 30.50 (s.e. 1.02) percentage points more likely to initially choose an amount less than the minimum, compared to a control mean of 33.74%, and adding the 50% option makes them 11.06 (s.e. 1.38) percentage points less likely to do so because the 50% option gives some consumers a one-click amount, so fewer type an amount at all.<sup>11</sup>

Appendix Figure A4 shows the full distributions of first and last payment choices in a cycle, pooled across cycles, with solid lines for first choices and dashed lines for last choices, in dollars (Panels A and B), relative to the minimum payment (Panels C and D), and as a percentage of the statement balance (Panels E and F). Consumers in T1 and T2 are much more likely to first choose zero, an amount close to zero, or nothing at all, which I classify as zero, and the large spike at exactly \$11, a common minimum payment amount, is much smaller than in the control group. In Panels C and D, values to the left of zero are choices below the minimum, and the spike near  $-\$11$  corresponds to consumers choosing zero or near-zero amounts while having an \$11 minimum due. The pattern is robust to excluding blank and zero-dollar clicks, which may be warranted if some of these are accidental rather than purposeful attempts to reveal the minimum (Panel B of Table 5; Panels B, D, and F of Appendix Figure A4). The effects are smaller in cycle zero, when some consumers had

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<sup>11</sup>One reading of this result is that consumers are so habituated to the minimum that shrouding it once cannot undo the habit. In the terms of Section 3, a number that continues to guide choices when it is no longer displayed is a target, not an anchor, so this reading is targeting under another name. It does imply that consumers with no prior experience of credit cards or minimum payments may respond differently, which the experiment cannot test.

already paid before the experiment launched, and stable from cycle one onward, so consumers learn to hack the nudge immediately (Appendix Figure A5).

Panel B of Figure 5 shows the same effects in regression form, using the initial payment choice as a percentage of the statement balance as the outcome in place of the actual payment in Panel A. T1 and T2 make consumers significantly more likely to choose a zero payment, with the mass coming from consumers who are less likely to choose initial payments of 5% to 25% of the statement balance; excluding zeros and no-choice clicks, the effects are positive for 0% to 5% and negative for 5% to 20% (Appendix Figure A6). Around 50% the two treatments differ: T2 makes consumers more likely to initially choose to pay in full, consistent with the full balance being more salient, whereas adding the 50% option makes them less likely to. By the last choice in a cycle, the treatment differences have closed except around 50% (dashed lines in Appendix Figure A4), consistent with the payment results in Panel A of Figure 5.

Shrouding the minimum payment also adds friction to the payment journey. T1 and T2 cause significant increases of 0.4 (s.e. 0.11) and 0.85 (s.e. 0.11) in the average number of payment choices made per statement cycle, from 3.8 in the control group, while adding the 50% option reduces the number of choices (Panel A of Table 5).

## 4.6 Asymmetric Bunching

Recent work in marketing (Bartels et al., 2024; Herzog et al., 2025) developed “value proximity analysis” to evaluate whether a focal value is an anchor or a target from the shape of the distribution around it: the share of observations at or above the value, within windows that expand from the closest 1% of observations to all of them. An anchor gives a share near 50% in the narrowest window; a target gives a share near 100% that falls as the window widens. I regress the share on the window width, treatment indicators, and their interactions, separately for the minimum, the 50% amount, and the statement balance (Appendix D gives the specification; Appendix Table A7 and Figure A7 report the results).

At the minimum and the statement balance, the intercept alone cannot separate anchoring from targeting because the fee and the end of interest produce the asymmetry whether or not consumers are reference dependent (Proposition C1). What separates them is what happens when the amount is shrouded. Under anchoring, the shrouding treatments should move the intercept toward 50% and change the slope. Under targeting, neither should move.

The asymmetry around the minimum is nearly complete, with an intercept of 98.5% and a negative slope; T1 changes neither, and T2 lowers the intercept by 2.8 (s.e. 1.0) points with the slope unchanged. The statement balance shows the same pattern, with an intercept

near 90% and no treatment effects. The 50% amount, which has no notch, has an intercept of 77%, between the two benchmarks, and displaying it raises the intercept by 12.9 (s.e. 2.9) points and steepens the slope by 0.12 (s.e. 0.03): its shape depends on whether it is displayed, as the model predicts for an amount at which payoffs do not change.

## 4.7 Autopay

Both autopay and non-autopay enrollees are sensitive to the shrouding of minimum payment amounts in online experiments (Guttman-Kenney, 2024). Autopay enrollment does not explain the results of my field experiment. One potential reason for the null effects could be that consumers who pay by autopay may never view the manual payment screen, which would attenuate any effect among them. My main results already control for autopay enrollment, and even among the 41% of consumers enrolled in autopay before the experiment, 83% of the total value of their payments is made manually (as Guttman-Kenney et al., 2025 also find). Autopay enrollment choices are highly persistent (Guttman-Kenney et al., 2025), so I can re-estimate my main regressions on the consumers not enrolled in autopay before the experiment. Doing so, I find no significant increase in payments; T1 significantly reduces payments in dollars after six cycles, with no significant change in payments as a percentage of the statement balance (Appendix Table A8).

## 4.8 Portfolio Effects

All of my outcomes are measured on the card in the experiment, but consumers often hold other credit cards with different lenders. Ideally, as discussed in my pre-analysis plan, I would evaluate the effects of the nudges on total payments across the portfolio of credit cards held by a consumer. As documented in Guttman-Kenney and Shahidinejad (2025), however, consumer credit reporting data in the United States does not include credit card payments information for any of the six largest credit card lenders, and only 24% of credit cardholders have this information across all their cards, nor does it include interest rates on their credit card portfolio. Instead, I can only estimate the effects on the portfolio of statement balances observed in the linked Equifax credit reporting data, a noisy proxy for the portfolio of revolving debt, although the noise is mainly an issue for high (721+) credit score consumers (Guttman-Kenney and Shahidinejad, 2025); this is also the portfolio measure used by Keys and Wang (2019) (their Online Appendix Figure A7). Appendix Table A9 shows, as expected, no evidence of the treatments reducing the portfolio of statement balances, consistent with the nudges not reducing consumer debt.

Even though the nudges do not increase payments on the card in the experiment, con-

sumers could in principle have increased payments on other cards they hold. I cannot observe those payments, but three pieces of evidence make this unlikely. The nudging literature finds the strongest effects on the most proximate outcomes (e.g., Beshears and Kosowsky, 2020; Guttman-Kenney et al., 2025), which here would be the card in the experiment. Consumers tend to allocate payments across cards not to minimize interest costs but in proportion to their statement balances, the “balance-matching heuristic” (Gathergood et al., 2019; Ponce et al., 2017; Hirshman and Sussman, 2022), so unchanged payments and balances on the card in the experiment predict unchanged payments on other cards. And the credit card autopay nudges in Adams et al. (2022) and Guttman-Kenney et al. (2025), which also had no effect on payments on the cards in those experiments, had no effect on payments across the portfolio either.

## 5 Heterogeneity

In theory, the welfare effects of a nudge can be positive even if it has no average effect on behavior (Allcott et al., 2025; List et al., 2026), though a regulator would need to show that the benefits to some consumers outweigh the costs to others and the costs of compliance. In this experiment, the largest welfare gains would come from consumers who are able to pay more to reduce their credit card debt but do not, and anchoring to the minimum is one reason they might not, and the one the design can test. I therefore examine heterogeneous treatment effects by whether consumers appear to be constrained, motivated by the online experiment in Guttman-Kenney et al. (2023), where shrouding the minimum caused the largest increases in hypothetical payments among consumers who reported no financial distress. I measure whether a consumer is constrained in three ways: pre-experiment payment behavior on the card in the experiment, pre-experiment credit card utilization, and liquid cash available before the experiment.

### 5.1 Near-Minimum Payers

Not all consumers in my data are liquidity-constrained, since the majority pay more than the minimum (Figure 1 and Table 1), the corner a constrained consumer would attempt to reach. A more rigorous way to isolate consumers who do not appear constrained is to follow Keys and Wang (2019) in identifying near-minimum payers: consumers who pay slightly more than the minimum are not constrained at the minimum, and would be the most susceptible to the nudge if their choices reflect non-standard preferences. I classify 18% of consumers as near-minimum payers, those who paid more than the minimum but no more than \$50 above

it, without paying in full, in at least half of the six statement cycles before the experiment (or the cycles available for newer cards), close to the 20% that Keys and Wang (2019) classify the same way. I pool data across cycles to increase power.

If anchoring were the explanation, payments for near-minimum payers should increase under both T1 and T2. That is not what I find in Appendix Table A10. The estimates for this group are less precise than my main analysis and should be read with more caution, but there is no evidence that shrouding the minimum significantly increases their payments: the effect of T1 is 2.89 (95% C.I.  $-0.50, 6.28$ ) and of T2  $-0.56$  (95% C.I.  $-3.58, 2.46$ ) percentage points. There is also no significant effect for the remaining consumers, as expected, since that group includes consumers who pay only the minimum and may be constrained, and consumers who already pay in full. The results are similar when payments are measured in dollars (Panel B of Appendix Table A10).

## 5.2 Credit Card Utilization

A common way to evaluate liquidity constraints is whether a cardholder’s balance is at or close to the credit limit (e.g. Gross and Souleles, 2002), and lender-driven limit increases raise borrowing and spending even among cardholders whose balances were far below their limits before the increase (Agarwal et al., 2018; Aydin, 2022; Yin, 2026; Agarwal et al., 2025). I split consumers into tertiles of their pre-experiment credit card utilization, the statement balance divided by the credit limit, with full results in Appendix Table A11.

There are no significant effects of T1 in any tertile. For T2, there is one significant increase in payments, in the lowest utilization tertile, of 3.7 (95% C.I.  $0.5, 6.8$ ) percentage points; it is not robust to measuring payments in dollars, where the estimate for the same group is  $-\$1.5$  (95% C.I.  $-\$9.6, \$6.6$ ) relative to a control mean of  $\$80.5$ . Splitting instead by tertiles of portfolio utilization, observed in the linked Equifax credit reporting data, I find no significant effects of T1 or T2 in any tertile, including the least utilized. That the one significant increase appears for low-utilization cards rather than low-utilization consumers is weakly suggestive of payment choices not being fully coordinated across cards, as under mental accounting (Gelman and Roussanov, 2024).

Consumers in my experiment frequently have available credit across their portfolio of cards: median and mean portfolio utilization are 25% and 36%, and only 16% of consumers have portfolio utilization of 90% or more, a standard threshold for being effectively constrained (Gibbs et al., 2025). Consumers are therefore generally not at a binding borrowing constraint. This is the pattern that Lee and Maxted (2026) document in the Survey of Consumer Finances, where consumers with approximately zero liquid assets revolve intermediate

levels of credit card debt rather than sitting at their limits, contrary to how hand-to-mouth consumers are modeled in heterogeneous-agent consumption models (e.g. Kaplan and Violante, 2022). I find it in daily administrative data on liquid assets and credit cards, which avoid the single-point-in-time measurement and known underreporting of credit card debt in the SCF (Zinman, 2009).

### 5.3 Liquid Cash

The final source of heterogeneity I examine is consumers’ liquid cash balances before the start of the experiment, calculated from the linked Plaid data. Consumers with liquid cash available forgo the interest they would save by paying down more of their credit card debt, holding cash that earns little or no interest while paying 15% or more on their credit card; this is costly, as is how consumers allocate payments across debts (Gathergood et al., 2019; Hirshman and Sussman, 2022; Katz et al., 2026). I follow Guttman-Kenney et al. (2025) in using the *minimum* liquid cash balance over the 90 days before the experiment, which captures the mismatch in timing between income and expenditure: a consumer may temporarily hold cash that is committed to upcoming expenses such as rent, and is therefore not spare cash to pay down debt. I also show a point-in-time balance. Because what counts as “high” liquid cash is a judgment, I show results for three thresholds: \$1+, \$501+, and \$1,001+. When interpreting these splits, liquid cash is an equilibrium outcome that can reflect traditional economic reasons, financial illiteracy, and behavioral explanations (Gelman, 2022; Jørring, 2024).

Consumers in my experiment frequently have limited liquid cash. Only 4% have minimum liquid cash balances of at least \$1,001 over the 90 days, against 30% with a point-in-time balance of at least \$1,001; 7% have minimum balances of at least \$501, against 43% point-in-time; and 63% have a minimum balance of \$1 or less, that is, they are overdrawn at some point. This pattern of frequently holding limited liquid cash appears general among consumers who revolve credit card debt. Guttman-Kenney et al. (2025) find that only 11% of cardholders enrolling in autopay at a large U.K. bank have at least £1,001, daily data show far less co-holding of cash and overdraft debt in Iceland than monthly data suggest (Gathergood and Olafsson, 2026), and financial diaries document how few consumers hold more than a few months of liquid assets (Morduch and Schneider, 2017).

I find no subgroup defined by liquid cash that significantly increases its credit card payments when the minimum is shrouded (Appendix Table A12 for payments as a percentage of the statement balance; Appendix Table A13 for dollars). Consumers with low minimum liquid cash do not increase their payments, which is expected, since paying more would require

them to change other behavior, such as spending less or earning more. The small subset with high minimum liquid cash does not significantly increase payments either: for the 4% with minimum balances of at least \$1,001, the estimate for T2 is 5.9 (s.e. 4.6) percentage points, a noisy estimate whose 95% confidence interval contains economically meaningful effects of either sign. For the remaining 96%, the T2 estimate is 0.21 (s.e. 0.82) percentage points, and the 95% confidence interval for the difference between the two groups runs from  $-3.4$  to  $14.7$  percentage points, so the one margin the data do not settle is the small group where anchoring would predict the largest effect and where the experiment has the least power.

Across all three measures of being constrained, there is no subgroup with a systematic response to shrouding the minimum. Few consumers could pay more without changing other behavior: 96% of cardholders in this market never held more than \$1,001 in liquid cash over the ninety days before the experiment. Anchoring predicts the largest effects among consumers with cash to spare, whereas targeting predicts none anywhere.

## 6 Implications of Targeting

Sections 4 and 5 are consistent with the targeting framework set out in Section 3. In this section, I explain why, what present bias adds to it, and what does move payments.

### 6.1 Why Anchoring Appears in Hypotheticals but Not in the Field

I model anchoring as a psychological cost of straying from the displayed number. The weight the anchor receives is  $\kappa_A/(\kappa_A + |U''|)$ , where  $\kappa_A$  is the anchor's psychological strength and  $|U''|$  the curvature of the economic objective (Appendix C, Proposition C5): near one when the objective is flat, near zero when it is steep, and exactly zero at a kink or notch. In a hypothetical scenario the objective is flat, since there is no cash scarcity, no fee, and no minimum the respondent has to meet, so the displayed number largely determines the answer. By contrast, a cash-poor consumer in the field faces a steep objective and sits at a kink. The same psychology produces large anchoring effects on hypothetical payments and none on actual payments, consistent with anchoring surviving in low-stakes experiments and fading under incentives (Fudenberg et al., 2012; Maniadis et al., 2014; Alevy et al., 2015).

### 6.2 What Present Bias Adds

My framework takes  $\rho$  as given. For the bunching interval to hold consumers with cash to spare, as Section 5 finds,  $\rho$  must be low for many of them: a dollar in the pocket must be worth more than a dollar off a card charging 20%. Present bias, which is well documented

among cardholders (Meier and Sprenger, 2010; Kuchler and Pagel, 2021; Lee and Maxted, 2026; Laibson et al., 2026), does this by scaling  $\rho$  to  $\beta\delta(1+r)$ .

The two ingredients interact in both directions. Present bias fills the bunching interval with consumers who could pay more, and the target stops present bias from taking payments lower: with targeting alone, only cash-poor consumers would sit in the bunching interval, and with present bias alone, nothing but the fee would hold payments up.

In the dynamic version of the model (Appendix C.2), a stationary pattern of “pay the minimum, spend the rest, revolve the balance” exists if and only if  $\beta\delta(1+r)$  plus the reference pulls is at most one, and cannot exist for a time-consistent consumer, who pays down expensive debt whenever she has cash (Proposition C6). This is the hand-to-mouth-but-not-constrained pattern of Lee and Maxted (2026), who show the same force at the borrowing margin, and the coholding of cash and revolving debt documented in Section 5.

Present bias makes the minimum a commitment device, provided the higher payment is not re-borrowed. A consumer in the bunching interval does not respond to  $\rho$  at all, but responds one for one to  $m$  (Proposition C7). This holds whether  $\rho$  is low because she discounts the future or because she misperceives interest costs (Stango and Zinman, 2009; Han and Yin, 2026), which is why disclosures about interest costs and repayment horizons have had small and short-lived effects even where they reduce confusion (Agarwal et al., 2015; Keys and Wang, 2016; Seira et al., 2017; Adams et al., 2022).

As a commitment device, a minimum payment is a minimum-savings rule of the kind Amador et al. (2006) show is optimal for present-biased agents facing liquidity shocks: a floor that commits, with flexibility above it, at a cost to those for whom the floor binds.

The condition that the higher payment is not re-borrowed is the present-bias dilemma of Maxted (2025): where borrowing limits do not bind, a forced payment can be charged back, and a self-imposed financial commitment then cannot raise welfare. The minimum escapes the dilemma only where consumers cannot re-borrow the payment, because their limits bind, or do not, for example, through separate mental accounts for paying and spending.

Whether re-borrowing undoes the commitment is an empirical question, and the sufficient-statistics approach of Allen et al. (2026) to the welfare effects of higher minimum payments in Quebec answers it inside the statistic: the welfare gain is the internality per dollar,  $\gamma \approx 1 - \beta$ , times the change in revolving debt net of any charging back, and their estimates show revolvers with ample unused credit reducing debt by more than the mandate required. Remark C2 derives  $\gamma$  from the model and shows it is exact for interior payers and an upper bound for consumers at the kink.

Present bias also makes plans differ from actions. When a present-biased consumer plans next month’s payment, both the payment and its benefit lie in the future and  $\beta$  cancels:

she plans like a time-consistent consumer and pays like a present-biased one (Kuchler and Pagel, 2021). A hypothetical payment question elicits the plan, which is made against a flat objective and so is display-sensitive (Section 6.1). The field observes the action, which is made at a notch. This may explain why the largest hypothetical effects appear among the “unconstrained,” whose plans are least tied down by cash.

The goals literature shows that a present-biased agent can discipline herself with a self-set goal as a reference point (Heath et al., 1999; Hsiaw, 2013). This is not the kind of commitment the irrelevance result of Maxted (2025) rules out: that result concerns financial instruments in a smooth environment, and he identifies personal rules of exactly this kind as what breaks it. My model puts a limit on that idea: a self-set target is fully self-enforcing if and only if loss aversion around it is at least  $(1 - \beta)\delta(1 + r)$  (Proposition C8). Round-number payments above the minimum are such targets. The minimum is an externally set target of the same kind, backed by a fee, which is why it appears to hold even for consumers whose own loss aversion would not sustain a target.

### 6.3 What Moves Payments

In the model, raising the minimum moves payments through three channels that a nudge does not have. First, it moves every consumer in the bunching interval one for one, including near-minimum payers, whom the old minimum did not bind but the new one reaches, and it pushes those who cannot reach the new minimum into delinquency, as d’Astous and Shore (2017) find (Proposition C2(i)). Second, it shrinks the distance to the full balance, so that some target payers jump to paying in full (Proposition C2(iv)), in the way a closer finish line pulls marathon runners (Allen et al., 2017). Schwartz (2026) provides suggestive evidence of this channel: consumers appear more likely to pay in full when the gap is smaller between the full balance and the minimum.

Third, a higher minimum also moves consumers for whom it does not bind, because some above-minimum payers set their target in terms of the minimum rather than in dollars. A consumer whose rule is “double the minimum” pays \$22 when the minimum is \$11 and \$30 when it is \$15; the minimum never binds for her, yet her payment rises by \$8 provided she has the cash to keep to her rule, and a consumer whose rule is “the minimum plus \$20” moves one for one (Proposition C9). Medina and Negrin (2022) estimate that two-thirds of the response of consumers whom the minimum does not bind runs through such multiples of the minimum, and consumers describe such rules when asked (“I always pay at least \$50 to \$100 over the minimum,” “double the minimum”; Katz et al., 2026). A weaker form of this channel applies to consumers with no such rule: under diminishing sensitivity, gains above

the minimum are evaluated relative to it, so every interior payment rises with  $m$ , though by less than one for one and by most for those closest to it (Proposition C2(iii)).

My experiment also shows that targets are indexed to the minimum: when it is shrouded, payments at multiples of it fall and small round numbers take their place (Appendix Table A6), because a target indexed to an amount the consumer cannot see becomes a target in dollars. That implies that where the minimum is not displayed, a minimum increase should move above-minimum payers less, since their targets are no longer indexed to it. Together, the three channels are why unconstrained consumers respond to minimum increases in Keys and Wang (2019), Medina and Negrin (2022), Castellanos et al. (2026), Katz et al. (2026), and Allen et al. (2026), and why paydown exceeds what the mandate compels in Allen et al. (2026). These responses are consistent with targeting but cannot distinguish it from anchoring, as my experiment can.

The general principle is that reference points which coincide with payoff notches are robust to how they are displayed, while reference points that are only displayed are weak and act locally. The three-year payment scenario disclosed under the CARD Act is such an amount, between the two targets with no notch, and its small effects (Agarwal et al., 2015; Keys and Wang, 2016) are what the model predicts for the 50% option. What moves payments is the location of the notch, not the display of the number.

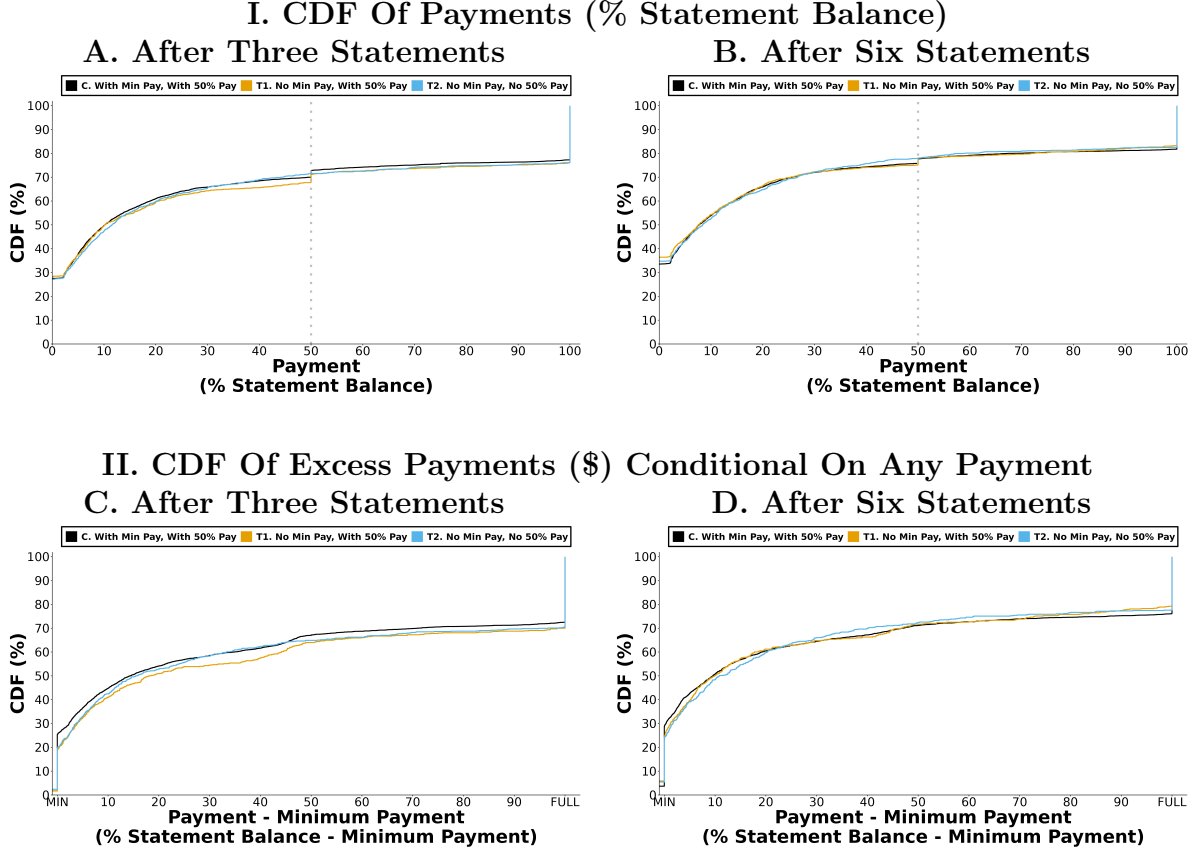
## 7 Conclusions

A large body of work across economics, finance, management, marketing, and psychology attributes low credit card payments to anchoring on the minimum payment. In a field experiment on 6,714 consumers over six statement cycles, shrouding the minimum payment amount does not increase payments, and adding a 50% payment option does not either, instead temporarily making consumers less likely to pay in full. The large increases in hypothetical payments found in online experiments do not extrapolate to the field.

My results are consistent with the minimum acting as a target, a reference point at a notch in economic payoffs, rather than as an anchor that lowers payments: when it is shrouded, consumers initially choose less than the minimum and then revise up to it. The conceptual framework I develop shows why the two accounts differ. A target at a payoff notch does not depend on whether it is displayed, anchoring matters only where the economic objective is flat, as in a hypothetical scenario, and intermediate amounts cannot compete with the minimum or the full balance. Consumers are not at their borrowing constraints but frequently have limited liquid cash, and their payments respond to changes in the minimum but not to changes in its display.

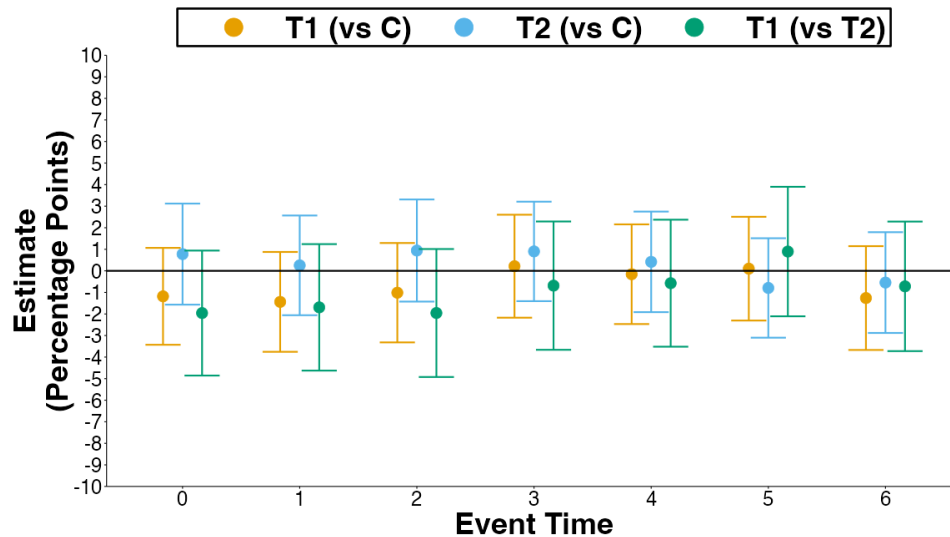
## 8 Figures & Tables

Figure 1: Distribution Of Credit Card Payments, After Three and Six Statements



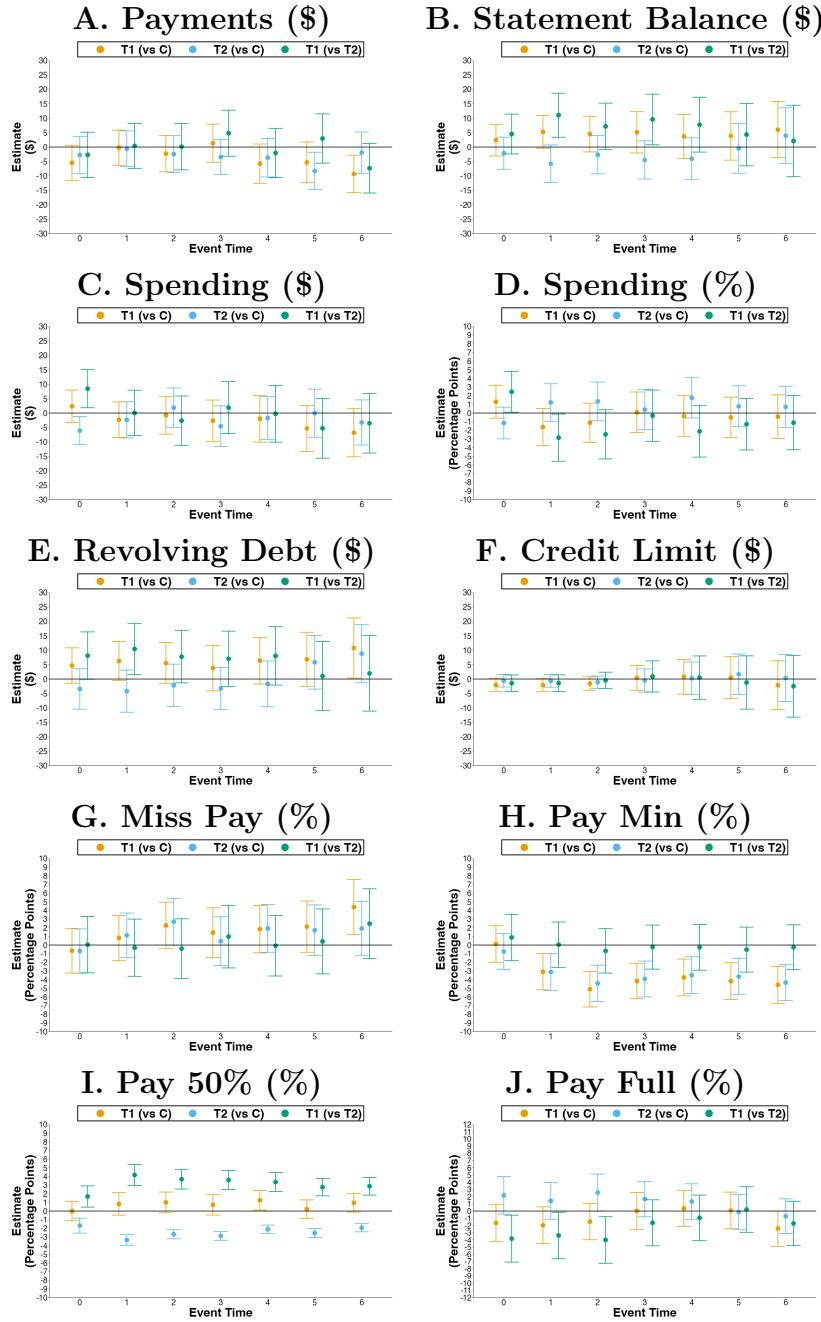
*Notes:* These panels show the CDFs of payments. Panels A and B show the CDF of payments as a percent of the statement balance. The dotted vertical line in Panels A and B denote 50% of the statement balance amount. Panels C and D show the CDF of excess payments, defined as the payments less the minimum payment (as a percent of the statement balance less minimum payment) conditional on consumers making any payment. In panels C and D, MIN denotes consumers that make exactly the minimum payment and FULL denotes those that pay their statement balance in full (or more). In all panels, each color shows the CDF for the control group in black, treatment group T1 (that shrouds the minimum payment option) in orange, and treatment group T2 (that shrouds both the minimum payment option and also the 50% payment option) in blue. In panels A and C, the outcome is measured after three completed statement cycles, and in panels B and D it is measured after six completed statement cycles. Panel A contains 6,693 consumers observed in cycle three, and Panel C is the subset of the consumers that make any payment in the third cycle, which is 4,592 consumers. Panel B contains 6,389 consumers observed in cycle six, and Panel D is the subset of the consumers that make any payment in the sixth cycle, which is 3,983 consumers.

**Figure 2: Dynamic Effects On Primary Outcome: Credit Card Payments (% Statement Balance)**



*Notes:* The outcome in this figure is payments as a percent of the statement balance. This figure shows the estimates from interactions between treatment indicators and time indicators, with 95% confidence intervals based on clustering standard errors at the consumer-level. In orange and blue are respectively the estimates from a regression where both T1 and T2 are each separately interacted with time indicators, with the control group as the omitted category. In green are the estimates from a T1 interaction, with the T2 group as the omitted category and group C not included in the regression. Both regressions also include time fixed effects for each statement end date and a vector of pre-experiment consumer controls listed in Section 2.3. The regressions use data on 6,714 consumers and  $N = 46,519$  observations (statement cycles 0 to 6).

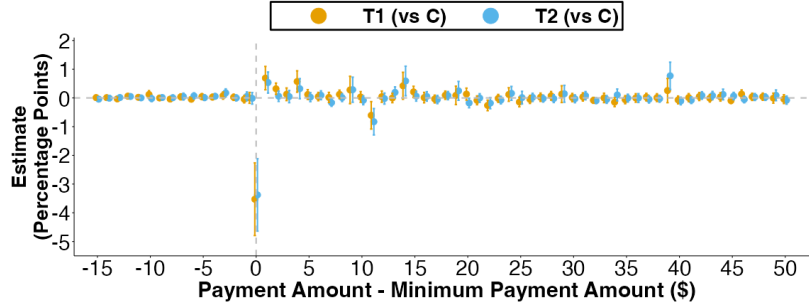
Figure 3: Dynamic Effects On Secondary Outcomes



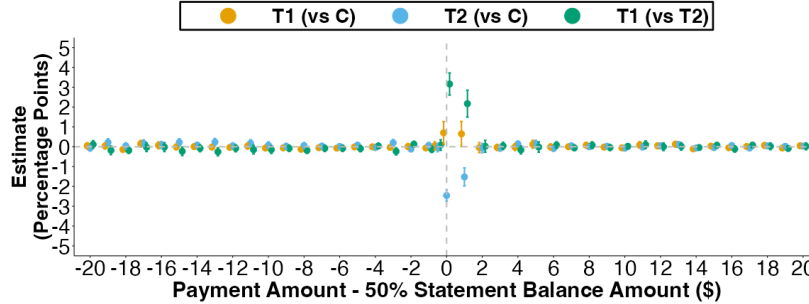
*Notes:* Each panel shows results for a different outcome. This figure shows the estimates from interactions between treatment indicators and time indicators, with 95% confidence intervals based on clustering standard errors at the consumer-level. In orange and blue are respectively the estimates from a regression where both T1 and T2 are each separately interacted with time indicators, with the control group as the omitted category. In green are the estimates from a T1 interaction, with the T2 group as the omitted category and data for group C is not used in these regressions. All regressions also include time fixed effects for each statement end date and a vector of pre-experiment consumer controls listed in Section 2.3. The regressions use data on 6,714 consumers and  $N = 46,519$  observations (statement cycles 0 to 6).

Figure 4: Effects On Payments Proximate To Focal Amounts

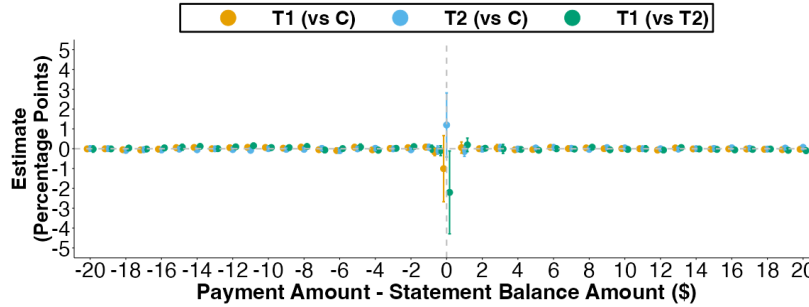
A. Payment Amount Less Minimum Payment Amount



B. Payment Amount Less 50% Statement Balance Amount

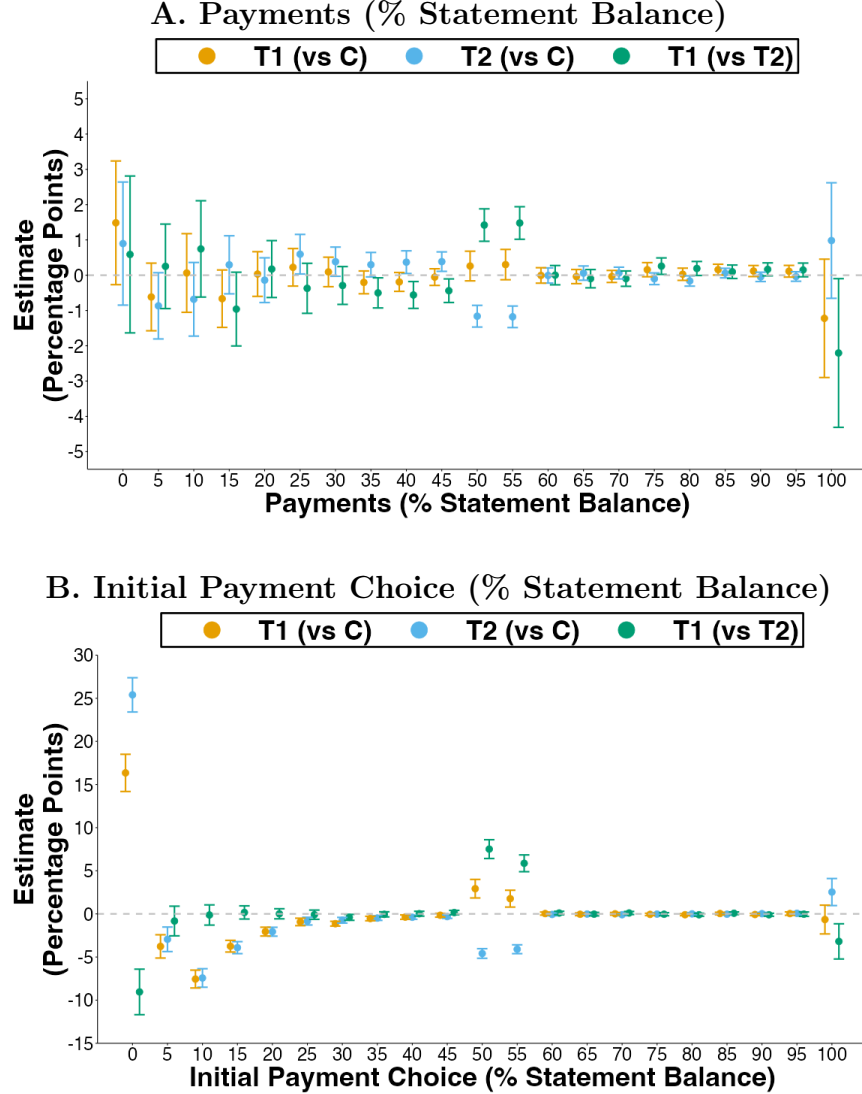


C. Payment Amount Less Statement Balance Amount



*Notes:* This figure shows the estimates on treatment indicators, showing the average effects pooled across statements, with 95% confidence intervals based on clustering standard errors at the consumer-level. In orange and blue are respectively the estimates from a regression with treatment indicators for each of T1 and T2, with the control group as the omitted category. In green are the estimates from a T1 interaction, with the T2 group as the omitted category and data for group C is not used in these regressions (green dots are excluded from Panel A to ease presentation since all of these estimates are insignificant from zero). All regressions also include time fixed effects for each statement end date and a vector of pre-experiment consumer controls listed in Section 2.3. The regressions use data on 6,714 consumers and  $N = 46,519$  observations (statement cycles 0 to 6). Each panel shows results for a different measure. Within each panel, the estimates corresponding with each x-axis level show results for a different binary outcome that takes a value of one if a consumer paid that amount, and zero otherwise. The estimates corresponding to where there is the vertical dashed line at zero represent an outcome where a consumer pays exactly that amount. The other estimates are \$1 intervals of payments.

Figure 5: Effects On Payment Distribution



*Notes:* Both panels of this figure show the estimates on treatment indicators, showing the average effects pooled across statements, with 95% confidence intervals based on clustering standard errors at the consumer-level. In orange and blue are respectively the estimates from a regression with separate indicators for T1 and T2, with the control group as the omitted category. In green are the estimates from a T1 indicator, with the T2 group as the omitted category and data for group C is not used in these regressions. All regressions also include time fixed effects for each statement end date and a vector of pre-experiment consumer controls listed in Section 2.3. The regressions in Panel A use data on 6,714 consumers and  $N = 46,519$  observations (statement cycles 0 to 6) for consumers ultimate payments that cycle. The regressions in Panel B use data on 5,644 consumers and  $N = 25,928$  observations, the initial payment choices of consumers in the app each cycle (statement cycles 0 to 6). The estimates for each x-axis level show results for a different binary outcome that takes a value of one if a consumer paid 5 percentage points up to that amount, and zero otherwise. The estimates corresponding to the x-axis value of zero are if a consumer made no payment, and those corresponding to the x-axis value of one are if a consumer paid in full.

**Table 1: Pre-Experiment Summary Statistics**

| Outcome                 | C Mean<br>(1) | T1 Mean<br>(2) | T2 Mean<br>(3) | T1 – C: Mean<br>(4) | [p-value]<br>(5) | T2 – C: Mean<br>(6) | [p-value]<br>(7) |
|-------------------------|---------------|----------------|----------------|---------------------|------------------|---------------------|------------------|
| Age                     | 45.51         | 45.68          | 45.84          | 0.17<br>(0.38)      | [0.652]          | 0.33<br>(0.39)      | [0.390]          |
| Credit Score            | 561.70        | 561.06         | 562.10         | −0.64<br>(2.06)     | [0.757]          | 0.40<br>(2.10)      | [0.847]          |
| APR (%)                 | 15.71         | 15.72          | 15.72          | 0<br>(0.01)         | [0.826]          | 0<br>(0.01)         | [0.850]          |
| Tenure (days)           | 265.58        | 257.53         | 262.24         | −8.05<br>(5.10)     | [0.114]          | −3.34<br>(5.10)     | [0.512]          |
| Autopayer (%)           | 41.19         | 43.59          | 39.69          | 2.39<br>(1.68)      | [0.154]          | −1.50<br>(1.65)     | [0.362]          |
| Payments (%)            | 45.06         | 48.00          | 44.34          | 2.95<br>(1.44)      | [0.040]          | −0.72<br>(1.39)     | [0.608]          |
| Payments                | 77.07         | 85.00          | 76.99          | 7.93<br>(4.21)      | [0.060]          | −0.08<br>(3.79)     | [0.984]          |
| Statement Balance       | 207.94        | 206.38         | 208.75         | −1.55<br>(7.57)     | [0.837]          | 0.82<br>(7.41)      | [0.912]          |
| Spending                | 98.13         | 99.81          | 101.30         | 1.68<br>(4.67)      | [0.719]          | 3.17<br>(4.82)      | [0.510]          |
| Spending (%)            | 50.11         | 51.95          | 50.25          | 1.84<br>(1.46)      | [0.207]          | 0.14<br>(1.44)      | [0.925]          |
| Revolving Debt          | 149.09        | 144.62         | 149.98         | −4.47<br>(7.15)     | [0.532]          | 0.89<br>(6.98)      | [0.898]          |
| Credit Limit            | 292.09        | 287.64         | 294.11         | −4.45<br>(8.16)     | [0.585]          | 2.02<br>(8.13)      | [0.803]          |
| Miss Pay (%)            | 17.00         | 14.59          | 15.88          | −2.41<br>(1.21)     | [0.047]          | −1.12<br>(1.24)     | [0.365]          |
| Pay Min (%)             | 17.48         | 17.38          | 16.79          | −0.10<br>(1.29)     | [0.937]          | −0.69<br>(1.26)     | [0.583]          |
| Pay 50% (%)             | 3.74          | 3.07           | 4.20           | −0.68<br>(0.60)     | [0.257]          | 0.45<br>(0.67)      | [0.497]          |
| Pay Full (%)            | 31.73         | 34.39          | 30.02          | 2.66<br>(1.61)      | [0.098]          | −1.71<br>(1.55)     | [0.270]          |
| Pay Current Balance (%) | 9.25          | 11.15          | 9.67           | 1.91<br>(1.05)      | [0.070]          | 0.42<br>(0.99)      | [0.668]          |
| Minimum Payment         | 14.92         | 15.08          | 15.43          | 0.16<br>(0.4)       | [0.692]          | 0.51<br>(0.41)      | [0.216]          |

*Notes:* Columns (1), (2), and (3) respectively show the means for the control group C of 4,542 consumers, the Treatment 1 (T1) group of 1,096 consumers, and the Treatment 2 (T2) group of 1,076 consumers. Columns (4) to (7) show the results of t-tests, showing the difference in the means between T1 and C in Column (4), with p-values in Column (5), and between T2 and C in Column (6), with p-values in Column (7). Standard errors displayed in the row below the difference in means in (.). Data for these calculations uses the last statement cycle before the start of the experiment. The units shown for the means and standard errors are in dollars, except for credit score (points), card tenure (days), and for outcomes with (%) are in percentage points.

**Table 2: T-Tests, After Three Statements**

| Outcome                 | C Mean<br>(1) | T1 Mean<br>(2) | T2 Mean<br>(3) | T1 – C<br>(4)                 | T2 – C<br>(5)                 | T1 – T2<br>(6)               |
|-------------------------|---------------|----------------|----------------|-------------------------------|-------------------------------|------------------------------|
| Payments (%)            | 32.92         | 34.40          | 33.83          | 1.48<br>(1.39)<br>[0.285]     | 0.92<br>(1.37)<br>[0.503]     | 0.57<br>(1.76)<br>[0.747]    |
| Payments                | 59.01         | 63.18          | 55.75          | 4.16<br>(4.00)<br>[0.297]     | –3.26<br>(3.61)<br>[0.367]    | 7.42<br>(4.81)<br>[0.123]    |
| Statement Balance       | 231.01        | 233.85         | 225.52         | 2.84<br>(7.93)<br>[0.720]     | –5.49<br>(7.65)<br>[0.473]    | 8.33<br>(9.89)<br>[0.400]    |
| Spending                | 71.94         | 73.64          | 67.50          | 1.70<br>(4.54)<br>[0.708]     | –4.43<br>(4.27)<br>[0.299]    | 6.14<br>(5.57)<br>[0.271]    |
| Spending (%)            | 36.03         | 38.04          | 36.55          | 2.01<br>(1.43)<br>[0.158]     | 0.52<br>(1.42)<br>[0.714]     | 1.49<br>(1.82)<br>[0.412]    |
| Revolving Debt          | 187.21        | 187.88         | 182.58         | 0.67<br>(7.70)<br>[0.931]     | –4.63<br>(7.48)<br>[0.536]    | 5.30<br>(9.65)<br>[0.583]    |
| Credit Limit            | 320.21        | 319.02         | 321.35         | –1.20<br>(8.75)<br>[0.891]    | 1.14<br>(8.64)<br>[0.895]     | –2.34<br>(11.02)<br>[0.832]  |
| Miss Pay (%)            | 28.82         | 29.58          | 29.09          | 0.76<br>(1.55)<br>[0.622]     | 0.28<br>(1.53)<br>[0.857]     | 0.49<br>(1.96)<br>[0.803]    |
| Pay Min (%)             | 16.20         | 11.63          | 11.62          | –4.57***<br>(1.12)<br>[0.000] | –4.58***<br>(1.11)<br>[0.000] | 0.01<br>(1.38)<br>[0.995]    |
| Pay 50% (%)             | 2.72          | 3.35           | 0              | 0.63<br>(0.60)<br>[0.293]     | –2.72***<br>(0.24)<br>[0.000] | 3.35***<br>(0.55)<br>[0.000] |
| Pay Full (%)            | 22.76         | 24.19          | 24.25          | 1.42<br>(1.45)<br>[0.326]     | 1.48<br>(1.44)<br>[0.303]     | –0.06<br>(1.84)<br>[0.974]   |
| Pay Current Balance (%) | 5.90          | 5.86           | 6.50           | –0.04<br>(0.80)<br>[0.960]    | 0.60<br>(0.82)<br>[0.470]     | –0.64<br>(1.03)<br>[0.539]   |

*Notes:* This table uses data for the consumers that are observed after three completed statement cycles. Columns (1), (2), and (3) respectively show the means for the control group C of 4,525 consumers, the Treatment 1 (T1) group of 1,093 consumers, and the Treatment 2 (T2) group of 1,075 consumers. Column (4) is the result of t-tests, showing the difference in the means between T1 and C, i.e., the effect of shrouding the minimum payment option while retaining the 50% payment option, with standard errors in the row below in (.) and p-values in the row below that in [.]. Similarly, column (5) is the result of t-tests, showing the difference in the means between T2 and C, i.e., the effect of shrouding both the minimum payment option and the 50% payment option, with standard errors in the row below in (.) and p-values in the row below that in [.]. Column (6) is the result of t-tests, showing the difference in the means between T1 and T2, i.e., the effect of displaying a 50% payment option, with standard errors in the row below in (.) and p-values in the row below that in [.]. The units shown for the means and standard errors are in dollars, or for outcomes with (%) are in percentage points. Statistical significance denoted at \* 5%, \*\* 1%, and \*\*\* 0.5% levels.

**Table 3: Regression Estimates On Payments (% Statement Balance)****A: Dynamic Estimates After Three and Six Statements**

| Comparison        | Dynamic ( $\delta_3^k$ ) |                |                 |                 | Dynamic ( $\delta_6^k$ ) |                 |                 |                 |
|-------------------|--------------------------|----------------|-----------------|-----------------|--------------------------|-----------------|-----------------|-----------------|
|                   | (1)                      | (2)            | (3)             | (4)             | (5)                      | (6)             | (7)             | (8)             |
| T1 (vs C)         | -0.21<br>(1.33)          | 1.47<br>(1.39) | 0.22<br>(1.22)  | -1.55<br>(1.57) | -7.09***<br>(1.26)       | 0.01<br>(1.30)  | -1.26<br>(1.23) | -2.83<br>(1.62) |
| T2 (vs C)         | -0.78<br>(1.31)          | 1.00<br>(1.37) | 0.90<br>(1.18)  | 1.68<br>(1.44)  | -7.42***<br>(1.24)       | -0.38<br>(1.29) | -0.54<br>(1.19) | 0.10<br>(1.56)  |
| T1 (vs T2)        | 0.57<br>(1.75)           | 0.47<br>(1.75) | -0.69<br>(1.52) | -3.23<br>(1.90) | 0.33<br>(1.64)           | 0.40<br>(1.64)  | -0.72<br>(1.53) | -2.93<br>(2.01) |
| Time F.E.         |                          | X              | X               | X               |                          | X               | X               | X               |
| Consumer Controls |                          |                | X               |                 |                          |                 | X               |                 |
| Consumer F.E.     |                          |                |                 | X               |                          |                 |                 | X               |

**B: Pooled Estimates Across Statements**

| Comparison        | Pooled ( $\delta^k$ ) |                |                 |                  |
|-------------------|-----------------------|----------------|-----------------|------------------|
|                   | (1)                   | (2)            | (3)             | (4)              |
| T1 (vs C)         | 0.49<br>(1.06)        | 0.55<br>(1.06) | -0.67<br>(0.81) | -2.39<br>(1.23)  |
| T2 (vs C)         | 0.35<br>(1.05)        | 0.37<br>(1.05) | 0.29<br>(0.80)  | 1.04<br>(1.16)   |
| T1 (vs T2)        | 0.14<br>(1.35)        | 0.18<br>(1.35) | -0.96<br>(1.02) | -3.44*<br>(1.51) |
| Time F.E.         |                       | X              | X               | X                |
| Consumer Controls |                       |                | X               |                  |
| Consumer F.E.     |                       |                |                 | X                |

*Notes:* The first four columns of Panel A of this table shows the estimates from interactions between treatment indicators and an indicator for the third statement cycle in the experiment, and the last four columns of Panel A show this for the interaction with an indicator for the sixth statement cycle. All of the Panel A regressions also include interactions between treatments and the other statement cycles. All columns of Panel B only include treatment indicators, showing the average effects pooled across statement cycles 0 to 6. All regressions in both panels use the same outcome, payments as a percent of the statement balance, and the units shown are percentage points. All columns except Panel A columns (1) and (5), and Panel B column (1) include time fixed effects for each statement end date in the regressions. Panel A columns (3) and (7), and Panel B column (3) include in the regressions a vector of pre-experiment consumer controls listed in Section 2.3. Panel A columns (4) and (8), and Panel B column (4) include an extra observation per consumer for the statement cycle before the experiment to enable including consumer fixed effects in the regressions. Panel B column (4) estimates Equation (3), and Panel A columns (4) and (8) estimate its dynamic version, with the statement cycle before the experiment as the omitted category. In all regressions, standard errors are clustered at the consumer-level and are shown in parenthesis. Rows *T1 (vs C)* and *T2 (vs C)* are from regressions where the control group (C) is the omitted group. The regressions use data on 6,714 consumers and  $N = 46,519$  observations (statement cycles 0 to 6), except for the models with consumer fixed effects that contain  $N = 53,233$  observations (statement cycles -1 to 6). Statistical significance denoted at \* 5%, \*\* 1%, and \*\*\* 0.5% levels.

**Table 4: Regression Estimates On Secondary Outcomes, After Three Statements**

| Outcome                 | T1 (vs C)<br>(1)   | T2 (vs C)<br>(2)   | T1 (vs T2)<br>(3) |
|-------------------------|--------------------|--------------------|-------------------|
| Payments                | 1.33<br>(3.37)     | −3.48<br>(3.08)    | 4.81<br>(4.08)    |
| Statement Balance       | 5.10<br>(3.65)     | −4.48<br>(3.38)    | 9.57*<br>(4.48)   |
| Spending                | −2.69<br>(3.68)    | −4.57<br>(3.60)    | 1.88<br>(4.59)    |
| Spending (%)            | 0.09<br>(1.20)     | 0.39<br>(1.19)     | −0.31<br>(1.51)   |
| Revolving Debt          | 3.79<br>(4.03)     | −3.25<br>(3.70)    | 7.03<br>(4.90)    |
| Credit Limit            | 0.39<br>(2.20)     | −0.50<br>(2.05)    | 0.89<br>(2.75)    |
| Miss Pay (%)            | 1.42<br>(1.47)     | 0.45<br>(1.44)     | 0.98<br>(1.85)    |
| Pay Min (%)             | −4.17***<br>(1.04) | −3.93***<br>(1.06) | −0.24<br>(1.3)    |
| Pay 50% (%)             | 0.71<br>(0.60)     | −2.88***<br>(0.26) | 3.59***<br>(0.56) |
| Pay Full (%)            | 0.02<br>(1.31)     | 1.66<br>(1.26)     | −1.64<br>(1.63)   |
| Pay Current Balance (%) | −0.32<br>(0.78)    | 0.60<br>(0.80)     | −0.92<br>(1.01)   |
| Time F.E.               | X                  | X                  | X                 |
| Consumer Controls       | X                  | X                  | X                 |

*Notes:* Each row of this table shows estimates for a different outcome. All of the estimates are from interactions between treatment indicators and an indicator for the third statement cycle in the experiment. All of these regressions also include interactions between treatments and the other statement cycles. All regressions include time fixed effects for each statement end date in the regressions and a vector of pre-experiment consumer controls listed in Section 2.3. In all regressions, standard errors are clustered at the consumer-level and are shown in parenthesis. The regressions use data on 6,714 consumers and  $N = 46,519$  observations (statement cycles 0 to 6). The units shown are in dollars, or for outcomes with (%) are in percentage points. Statistical significance denoted at \* 5%, \*\* 1%, and \*\*\* 0.5% levels.

**Table 5: Regression Estimates On Initial Payment Choices****A: Initial Payment Choices**

| Outcome                               | T1 (vs C)<br>(1)   | T2 (vs C)<br>(2)   | T1 (vs T2)<br>(3)   | Control<br>Mean |
|---------------------------------------|--------------------|--------------------|---------------------|-----------------|
| Initial Payment Choice (% Balance)    | −0.90<br>(0.87)    | −4.33***<br>(0.81) | 3.43***<br>(1.07)   | 26.83%          |
| Pr (Initial Payment Choice < Minimum) | 19.44***<br>(1.16) | 30.50***<br>(1.02) | −11.06***<br>(1.38) | 33.74%          |
| Number of Payment Choices             | 0.40***<br>(0.11)  | 0.85***<br>(0.11)  | −0.46***<br>(0.14)  | 3.82            |
| Time F.E.                             | X                  | X                  | X                   |                 |
| Consumer Controls                     | X                  | X                  | X                   |                 |

**B: Initial Payment Choices, Excluding Blank Choices And Choices Of Zero Dollars**

| Outcome                               | T1 (vs C)<br>(1)   | T2 (vs C)<br>(2)   | T1 (vs T2)<br>(3)  | Control<br>Mean |
|---------------------------------------|--------------------|--------------------|--------------------|-----------------|
| Initial Payment Choice (% Balance)    | −0.22<br>(0.86)    | −1.82*<br>(0.83)   | 1.60<br>(1.07)     | 30.19%          |
| Pr (Initial Payment Choice < Minimum) | 15.59***<br>(1.08) | 23.57***<br>(1.04) | −7.98***<br>(1.38) | 21.73%          |
| Number of Payment Choices             | 0.10<br>(0.11)     | 0.35***<br>(0.11)  | −0.25*<br>(0.14)   | 3.01            |
| Time F.E.                             | X                  | X                  | X                  |                 |
| Consumer Controls                     | X                  | X                  | X                  |                 |

*Notes:* Each row of this table shows estimates for a different outcome. All of the estimates are from treatment indicators, showing the average effects pooled across statement cycles 0 to 6. All regressions include time fixed effects for each statement end date in the regressions and a vector of pre-experiment consumer controls listed in Section 2.3. In all regressions, standard errors are clustered at the consumer-level and are shown in parenthesis. The regressions in Panel A use data on 5,644 consumers and  $N = 25,928$  observations, the initial payment choices of consumers in the app each cycle (statement cycles 0 to 6). The regressions in Panel B use data on 5,635 consumers and  $N = 25,820$  observations, the initial payment choices of consumers in the app each cycle (statement cycles 0 to 6) after excluding blank choices or choices of zero dollars. The units shown are in percentage points, except for the number of payment choices which shows the total number of payment choices by a consumer during a cycle. Statistical significance denoted at \* 5%, \*\* 1%, and \*\*\* 0.5% levels.

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# Supplemental Appendix Accompanying “*Targeting Higher Credit Card Payments*”

## A. Pre-Analysis Plan (March 20, 2025)

### A.1 Research Question

Does nudging credit card payment options increase credit card payments? A large body of lab experimental evidence exists showing that a nudge shrouding the credit card minimum payment option increases hypothetical manual payments.<sup>12</sup> This new field experiment provides an important test of whether this finding extrapolates to the field. I test two nudges that help to understand the credit card payment amounts that consumers target. Prior empirical studies have shown, across countries, other efforts to nudge consumers to reduce their debt have been ineffective.<sup>13</sup>

### A.2 Experimental Sample

The experiment studies a card lender’s customers in the USA. Consumers can only hold one credit card with the lender, so therefore studying results at the credit card account-level also correspond to the consumer-level. The sample includes consumers who have had at least one statement issued, a credit limit of over \$40, a non-zero statement balance and have never been over 35 days past due. We also exclude 2,337 consumers who were incorrectly randomized to treatments in December 2024. At least 1,100 consumers are randomly selected for each of the two treatment groups. The control group consists of the remaining consumers and will contain at least 1,100 consumers.

### A.3 Experimental Design

The experiment varies the presentation of the credit card manual payment options a consumer is presented with on the lender’s mobile app. This experiment was approved by Rice University’s Institutional Review Board (IRB-FY2024-399 on 21st June 2024). The sample will be randomly split across three groups:

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<sup>12</sup>Stewart (2009), Navarro-Martinez et al. (2011), Jiang and Dunn (2013), Salisbury (2014), Salisbury and Zhao (2020), Hendy et al. (2021), Sakaguchi et al. (2022), Guttman-Kenney et al. (2023). Empirical evidence from historical data consistent with “anchoring” or “targeting” to the minimum payment has also been shown in Keys and Wang (2019), Medina and Negrin (2022), and Vihriälä (2025). Thaler and Sunstein (2008) write a credit card’s minimum payment “can serve as an anchor, and as a nudge that this minimum payment is an appropriate amount.”

<sup>13</sup>Agarwal et al. (2015), Seira et al. (2017), Keys and Wang (2019), Adams et al. (2022).

1. 68% of consumers (4,674) are in the control group (C). This has these payment options: current balance, statement balance, 50% of statement balance, minimum payment, and a free text box to choose any payment amount. These are the status quo options customers experience in the absence of the experiment.
2. 16% of consumers (1,103) are in Treatment 1 (T1). This does not show consumers the minimum payment option. It has these payment options: current balance, statement balance, 50% of statement balance, and a free text box to choose any payment amount.
3. 16% of consumers (1,128) are in Treatment 2 (T2). This does not show consumers the minimum payment option and it also does not show consumers the 50% of statement balance option. It has these payment options: current balance, statement balance, and a free text box to choose any payment amount.

Importantly, in all the groups (control, T1, and T2), if a consumer attempts to pay an amount less than the minimum payment then a warning comes up showing the minimum payment amount. This enables the consumer to revise their initial payment choice to prevent the consumer accidentally paying less than the minimum. Sakaguchi et al. (2022) shows in a lab experiment how such prompts appear effective at preventing such choices.

The motivation behind T1 is that the minimum payment amount acts as a psychological default (see cites in the preceding footnote), potentially reducing some consumer payment choices to exactly this amount or amounts just above it. By shrouding this option, it may increase payments by increasing the salience of the statement balance for consumers to use as a target and enables consumers to make an active choice without being distorted by the presence of the minimum.<sup>14</sup>

The motivation behind T2 is that the 50% of statement balance option can also act as a target, potentially reducing some consumer payment choices to exactly 50%, or amounts close to it, that would otherwise be higher. Prior research has indicated more payment options may unintentionally reduce payments for some consumers.<sup>15</sup> By shrouding the 50% option, along with the minimum payment option, it may increase payments by increasing the salience of the statement balance for consumers to use as a target (see cites in the preceding footnote).

Comparing C to T1 shows the effect of shrouding the minimum payment option, to target the higher payment options. Comparing C to T2 shows the effect of shrouding both the minimum payment and 50% of statement balance options, to target the statement balance

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<sup>14</sup>Bartels et al. (2024) and Schwartz (2026).

<sup>15</sup>Agarwal et al. (2015), Hershfield and Roese (2015), and Keys and Wang (2016).

as a higher payment option. Comparing T1 to T2 shows the marginal effect of including the 50% of statement balance option.

## **A.4 Data**

Outcomes are evaluated using the credit card lender’s anonymized administrative data. The includes data on each credit card statement and each credit card payment a consumer makes.

### **Data Preparation**

I will winsorize balances, debt, payments, and spending variables, when measured in dollars, at their 99th percentiles. This is done as these variables are highly non-normal in their distributions, with a mass at zero and fat right tails to their distributions. I will check for balance between the control and treatment groups using covariates observed before the experiment’s start to ensure the randomization was successfully implemented and report t-statistics comparing differences in these. I do not expect the treatments to affect card closure rates. However, if they did it affects how to deal with missing data. Therefore, I plan to examine card closing rates. Assuming there is no significant difference that is economically meaningful, more than 10%, I proceed with studying outcomes across observed card payment cycles.

## **A.5 Experiment Dates**

The experiment was planned to start from 11th September 2024 but, due to operational considerations at the lender it was delayed and was scheduled to go live on 12th December 2024, however, the randomization was incorrectly done between treatments and therefore the experiment is scheduled to re-launch after correct randomization in late March 2025. The experiment is planned to end after at least three completed credit card payment cycles. Payments against the third credit card statement are expected to be due by July 2025.

I intend to report outcomes using data after at least three completed statement cycles. If after three completed statement cycles, the experiment is having a positive impact on consumer outcomes, it will be continued for another three completed statement cycles, and primary results will be evaluated after six completed statement cycles. If the experiment is having no effect on consumer outcomes after three completed statement cycles, the experiment will end with primary outcomes evaluated at this time. If there is a severe negative impact, the lender may stop the experiment prior to three completed statement cycles.

## A.6 Experimental Analysis

### Descriptive Analysis

I will show the cumulative distribution function (CDF) of payments for control and treatment groups. This is designed to examine how the nudge affects the distribution of payments, and whether it changes the bunching of payments around the minimum and the 50% statement balance options, as would be consistent with consumers targeting these payment amounts.

1. Distribution of payments (% statement balance).
2. Distribution of excess payments (%). Excess payments are calculated by dividing payments by balances, both excess of the minimum payment amount. The numerator is payments less the minimum payment amount, and the denominator is statement balance less the minimum payment amount.

### Significance Testing

I will primarily use t-tests to evaluate significance, using the latest payment cycle of data (at least 3 payment cycles) to evaluate the effects of the nudges. I will use 5% as a threshold for statistical significance. I will also report Minimum Detectable Effects (MDE) for primary outcomes assuming 80%, with significance thresholds of 0.5%, 1%, and 5% (5% is the threshold for significance used for analysis).

### Regression Specification

To understand the dynamics of the treatment effects, I will use the dynamic regression specification shown below that uses data across all completed statement cycles during the experiment:

$$Y_{i,t} = \alpha + \sum_{\tau} \left[ \delta_{\tau}^{T1} (D_{\tau} \times T1_i) + \delta_{\tau}^{T2} (D_{\tau} \times T2_i) \right] + \varepsilon_{i,t}$$

In this regression,  $Y_{i,t}$  denotes outcome  $Y$  for consumer  $i$  at time  $t$ , and  $\varepsilon_{i,t}$  is the error term. The term  $\alpha$  is a constant. The term  $T1_i$  is an indicator for Treatment 1 ( $T1$ ) and the term  $T2_i$  is an indicator for Treatment 2 ( $T2$ ). The term  $D_{\tau}$  is an indicator for time  $\tau$ . I focus on the coefficients  $\delta_{\tau}^{T1}$  and  $\delta_{\tau}^{T2}$  from the latest period observed, but will also show results from earlier periods to enable readers to see the dynamics of the effects, and will also show results from a static regression that pools data across cycles as a robustness exercise. Standard errors are clustered at consumer level in both regressions.

$$Y_{i,t} = \alpha + \delta^{T1} T1_i + \delta^{T2} T2_i + \varepsilon_{i,t}$$

## A.7 Experimental Outcomes

### Primary Outcome

1. Payment (%): Continuous outcome defined as the sum of payments (\$) divided by the statement balance (\$). Payment (%) is given a value of one if statement balance is zero. If Payment (%) is greater than one, it is winsorized at a value of one.

### Secondary Outcomes

2. Payments (\$): Continuous outcome.
3. Statement Balance (\$): Continuous outcome.
4. Spending (\$): Continuous outcome defined as the sum of purchases (\$).
5. Spending (%): Continuous outcome defined as the sum of purchases (\$) divided by the statement balance (\$). Spending (%) is given a value of one if statement balance is zero. If Spending (%) is greater than one, it is winsorized at a value of one.
6. Debt (\$): Continuous outcome measuring revolving credit card debt. This is defined as the statement balance (\$) less payments (\$). Debt is given a value of zero if payments exceed the statement balance.
7. Credit Limit (\$): Integer outcome.
8. Miss Pay: Binary outcome. Takes a value of one if the consumer's payments are zero or less than the minimum payment (and the minimum payment is non-zero, and the payments are less than the statement balance).
9. Pay Min: Binary outcome. Takes a value of one if the consumer's payments are only exactly the minimum payment (and the minimum payment is non-zero, and the payments are less than the statement balance).
10. Pay 50%: Binary outcome. Takes a value of one if the consumer's payments are only exactly the 50% of the statement balance (and the minimum payment is non-zero).
11. Pay Full: Binary outcome. Takes a value of one if the consumer's payments are for the full statement balance or more (or if a zero minimum payment is due, including cases of zero statement balances).
12. Pay Current: Binary outcome. Takes a value of one if the consumer's payments are for the current balance or more (if current balance is non-zero).

## **Power Calculations**

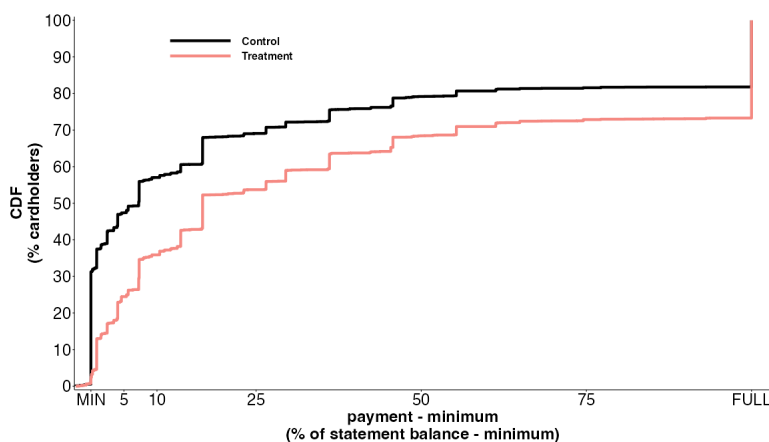
Based on data from June 2024, a sample of 2,200 consumers evenly split between a control and a treatment group would have sufficient power to detect a change in payments of 0.045, given a baseline mean of 0.235 and a standard deviation of 0.372. It would have sufficient power to detect a change in revolving debt of \$26, given a baseline mean of \$214 and a standard deviation of \$221. This appears sufficiently powerful given the effect sizes of nudges shrouding minimum payments that I have previously tested in lab experiments on hypothetical credit card payment scenarios. In Guttman-Kenney (2024), the nudge has a 0.124 increase in payments (% statement balance) on a mean of 0.378. In Sakaguchi et al. (2022), the nudge has a 0.127 increase in payments (% statement balance) on a mean of 0.385.

## **Consumer Credit Reporting Data**

Ideally, I would want to examine the effects of the treatments across the portfolio of credit cards held by a consumer. This would enable me to evaluate whether effects such as increasing payments on the credit card in the experiment are offset by consumers reducing payments on other credit cards. See Guttman-Kenney et al. (2025) for an example of such an approach when studying a different nudge treatment using UK consumer credit reporting data. Unfortunately, as documented in Guttman-Kenney and Shahidinejad (2025), consumer credit reporting data (reviewed in Gibbs et al., 2025) in the USA does not include credit card payments information for any of the six largest US credit card lenders, and only 24% of credit cardholders have this information across all their cards held. This means researchers cannot accurately distinguish between revolving debt and new credit card spending. Another issue is, even if consumer credit reporting data included credit card payments data, it would be challenging to have sufficient statistical power to detect offsetting effects. This is because the credit card lender's business model offers low credit limit cards, and the size of payment amounts may be too small to precisely estimate whether higher payments on the card in the experiment have been offset by lower payments on other cards held.

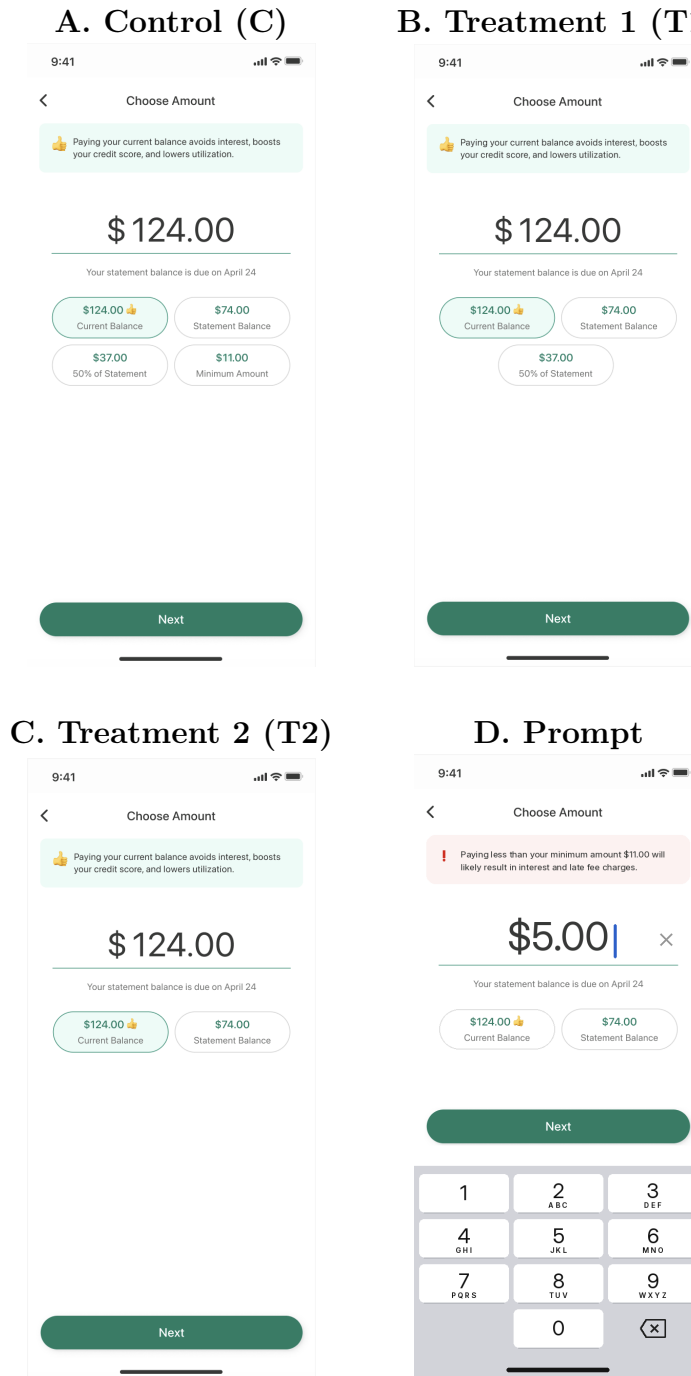
## B. Supplementary Figures And Tables

**Figure A1: Example Of Effects Of Shrouding The Minimum Payment In A Hypothetical Credit Card Payment Experiment**



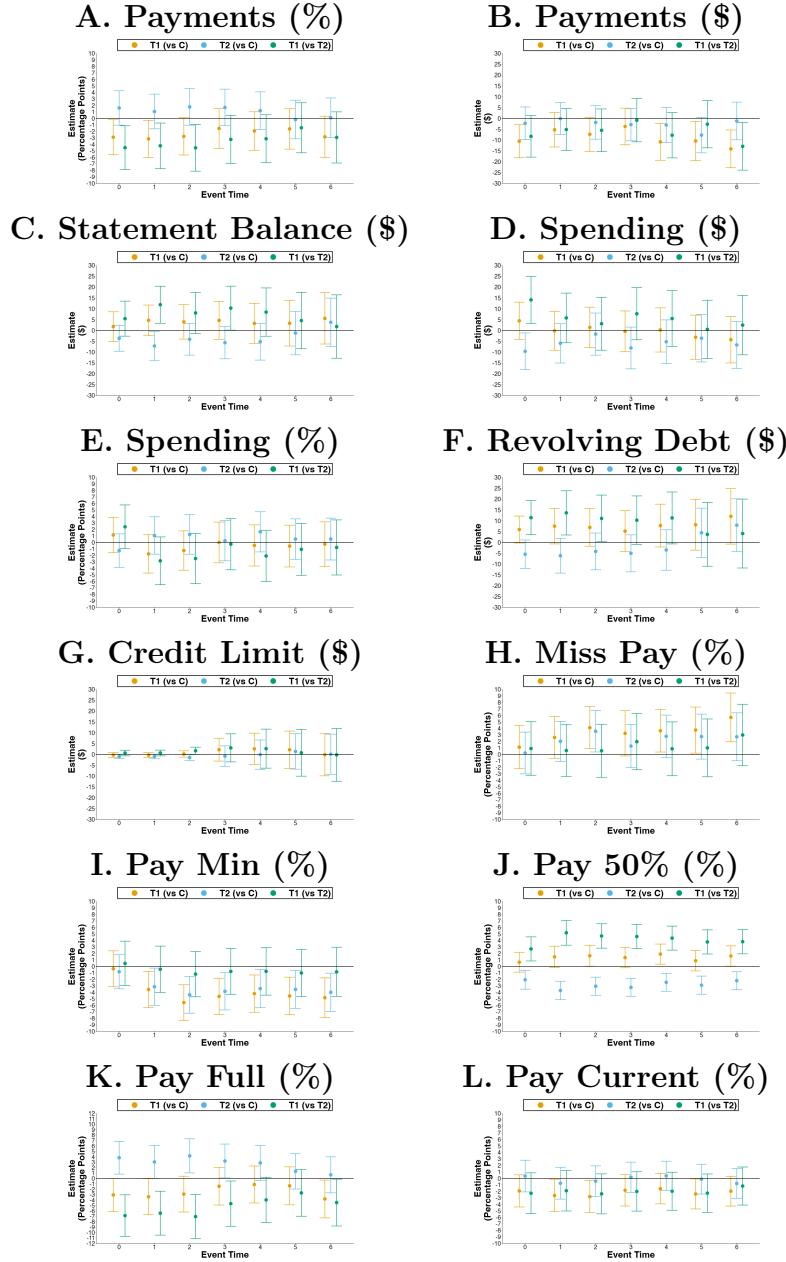
*Notes:* This figure is reprinted from Guttman-Kenney (2024), and shows the results of an online experiment testing the effect of shrouding the minimum payment amount on a sample of  $N = 7,938$  consumers who have credit cards in the U.K. The figure displays the CDFs of the payment amount less the minimum payment amount (as a percent of statement balance less than the minimum payment amount). Payments of exactly the minimum are denoted by “MIN” and payments of the full amount are denoted by “FULL”. The black line is the control group where the minimum payment amount is displayed. The orange line is the treatment group where the minimum payment amount is shrouded. The average effect on payments is to significantly increase average hypothetical credit card payments as a percent of the statement balance by 12.4 (s.e. 0.8) percentage points.

**Figure A2: Credit Card Payment App Choice Architectures**



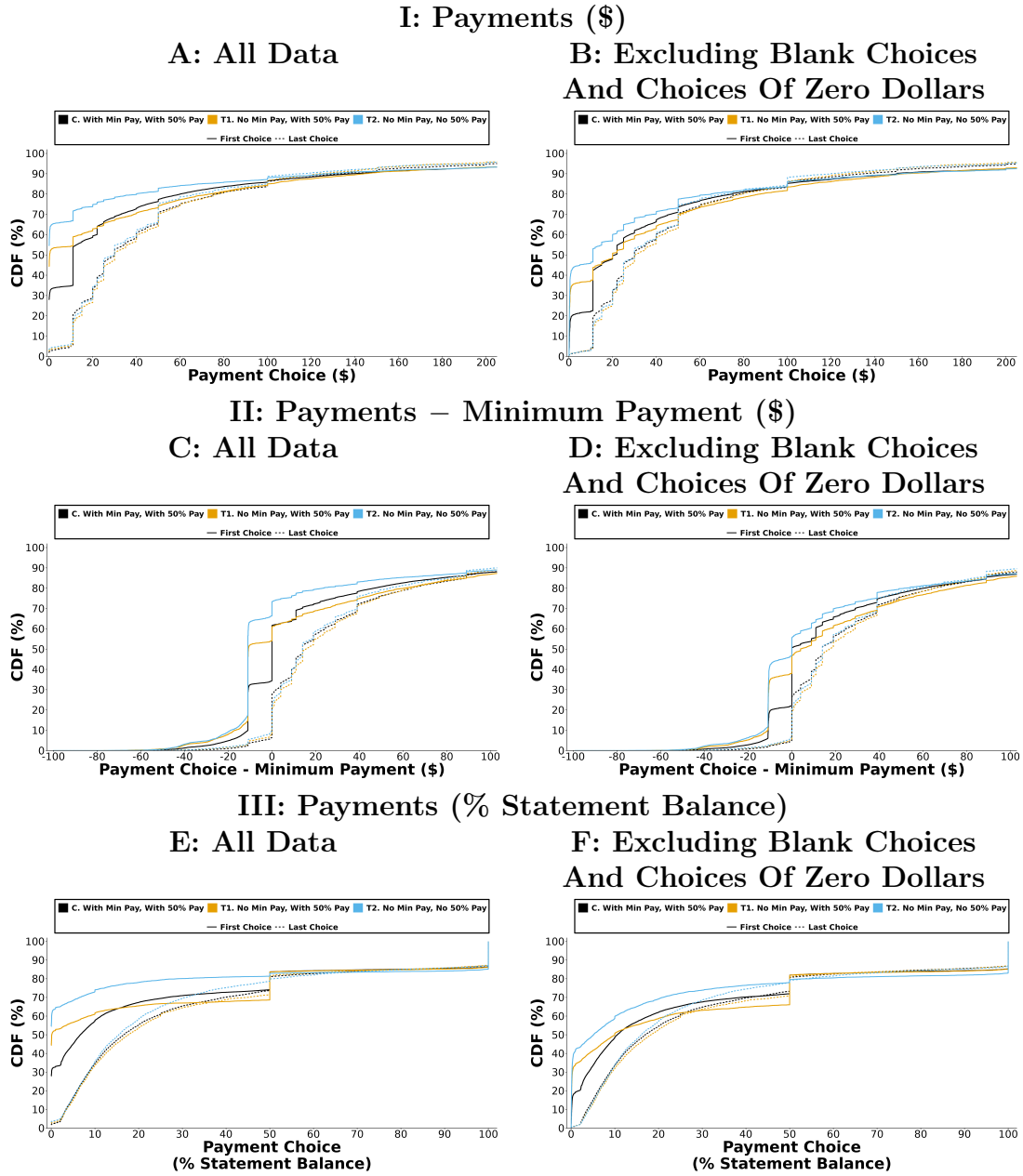
*Notes:* These screenshots show an example where the consumer has provisionally selected in the app to pay the current balance amount of \$124.00, however, for clarity there are no options pre-selected as a default. The minimum payment warning prompt in Panel D only appears if a consumer enters an amount less than the minimum payment amount. Panel D displays this warning prompt for the Treatment 2 (T2) payment options, such a warning prompt also appears for the Control (C) group and for the Treatment 1 (T1) group.

Figure A3: Dynamic Effects On Outcomes, With Time F.E. And Consumer F.E.



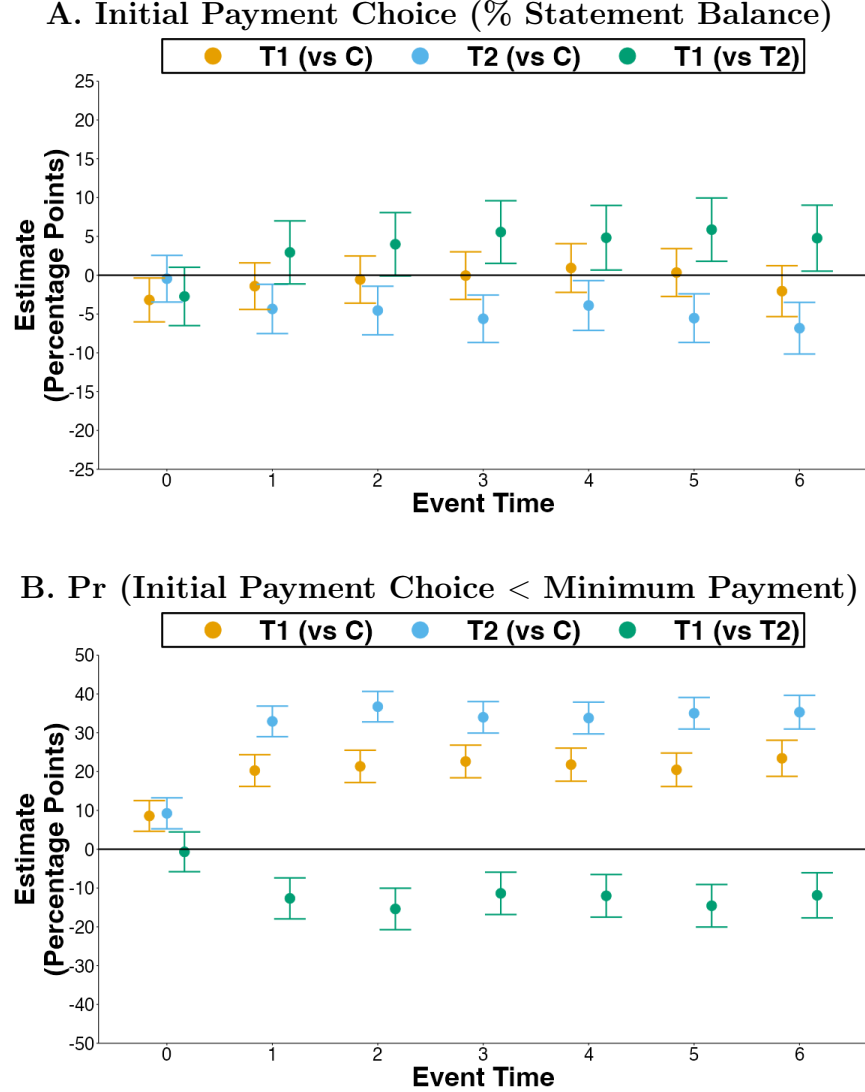
*Notes:* Each panel shows results for a different outcome. This figure shows the estimates from interactions between treatment indicators and time indicators, with 95% confidence intervals based on clustering standard errors at the consumer-level. In orange and blue are respectively the estimates from a regression where both T1 and T2 are each separately interacted with time indicators, with the control group as the omitted category. In green are the estimates from a T1 interaction, with the T2 group as the omitted category and data for group C is not used in these regressions. All regressions include time fixed effects for each statement end date and consumer fixed effects, estimating the dynamic version of Equation (3) with the statement cycle before the experiment as the omitted category. The regressions use data on 6,714 consumers and  $N = 53,233$  observations (statement cycles -1 to 6).

Figure A4: CDFs Of First And Last Payment Choices, With and Without Blank Choices And Choices Of Zero Dollars



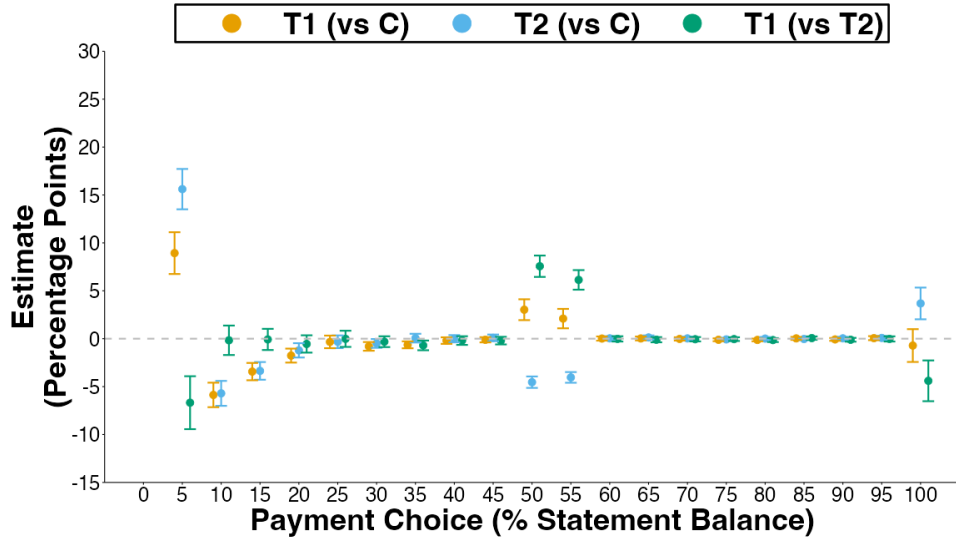
*Notes:* This includes consumers who make at least one payment and have a non-zero statement balance and non-zero minimum payment due in that cycle. Panels A, C, and E isolate first and last choices in a cycle using data for all choices whereas B, D, and F exclude blank choices and choices of zero dollars to isolate first and last choices in a cycle. Each panel shows CDFs calculated a different way, Panel I shows CDFs for payments in dollars, Panel II shows for excess payments (i.e., payments less the minimum payment), and Panel III shows for payments as a percentage of the statement balance. The colors show the CDFs for consumers in the control group, C in black, and the two treatment groups, T1 in orange and T2 in blue. The solid lines show the CDF of consumers' first payment choices in a payment cycle and the dashed lines show the CDF of consumers' last payment choices in a payment cycle. All CDFs use data pooled across cycles zero to six.

Figure A5: Dynamic Effects On Initial Payment Choices



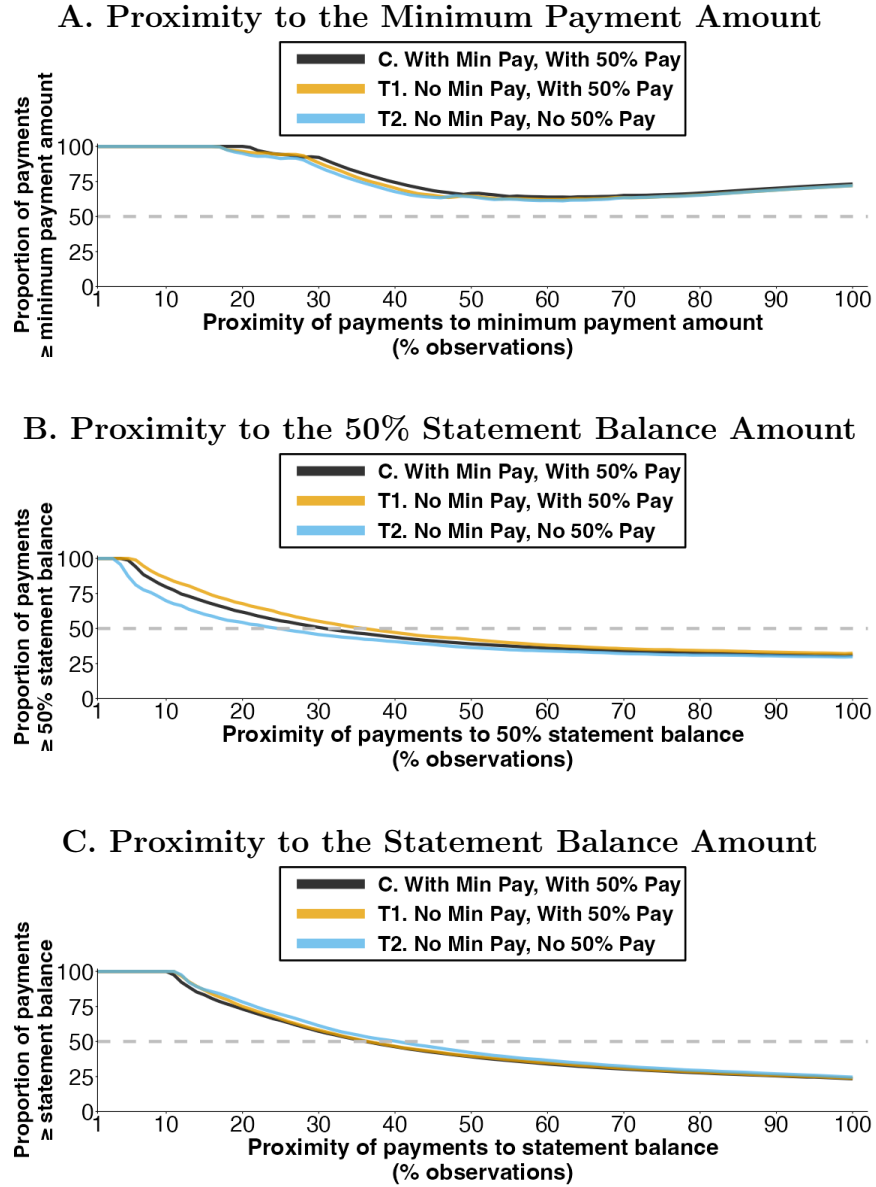
*Notes:* Each panel shows results for a different outcome. This figure shows the estimates from interactions between treatment indicators and time indicators, with 95% confidence intervals based on clustering standard errors at the consumer-level. In orange and blue are respectively the estimates from a regression with separate indicators for T1 and T2, with the control group as the omitted category. In green are the estimates from a T1 interaction, with the T2 group as the omitted category and data for group C is not used in these regressions. All regressions also include time fixed effects for each statement end date and a vector of pre-experiment consumer controls listed in Section 2.3. The regressions use data on 5,644 consumers and  $N = 25,928$  observations, the initial payment choices of consumers in the app each cycle (statement cycles 0 to 6). The units are percentage points.

**Figure A6: Effects On Payment Distribution Of Initial Payment Choice (% Statement Balance), Excluding Blank Choices And Choices Of Zero Dollars**



*Notes:* This figure shows the estimates on treatment indicators, showing the average effects pooled across statements, with 95% confidence intervals based on clustering standard errors at the consumer-level. In orange and blue are respectively the estimates from a regression where both T1 and T2 are each separately interacted with time indicators, with the control group as the omitted category. In green are the estimates from a T1 indicator, with the T2 group as the omitted category and data for group C is not used in these regressions. All regressions also include time fixed effects for each statement end date and also a vector of pre-experiment consumer controls listed in Section 2.3. The regressions use data on 5,635 consumers and  $N = 25,820$  observations, the initial payment choices of consumers in the app each cycle (statement cycles 0 to 6). This excludes blank choices and choices of zero dollars. The estimates for each x-axis level show results for a different binary outcome that takes a value of one if a consumer paid 5 percentage points up to that amount, and zero otherwise. The estimates corresponding to the x-axis value of one are if a consumer paid in full.

Figure A7: Value Proximity Analysis (VPA)



*Notes:* Each panel is calculated by collapsing the raw data for 6,714 consumers containing payments across all cycles from zero to six,  $N = 46,519$  consumer cycles, to 300 observations. In each panel, there is one observation for each proximity window (denoted by the x-axis) that contains from 1% to 100% of observations in windows that incrementally include an additional one percentage point of observations, and this is calculated separately for each of the three groups: C, T1, and T2. The black lines denotes group C, the orange lines denotes group T1, and the blue lines denote group T2. In each panel, the y-axis shows for a given proximity window, the fraction of observations that are greater than or equal to a particular payment amount. Each panel shows the results for a different outcome payment amount. Panel A shows the fraction of observations that are greater than or equal to the minimum payment amount. Panel B shows the fraction of observations that are greater than or equal to 50% of the statement balance amount. Panel C shows the fraction of observations that are greater than or equal to the statement balance amount.

**Table A1: Regression Estimates On Attrition, After Three and Six Statements**

|                   | T1 (vs C)<br>(1) | T2 (vs C)<br>(2) | T1 (vs T2)<br>(3) | Control<br>Mean |
|-------------------|------------------|------------------|-------------------|-----------------|
| 3                 | −0.23<br>(0.14)  | −0.12<br>(0.18)  | −0.12<br>(0.18)   | 0.37%           |
| 6                 | −0.09<br>(0.64)  | 0.00<br>(0.70)   | −0.09<br>(0.83)   | 5.00%           |
| Time F.E.         | X                | X                | X                 |                 |
| Consumer Controls | X                | X                | X                 |                 |

*Notes:* This table shows estimates in percentage points for the effects on attrition as of the third cycle (denoted by row labeled 3), and attrition as of the sixth cycle (denoted by row labeled 6) estimated from separate regressions. All of the estimates are from treatment indicators, showing the average effects. All regressions include time fixed effects for each statement end date and a vector of pre-experiment consumer controls listed in Section 2.3. Heteroskedasticity-robust standard errors are shown in parenthesis. The regressions use data on 6,714 consumers with one observation per consumer. Statistical significance denoted at \* 5%, \*\* 1%, and \*\*\* 0.5% levels.

**Table A2: T-Tests, After Six Statements**

| Outcome                 | C Mean<br>(1) | T1 Mean<br>(2) | T2 Mean<br>(3) | T1 – C<br>(4)                 | T2 – C<br>(5)                 | T1 – T2<br>(6)               |
|-------------------------|---------------|----------------|----------------|-------------------------------|-------------------------------|------------------------------|
| Payments (%)            | 27.75         | 27.52          | 27.19          | –0.23<br>(1.31)<br>[0.864]    | –0.56<br>(1.29)<br>[0.664]    | 0.33<br>(1.65)<br>[0.839]    |
| Payments                | 58.41         | 51.53          | 57.07          | –6.89<br>(3.62)<br>[0.057]    | –1.34<br>(3.95)<br>[0.734]    | –5.55<br>(4.73)<br>[0.241]   |
| Statement Balance       | 278.41        | 282.85         | 283.45         | 4.44<br>(8.81)<br>[0.614]     | 5.04<br>(8.52)<br>[0.554]     | –0.60<br>(11.05)<br>[0.957]  |
| Spending                | 65.25         | 62.05          | 62.15          | –3.20<br>(4.59)<br>[0.486]    | –3.10<br>(4.28)<br>[0.468]    | –0.10<br>(5.59)<br>[0.986]   |
| Spending (%)            | 27.32         | 28.59          | 27.99          | 1.27<br>(1.34)<br>[0.346]     | 0.66<br>(1.32)<br>[0.616]     | 0.61<br>(1.70)<br>[0.721]    |
| Revolving Debt          | 230.91        | 239.63         | 240.26         | 8.72<br>(8.59)<br>[0.310]     | 9.35<br>(8.25)<br>[0.257]     | –0.63<br>(10.74)<br>[0.953]  |
| Credit Limit            | 366.11        | 362.14         | 369.84         | –3.97<br>(9.86)<br>[0.687]    | 3.73<br>(9.76)<br>[0.702]     | –7.70<br>(12.47)<br>[0.537]  |
| Miss Pay (%)            | 35.97         | 39.83          | 37.91          | 3.86*<br>(1.69)<br>[0.023]    | 1.94<br>(1.67)<br>[0.246]     | 1.92<br>(2.14)<br>[0.371]    |
| Pay Min (%)             | 15.62         | 10.56          | 10.46          | –5.06***<br>(1.11)<br>[0.000] | –5.16***<br>(1.10)<br>[0.000] | 0.10<br>(1.35)<br>[0.940]    |
| Pay 50% (%)             | 1.85          | 2.71           | 0.10           | 0.86<br>(0.55)<br>[0.116]     | –1.76***<br>(0.23)<br>[0.000] | 2.62***<br>(0.52)<br>[0.000] |
| Pay Full (%)            | 18.24         | 16.96          | 17.37          | –1.28<br>(1.31)<br>[0.328]    | –0.87<br>(1.31)<br>[0.509]    | –0.41<br>(1.66)<br>[0.803]   |
| Pay Current Balance (%) | 4.06          | 3.78           | 3.74           | –0.28<br>(0.67)<br>[0.678]    | –0.31<br>(0.66)<br>[0.636]    | 0.04<br>(0.84)<br>[0.965]    |

*Notes:* This table uses data for the consumers that are observed after six completed statement cycles. Columns (1), (2), and (3) respectively show the means for the control group C of 4,315 consumers, the Treatment 1 (T1) group of 1,042 consumers, and the Treatment 2 (T2) group of 1,032 consumers. Column (4) is the result of t-tests, showing the difference in the means between T1 and C, i.e., the effect of shrouding the minimum payment option while retaining the 50% payment option, with standard errors in the row below in (.) and p-values in the row below that in [.]. Similarly, column (5) is the result of t-tests, showing the difference in the means between T2 and C, i.e., the effect of shrouding both the minimum payment option and the 50% payment option, with standard errors in the row below in (.) and p-values in the row below that in [.]. Column (6) is the result of t-tests, showing the difference in the means between T1 and T2, i.e., the effect of displaying a 50% payment option, with standard errors in the row below in (.) and p-values in the row below that in [.]. The units shown for the means and standard errors are in dollars, or for outcomes with (%) are in percentage points. Statistical significance denoted at \* 5%, \*\* 1%, and \*\*\* 0.5% levels.

**Table A3: Regression Estimates On Secondary Outcomes, Pooled Across Statements**

| Outcome                 | T1 (vs C)          |                    | T2 (vs C)          |                    | T1 (vs T2)        |                    |
|-------------------------|--------------------|--------------------|--------------------|--------------------|-------------------|--------------------|
|                         | (1)                | (2)                | (3)                | (4)                | (5)               | (6)                |
| Payments                | -3.83<br>(2.19)    | -8.85*<br>(3.43)   | -3.31<br>(2.13)    | -2.73<br>(3.12)    | -0.52<br>(2.74)   | -6.12<br>(4.15)    |
| Statement Balance       | 4.39<br>(2.42)     | 3.85<br>(3.63)     | -2.29<br>(2.35)    | -3.40<br>(3.08)    | 6.68<br>(3.04)    | 7.25<br>(4.21)     |
| Spending                | -2.49<br>(2.13)    | -0.25<br>(3.93)    | -2.32<br>(2.10)    | -5.87<br>(4.11)    | -0.17<br>(2.70)   | 5.61<br>(5.12)     |
| Spending (%)            | -0.38<br>(0.71)    | -0.46<br>(1.29)    | 0.72<br>(0.71)     | 0.56<br>(1.25)     | -1.10<br>(0.9)    | -1.02<br>(1.61)    |
| Revolving Debt          | 6.29*<br>(2.78)    | 7.61<br>(3.95)     | -0.10<br>(2.72)    | -1.81<br>(3.64)    | 6.39<br>(3.50)    | 9.42*<br>(4.77)    |
| Credit Limit            | -0.89<br>(1.72)    | 0.91<br>(2.26)     | -0.09<br>(1.64)    | -0.41<br>(2.11)    | -0.80<br>(2.17)   | 1.32<br>(2.80)     |
| Miss Pay (%)            | 1.72<br>(0.90)     | 3.43*<br>(1.34)    | 1.29<br>(0.90)     | 2.18<br>(1.30)     | 0.43<br>(1.15)    | 1.26<br>(1.66)     |
| Pay Min (%)             | -3.53***<br>(0.65) | -3.94***<br>(1.17) | -3.38***<br>(0.64) | -3.29**<br>(1.18)  | -0.15<br>(0.80)   | -0.65<br>(1.49)    |
| Pay 50% (%)             | 0.70*<br>(0.29)    | 1.35*<br>(0.62)    | -2.46***<br>(0.15) | -2.80***<br>(0.67) | 3.16***<br>(0.28) | 4.15***<br>(0.81)  |
| Pay Full (%)            | -1.00<br>(0.85)    | -2.43<br>(1.42)    | 1.20<br>(0.82)     | 2.75*<br>(1.30)    | -2.20*<br>(1.07)  | -5.18***<br>(1.72) |
| Pay Current Balance (%) | -0.67<br>(0.40)    | -2.17*<br>(1.01)   | 0.26<br>(0.42)     | -0.16<br>(0.97)    | -0.93<br>(0.52)   | -2.01<br>(1.26)    |
| Time F.E.               | X                  | X                  | X                  | X                  | X                 | X                  |
| Consumer Controls       | X                  |                    | X                  |                    | X                 |                    |
| Consumer F.E.           |                    | X                  |                    | X                  |                   | X                  |

*Notes:* Each row of this table shows estimates for a different outcome. All of the estimates are from treatment indicators, showing the average effects pooled across statement cycles 0 to 6. All columns include time fixed effects for each statement end date in the regressions. Columns (1), (3), and (5) include in the regressions a vector of pre-experiment consumer controls listed in Section 2.3. Columns (2), (4), and (6) include an extra observation per consumer for the statement cycle before the experiment to enable including consumer fixed effects in the regressions, and estimate Equation (3). In all regressions, standard errors are clustered at the consumer-level and are shown in parenthesis. The regressions in columns (1), (3), and (5) use data on 6,714 consumers and  $N = 46,519$  observations (statement cycles 0 to 6), and the regressions in columns (2), (4), and (6) contain  $N = 53,233$  observations (statement cycles -1 to 6). The units shown are in dollars, or for outcomes with (%) are in percentage points. Statistical significance denoted at \* 5%, \*\* 1%, and \*\*\* 0.5% levels.

**Table A4: Regression Estimates On Secondary Outcomes, After Three Statements**

| Outcome                 | T1 (vs C)          |                    |                    | T2 (vs C)          |                    |                    | T1 (vs T2)        |                   |                   |
|-------------------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|-------------------|-------------------|-------------------|
|                         | (1)                | (2)                | (3)                | (4)                | (5)                | (6)                | (7)               | (8)               | (9)               |
| Payments                | -0.07<br>(3.84)    | 4.03<br>(4.00)     | -3.72<br>(4.32)    | -7.50*<br>(3.44)   | -3.09<br>(3.61)    | -2.87<br>(3.81)    | 7.42<br>(4.81)    | 7.12<br>(4.82)    | -0.85<br>(5.10)   |
| Statement Balance       | -3.34<br>(7.84)    | 2.45<br>(7.89)     | 4.61<br>(4.45)     | -11.67<br>(7.55)   | -6.00<br>(7.63)    | -5.67<br>(3.79)    | 8.33<br>(9.89)    | 8.45<br>(9.87)    | 10.28*<br>(5.19)  |
| Spending                | 1.07<br>(4.35)     | 1.07<br>(4.53)     | -0.42<br>(4.80)    | -5.07<br>(4.06)    | -4.77<br>(4.27)    | -8.11<br>(4.92)    | 6.14<br>(5.57)    | 5.84<br>(5.58)    | 7.69<br>(6.14)    |
| Spending (%)            | 2.46<br>(1.37)     | 1.93<br>(1.43)     | -0.01<br>(1.60)    | 0.97<br>(1.36)     | 0.42<br>(1.42)     | 0.26<br>(1.58)     | 1.49<br>(1.82)    | 1.51<br>(1.81)    | -0.27<br>(2.02)   |
| Revolving Debt          | -1.39<br>(7.58)    | 0.23<br>(7.65)     | 5.18<br>(4.82)     | -6.69<br>(7.36)    | -5.17<br>(7.46)    | -5.06<br>(4.36)    | 5.30<br>(9.65)    | 5.39<br>(9.62)    | 10.24<br>(5.73)   |
| Credit Limit            | -5.61<br>(8.72)    | -2.84<br>(8.68)    | 2.18<br>(2.68)     | -3.28<br>(8.61)    | -0.26<br>(8.56)    | -0.84<br>(2.45)    | -2.34<br>(11.02)  | -2.58<br>(10.94)  | 3.02<br>(3.31)    |
| Miss Pay (%)            | 2.79<br>(1.46)     | 0.84<br>(1.53)     | 3.24<br>(1.78)     | 2.30<br>(1.44)     | 0.38<br>(1.51)     | 1.29<br>(1.69)     | 0.49<br>(1.96)    | 0.46<br>(1.94)    | 1.95<br>(2.20)    |
| Pay Min (%)             | -4.76***<br>(1.05) | -4.70***<br>(1.12) | -4.62***<br>(1.42) | -4.77***<br>(1.04) | -4.69***<br>(1.11) | -3.84**<br>(1.46)  | 0.01<br>(1.38)    | -0.01<br>(1.37)   | -0.78<br>(1.80)   |
| Pay 50% (%)             | 0.75<br>(0.56)     | 0.68<br>(0.60)     | 1.36<br>(0.8)      | -2.59***<br>(0.12) | -2.74***<br>(0.25) | -3.23***<br>(0.71) | 3.35***<br>(0.55) | 3.43***<br>(0.56) | 4.59***<br>(0.95) |
| Pay Full (%)            | 0.36<br>(1.39)     | 1.33<br>(1.45)     | -1.45<br>(1.77)    | 0.42<br>(1.38)     | 1.57<br>(1.44)     | 3.22*<br>(1.60)    | -0.06<br>(1.84)   | -0.24<br>(1.84)   | -4.66*<br>(2.13)  |
| Pay Current Balance (%) | -0.65<br>(0.74)    | -0.08<br>(0.80)    | -1.83<br>(1.23)    | -0.01<br>(0.77)    | 0.55<br>(0.83)     | 0.18<br>(1.19)     | -0.64<br>(1.03)   | -0.63<br>(1.04)   | -2.01<br>(1.55)   |
| Time F.E.               |                    | X                  | X                  |                    | X                  | X                  |                   | X                 | X                 |
| Consumer F.E.           |                    |                    | X                  |                    |                    | X                  |                   |                   | X                 |

*Notes:* Each row of this table shows estimates for a different outcome. All of the estimates are from interactions between treatment indicators and an indicator for the third statement cycle in the experiment. All of these regressions also include interactions between treatments and the other statement cycles. All columns except (1), (4), and (7) include time fixed effects for each statement end date in the regressions. Columns (3), (6), and (9) include an extra observation per consumer for the statement cycle before the experiment to enable including consumer fixed effects in the regressions, and estimate the dynamic version of Equation (3), with the statement cycle before the experiment as the omitted category. In all regressions, standard errors are clustered at the consumer-level and are shown in parenthesis. The regressions use data on 6,714 consumers and  $N = 46,519$  observations (statement cycles 0 to 6), except for the models with consumer fixed effects that contain 53,233 observations (statement cycles -1 to 6). The units shown are in dollars, or for outcomes with (%) are in percentage points. Statistical significance denoted at \* 5%, \*\* 1%, and \*\*\* 0.5% levels.

**Table A5: Regression Estimates On Secondary Outcomes, After Six Statements**

| Outcome                 | T1 (vs C)          |                     | T2 (vs C)          |                    | T1 (vs T2)        |                   |
|-------------------------|--------------------|---------------------|--------------------|--------------------|-------------------|-------------------|
|                         | (1)                | (2)                 | (3)                | (4)                | (5)               | (6)               |
| Payments                | −9.31**<br>(3.32)  | −14.08***<br>(4.45) | −1.95<br>(3.65)    | −1.19<br>(4.44)    | −7.36<br>(4.38)   | −12.89*<br>(5.62) |
| Statement Balance       | 6.03<br>(4.96)     | 5.54<br>(6.04)      | 3.92<br>(4.94)     | 3.77<br>(5.69)     | 2.11<br>(6.33)    | 1.78<br>(7.46)    |
| Spending                | −6.83<br>(4.27)    | −4.28<br>(5.48)     | −3.27<br>(3.99)    | −6.73<br>(5.51)    | −3.56<br>(5.27)   | 2.45<br>(7.00)    |
| Spending (%)            | −0.41<br>(1.28)    | −0.26<br>(1.75)     | 0.71<br>(1.22)     | 0.52<br>(1.64)     | −1.12<br>(1.6)    | −0.78<br>(2.16)   |
| Revolving Debt          | 10.76*<br>(5.32)   | 12.02<br>(6.53)     | 8.79<br>(5.14)     | 7.94<br>(6.22)     | 1.97<br>(6.68)    | 4.09<br>(8.12)    |
| Credit Limit            | −2.15<br>(4.33)    | −0.14<br>(4.97)     | 0.35<br>(4.17)     | 0.08<br>(4.74)     | −2.50<br>(5.47)   | −0.22<br>(6.22)   |
| Miss Pay (%)            | 4.39**<br>(1.63)   | 5.70***<br>(1.91)   | 1.92<br>(1.61)     | 2.71<br>(1.88)     | 2.47<br>(2.06)    | 2.99<br>(2.41)    |
| Pay Min (%)             | −4.60***<br>(1.09) | −4.82***<br>(1.56)  | −4.34***<br>(1.06) | −3.99**<br>(1.50)  | −0.26<br>(1.32)   | −0.84<br>(1.93)   |
| Pay 50% (%)             | 0.94<br>(0.54)     | 1.60*<br>(0.80)     | −1.93***<br>(0.24) | −2.20***<br>(0.71) | 2.87***<br>(0.52) | 3.80***<br>(0.95) |
| Pay Full (%)            | −2.44<br>(1.25)    | −3.80*<br>(1.78)    | −0.72<br>(1.23)    | 0.66<br>(1.70)     | −1.72<br>(1.57)   | −4.46*<br>(2.21)  |
| Pay Current Balance (%) | −0.48<br>(0.66)    | −1.96<br>(1.17)     | −0.33<br>(0.66)    | −0.79<br>(1.17)    | −0.16<br>(0.84)   | −1.18<br>(1.49)   |
| Time F.E.               | X                  | X                   | X                  | X                  | X                 | X                 |
| Consumer Controls       | X                  |                     | X                  |                    | X                 |                   |
| Consumer F.E.           |                    | X                   |                    | X                  |                   | X                 |

*Notes:* Each row of this table shows estimates for a different outcome. All of the estimates are from interactions between treatment indicators and an indicator for the sixth statement cycle in the experiment. All of these regressions also include interactions between treatments and the other statement cycles. All columns include time fixed effects for each statement end date in the regressions. Columns (1), (3), and (5) include in the regressions a vector of pre-experiment consumer controls listed in Section 2.3. Columns (2), (4), and (6) include an extra observation per consumer for the statement cycle before the experiment to enable including consumer fixed effects in the regressions, and estimate the dynamic version of Equation (3), with the statement cycle before the experiment as the omitted category. In all regressions, standard errors are clustered at the consumer-level and are shown in parenthesis. The regressions use data on 6,714 consumers and  $N = 46,519$  observations (statement cycles 0 to 6), except for the models with consumer fixed effects that contain 53,233 observations (statement cycles -1 to 6). The units shown are in dollars, or for outcomes with (%) are in percentage points. Statistical significance denoted at \* 5%, \*\* 1%, and \*\*\* 0.5% levels.

**Table A6: Regression Estimates On Heuristic Payment Amounts, Pooled Across Statements**

| Outcome                  | T1 (vs C)<br>(1)   | T2 (vs C)<br>(2)   | T1 (vs T2)<br>(3) | Control<br>Mean |
|--------------------------|--------------------|--------------------|-------------------|-----------------|
| Pay Multiples Of Minimum | −0.76***<br>(0.25) | −0.94***<br>(0.23) | 0.18<br>(0.29)    | 3.48%           |
| Pay Round Number         | 2.51***<br>(0.62)  | 3.61***<br>(0.62)  | −1.10<br>(0.80)   | 9.58%           |
| Time F.E.                | X                  | X                  | X                 |                 |
| Consumer Controls        | X                  | X                  | X                 |                 |

*Notes:* This table shows the estimates from treatment indicators, showing the average effects, in percentage points, pooled across statement cycles 0 to 6. Each row of this table shows estimates for a different outcome. *Pay Multiples Of Minimum* is a binary outcome that takes a value of one if the consumer pays 2, 3, 4, or 5 times the minimum payment amount (after dollar rounding). *Pay Round Number* is a binary outcome that takes a value of one if the consumer pays exactly a round number  $\in \{\$15, \$20, \dots, \$45, \$50\}$ . These outcomes only take a value of one for the cases where the minimum payment is positive and the consumer pays more than the minimum and less than the full balance. All regressions include time fixed effects for each statement end date and a vector of pre-experiment consumer controls listed in Section 2.3. In all regressions, standard errors are clustered at the consumer-level and are shown in parenthesis. The regressions use data on 6,714 consumers and  $N = 46,519$  observations (statement cycles 0 to 6). Statistical significance denoted at \* 5%, \*\* 1%, and \*\*\* 0.5% levels.

**Table A7: Value Proximity Analysis (VPA) Regression Estimates**

|                       | Minimum Payment      | 50% Statement Balance | Statement Balance    |
|-----------------------|----------------------|-----------------------|----------------------|
|                       | (1)                  | (2)                   | (3)                  |
| Intercept             | 98.510***<br>(0.383) | 76.831***<br>(0.940)  | 89.091***<br>(0.822) |
| T1                    | -1.403<br>(0.952)    | 5.290**<br>(2.034)    | 1.380<br>(1.875)     |
| T2                    | -2.840***<br>(0.995) | -7.568***<br>(2.410)  | 3.377<br>(1.767)     |
| Proximity             | -0.406***<br>(0.006) | -0.584***<br>(0.009)  | -0.798***<br>(0.005) |
| Proximity $\times$ T1 | -0.005<br>(0.015)    | -0.044*<br>(0.018)    | -0.012<br>(0.011)    |
| Proximity $\times$ T2 | 0.008<br>(0.015)     | 0.078***<br>(0.023)   | -0.016<br>(0.010)    |
| T2 - T1: Intercept    | -1.438<br>(1.253)    | -12.858***<br>(2.862) | 1.997<br>(2.298)     |
| T2 - T1: Proximity    | 0.013<br>(0.019)     | 0.122***<br>(0.027)   | -0.004<br>(0.013)    |

*Notes:* Each column shows the estimates from a separate OLS regression examining a different outcome. Column (1) shows the fraction of observations that are greater than or equal to the minimum payment amount. Column (2) shows the fraction of observations that are greater than or equal to 50% of the statement balance amount. Column (3) shows the fraction of observations that are greater than or equal to the statement balance amount. The units are percentage points. Each regression is run by collapsing the raw data for 6,714 consumers containing payments across all cycles from zero to six ( $N = 46,519$  consumer cycles) to 300 observations. There is one observation for each proximity window that contains from 1% to 100% of observations in windows that incrementally include an additional one percentage point of observations, and this is calculated separately for each of the three groups: C, T1, and T2. Each regression includes an intercept, indicators for each of T1 and T2, a linear term for the Proximity, and interactions between the Proximity term and the T1 indicator and also separately with the T2 indicator. Standard errors, in parentheses, are from a bootstrap resampling consumers with replacement (10,000 draws). The last two rows are the T2 - T1 differences in intercept and slope. Statistical significance denoted at \* 5%, \*\* 1%, and \*\*\* 0.5% levels.

**Table A8: Regression Estimates For Subsample Without Autopay, After Six Statements**

| Outcome                 | T1 (vs C)          |                    | T2 (vs C)          |                    | T1 (vs T2)        |                   |
|-------------------------|--------------------|--------------------|--------------------|--------------------|-------------------|-------------------|
|                         | (1)                | (2)                | (3)                | (4)                | (5)               | (6)               |
| Payments (%)            | -2.18<br>(1.61)    | -2.97<br>(2.08)    | -0.94<br>(1.55)    | -0.24<br>(2.02)    | -1.24<br>(1.99)   | -2.73<br>(2.58)   |
| Payments                | -9.58*<br>(3.97)   | -10.48*<br>(5.08)  | 1.85<br>(4.50)     | 3.84<br>(5.37)     | -11.44*<br>(5.36) | -14.31*<br>(6.51) |
| Statement Balance       | 5.66<br>(5.8)      | 3.30<br>(6.93)     | 8.07<br>(5.75)     | 8.87<br>(6.78)     | -2.41<br>(7.42)   | -5.57<br>(8.82)   |
| Spending                | -9.12<br>(5.10)    | -7.76<br>(6.94)    | 1.15<br>(5.00)     | 0.10<br>(6.69)     | -10.27<br>(6.48)  | -7.86<br>(8.65)   |
| Spending (%)            | -1.12<br>(1.73)    | -0.42<br>(2.37)    | 0.03<br>(1.58)     | -0.39<br>(2.15)    | -1.15<br>(2.11)   | -0.03<br>(2.89)   |
| Revolving Debt          | 10.31<br>(6.09)    | 9.85<br>(7.46)     | 9.13<br>(6.04)     | 7.11<br>(7.43)     | 1.18<br>(7.76)    | 2.74<br>(9.47)    |
| Credit Limit            | -3.17<br>(5.2)     | -2.24<br>(5.78)    | 5.93<br>(5.20)     | 6.46<br>(5.89)     | -9.09<br>(6.73)   | -8.70<br>(7.50)   |
| Miss Pay (%)            | 5.03*<br>(2.21)    | 5.38*<br>(2.55)    | 2.01<br>(2.12)     | 2.77<br>(2.42)     | 3.02<br>(2.75)    | 2.61<br>(3.16)    |
| Pay Min (%)             | -4.99***<br>(1.24) | -5.17***<br>(1.93) | -5.63***<br>(1.18) | -6.05***<br>(1.77) | 0.64<br>(1.46)    | 0.88<br>(2.33)    |
| Pay 50% (%)             | 0.81<br>(0.81)     | 2.22<br>(1.15)     | -2.66***<br>(0.38) | -3.05***<br>(1.08) | 3.47***<br>(0.78) | 5.26***<br>(1.41) |
| Pay Full (%)            | -3.84*<br>(1.62)   | -3.81<br>(2.35)    | -1.33<br>(1.58)    | 0.43<br>(2.22)     | -2.51<br>(2.01)   | -4.24<br>(2.89)   |
| Pay Current Balance (%) | -0.29<br>(0.92)    | -0.96<br>(1.53)    | -0.45<br>(0.88)    | -0.42<br>(1.52)    | 0.16<br>(1.13)    | -0.54<br>(1.94)   |
| Time F.E.               | X                  | X                  | X                  | X                  | X                 | X                 |
| Consumer Controls       | X                  |                    | X                  |                    | X                 |                   |
| Consumer F.E.           |                    | X                  |                    | X                  |                   | X                 |

*Notes:* This table uses data for the subsample of consumers not enrolled in autopay in the month before the launch of the experiment. Each row of this table shows estimates for a different outcome. All of the estimates are from interactions between treatment indicators and an indicator for the sixth statement cycle in the experiment. All of these regressions also include interactions between treatments and the other statement cycles. All columns include time fixed effects for each statement end date in the regressions. Columns (1), (3), and (5) include in the regressions a vector of pre-experiment consumer controls listed in Section 2.3. Columns (2), (4), and (6) include an extra observation per consumer for the statement cycle before the experiment to enable including consumer fixed effects in the regressions, and estimate the dynamic version of Equation (3), with the statement cycle before the experiment as the omitted category. In all regressions, standard errors are clustered at the consumer-level and are shown in parenthesis. The regressions use data on 3,939 consumers and  $N = 27,206$  observations (statement cycles 0 to 6), except for the models with consumer fixed effects that contain  $N = 31,145$  observations (statement cycles -1 to 6). The units shown are in dollars, or for outcomes with (%) are in percentage points. Statistical significance denoted at \* 5%, \*\* 1%, and \*\*\* 0.5% levels.

**Table A9: Regression Estimates On The Portfolio Of Credit Card Statement Balances, Pooled Across Statements**

| Outcome                      | T1 (vs C)<br>(1) | T2 (vs C)<br>(2) | T1 (vs T2)<br>(3) | Control<br>Mean |
|------------------------------|------------------|------------------|-------------------|-----------------|
| Portfolio Statement Balances | 36.55<br>(51.85) | 6.60<br>(53.41)  | 29.94<br>(66.98)  | \$3,681         |
| Time F.E.                    | X                | X                | X                 |                 |
| Consumer Controls            | X                | X                | X                 |                 |

*Notes:* This table shows estimates in dollars for the effects on the portfolio of credit card statement balances, measured in credit reporting data. All of the estimates are from treatment indicators, showing the average effects pooled across months. All regressions include time fixed effects for each credit reporting month and a vector of pre-experiment consumer controls listed in Section 2.3. In all regressions, standard errors are clustered at the consumer-level and are shown in parenthesis. The regressions use data on 6,712 consumers and  $N = 46,280$  observations (months 0 to 6). Statistical significance denoted at \* 5%, \*\* 1%, and \*\*\* 0.5% levels.

**Table A10: Heterogeneous Treatment Effects On Payments By Pre-Experiment Payment Behavior, Pooled Across Statements**

**A: Payments (% Statement Balance)**

| Sample            | T1 (vs C)<br>(1) | T2 (vs C)<br>(2) | T1 (vs T2)<br>(3) | Control<br>Mean | Consumers |
|-------------------|------------------|------------------|-------------------|-----------------|-----------|
| Near-Minimum      | 2.89<br>(1.73)   | −0.56<br>(1.54)  | 3.45<br>(2.07)    | 22.18%          | 1,226     |
| Not-Near-Minimum  | −1.44<br>(0.92)  | 0.28<br>(0.92)   | −1.72<br>(1.17)   | 37.42%          | 5,488     |
| Time F.E.         | X                | X                | X                 |                 |           |
| Consumer Controls | X                | X                | X                 |                 |           |

**B: Payments (\$)**

| Sample            | T1 (vs C)<br>(1) | T2 (vs C)<br>(2) | T1 (vs T2)<br>(3) | Control<br>Mean | Consumers |
|-------------------|------------------|------------------|-------------------|-----------------|-----------|
| Near-Minimum      | 0.43<br>(2.56)   | −2.41<br>(2.39)  | 2.84<br>(3.00)    | \$36.76         | 1,226     |
| Not-Near-Minimum  | −4.05<br>(2.56)  | −3.79<br>(2.53)  | −0.26<br>(3.23)   | \$69.24         | 5,488     |
| Time F.E.         | X                | X                | X                 |                 |           |
| Consumer Controls | X                | X                | X                 |                 |           |

*Notes:* Each panel shows estimates for a different outcome. In Panel A the outcome is payments as a percent of the statement balance, with the units in percentage point. In Panel B the outcome is payments in dollars, which are also the units. This table shows the estimates from treatment indicators, showing the heterogeneous effects pooled across statement cycles 0 to 6. Each row of this table shows estimates for a different subsample. *Near-Minimum* is the 18.3% sample of consumers who paid more than the minimum but less than or equal to \$50 more than the minimum (but not paid in full) for greater than or equal to 50% of the six cycles before the start of the experiment (for cards opened less than six cycles before the experiment, we calculate this based on cycles that occur). *Not-Near-Minimum* is the sample of all other consumers: 81.7%. The outcome in both regressions is the primary outcome: payments (% statement balance). All regressions include time fixed effects for each statement end date and a vector of pre-experiment consumer controls listed in Section 2.3. In all regressions, standard errors are clustered at the consumer-level and are shown in parenthesis. The *Near-Minimum* regression uses data on 1,226 consumers and  $N = 8,502$  observations (statement cycles 0 to 6). The *Not Near-Minimum* regression uses data on 5,488 consumers and  $N = 38,017$  observations (statement cycles 0 to 6). Statistical significance denoted at \* 5%, \*\* 1%, and \*\*\* 0.5% levels.

**Table A11: Heterogeneous Treatment Effects On Payments By Tertiles Of Pre-Experiment Credit Card Utilization, Pooled Across Statements**

**A: Payments (% Statement Balance)**

| Sample                            | T1 (vs C)<br>(1) | T2 (vs C)<br>(2) | T1 (vs T2)<br>(3) | Control<br>Mean | Consumers |
|-----------------------------------|------------------|------------------|-------------------|-----------------|-----------|
| Q1 Card Utilization (< 60%)       | -1.89<br>(1.65)  | 3.70*<br>(1.60)  | -5.59**<br>(2.08) | 52.23%          | 2,238     |
| Q2 Card Utilization (60–91%)      | 1.57<br>(1.40)   | -1.19<br>(1.31)  | 2.76<br>(1.72)    | 30.08%          | 2,238     |
| Q3 Card Utilization (91%+)        | -1.45<br>(1.15)  | -1.30<br>(1.17)  | -0.15<br>(1.44)   | 21.38%          | 2,238     |
| Q1 Portfolio Utilization (< 12%)  | -1.68<br>(1.43)  | 0.37<br>(1.47)   | -2.05<br>(1.86)   | 33.24%          | 2,238     |
| Q2 Portfolio Utilization (12–45%) | -2.31<br>(1.43)  | 0.94<br>(1.40)   | -3.25<br>(1.83)   | 33.72%          | 2,238     |
| Q3 Portfolio Utilization (45%+)   | 1.66<br>(1.45)   | -0.96<br>(1.36)  | 2.62<br>(1.78)    | 36.85%          | 2,238     |
| Time F.E.                         | X                | X                | X                 |                 |           |
| Consumer Controls                 | X                | X                | X                 |                 |           |

**B: Payments (\$)**

| Sample                            | T1 (vs C)<br>(1) | T2 (vs C)<br>(2) | T1 (vs T2)<br>(3) | Control<br>Mean | Consumers |
|-----------------------------------|------------------|------------------|-------------------|-----------------|-----------|
| Q1 Card Utilization (< 60%)       | -8.30*<br>(4.15) | -1.53<br>(4.14)  | -6.78<br>(5.30)   | \$80.49         | 2,238     |
| Q2 Card Utilization (60–91%)      | 0.01<br>(3.31)   | -5.73<br>(2.99)  | 5.74<br>(3.90)    | \$58.93         | 2,238     |
| Q3 Card Utilization (91%+)        | -2.06<br>(3.12)  | -0.13<br>(3.33)  | -1.93<br>(4.02)   | \$50.19         | 2,238     |
| Q1 Portfolio Utilization (< 12%)  | -2.93<br>(3.83)  | 2.13<br>(4.07)   | -5.06<br>(5.00)   | \$64.13         | 2,238     |
| Q2 Portfolio Utilization (12–45%) | -4.57<br>(3.26)  | -1.15<br>(3.24)  | -3.43<br>(4.26)   | \$60.13         | 2,238     |
| Q3 Portfolio Utilization (45%+)   | -0.96<br>(3.95)  | -7.82*<br>(3.61) | 6.86<br>(4.81)    | \$65.48         | 2,238     |
| Time F.E.                         | X                | X                | X                 |                 |           |
| Consumer Controls                 | X                | X                | X                 |                 |           |

*Notes:* Each panel shows estimates for a different outcome. In Panel A the outcome is payments as a percent of the statement balance, with the units in percentage point. In Panel B the outcome is payments in dollars, which are also the units. This table shows the estimates from treatment indicators, showing the heterogeneous effects, in percentage points, pooled across statement cycles 0 to 6. Each row of this table shows estimates for a different subsample. The rows with ‘Card Utilization’ segment consumers by tertiles of their credit card utilization (credit card statement balance divided by credit card limit) calculated only using the card in the experiment based on the statement before the experiment went into the field. The rows with ‘Portfolio Utilization’ segment consumers by tertiles of their portfolio credit card utilization (sum of credit card statement balances divided by sum of credit card limits) using Equifax credit reporting data which uses the portfolio of credit cards held by a consumer in the month before the the experiment went into the field. Q1 denotes the lowest tertile, Q2 is the intermediate tertile, and Q3 is the highest tertile. All regressions include time fixed effects for each statement end date and a vector of pre-experiment consumer controls listed in Section 2.3. In all regressions, standard errors are clustered at the consumer-level and are shown in parenthesis. Statistical significance denoted at \* 5%, \*\* 1%, and \*\*\* 0.5% levels.

**Table A12: Heterogeneous Treatment Effects On Payments (%) By Pre-Experiment Liquid Cash Balances, Pooled Across Statements**

| Sample                        | T1 (vs C)<br>(1) | T2 (vs C)<br>(2) | T1 (vs T2)<br>(3) | Control<br>Mean | Consumers |
|-------------------------------|------------------|------------------|-------------------|-----------------|-----------|
| Liquid Cash < \$1             | 1.32<br>(2.28)   | 1.01<br>(2.22)   | 0.31<br>(2.97)    | 25.15%          | 1,025     |
| Liquid Cash \$1+              | -1.36<br>(0.88)  | 0.20<br>(0.86)   | -1.56<br>(1.10)   | 36.33%          | 5,689     |
| Liquid Cash < \$501           | -0.46<br>(1.06)  | 0.70<br>(1.06)   | -1.16<br>(1.34)   | 31.37%          | 3,796     |
| Liquid Cash \$501+            | -0.80<br>(1.3)   | -0.45<br>(1.23)  | -0.35<br>(1.60)   | 38.78%          | 2,918     |
| Liquid Cash < \$1,001         | -1.11<br>(0.95)  | 0.42<br>(0.95)   | -1.53<br>(1.21)   | 32.11%          | 4,726     |
| Liquid Cash \$1,001+          | 0.25<br>(1.58)   | -0.22<br>(1.50)  | 0.47<br>(1.96)    | 40.60%          | 1,988     |
| Minimum Liquid Cash < \$1     | -1.47<br>(1.02)  | 0.68<br>(1.03)   | -2.15<br>(1.31)   | 32.10%          | 4,230     |
| Minimum Liquid Cash \$1+      | -0.25<br>(1.37)  | -0.15<br>(1.33)  | -0.09<br>(1.70)   | 38.99%          | 2,484     |
| Minimum Liquid Cash < \$501   | -0.88<br>(0.83)  | 0.61<br>(0.83)   | -1.49<br>(1.06)   | 33.64%          | 6,252     |
| Minimum Liquid Cash \$501+    | 1.50<br>(3.33)   | 0.43<br>(3.34)   | 1.07<br>(4.07)    | 48.04%          | 462       |
| Minimum Liquid Cash < \$1,001 | -0.95<br>(0.82)  | 0.21<br>(0.82)   | -1.16<br>(1.04)   | 33.94%          | 6,422     |
| Minimum Liquid Cash \$1,001+  | 1.77<br>(4.46)   | 5.87<br>(4.55)   | -4.10<br>(5.77)   | 49.24%          | 292       |
| Time F.E.                     | X                | X                | X                 |                 |           |
| Consumer Controls             | X                | X                | X                 |                 |           |

*Notes:* The outcome in all regressions is the primary outcome: payments (% statement balance). This table shows the estimates from treatment indicators, showing the heterogeneous effects, in percentage points, pooled across statement cycles 0 to 6. Each row of this table shows estimates for a different subsample, with the fourth column showing the number of consumers in each subsample. The rows starting with the prefix ‘Liquid Cash’ segment consumers by their liquid cash balance the day before the experiment went into the field, for example ‘Liquid Cash <\$1’ denotes consumers who had a liquid cash balance of less than \$1 at this point-in-time, and ‘Liquid Cash \$1+’ is the remaining consumers who had a balance of \$1 or more. The rows starting with the prefix ‘Minimum Liquid Cash’ segment consumers by their *minimum* liquid cash balance in the 90 days up to the day before the experiment went into the field, for example ‘Minimum Liquid Cash <\$1’ denotes consumers who had a minimum liquid cash balance of less than \$1 at any point in those 90 days, and ‘Minimum Liquid Cash \$1+’ is the remaining consumers who had a minimum balance of \$1 or more. For consumers who opened credit cards less than 90 days before the start of the experiment, we calculate this using 60 or 30 days, as available. All regressions include time fixed effects for each statement end date and a vector of pre-experiment consumer controls listed in Section 2.3. In all regressions, standard errors are clustered at the consumer-level and are shown in parenthesis. Statistical significance denoted at \* 5%, \*\* 1%, and \*\*\* 0.5% levels.

**Table A13: Heterogeneous Treatment Effects On Payments (\$) By Pre-Experiment Liquid Cash Balances, Pooled Across Statements**

| Sample                       | T1 (vs C)<br>(1)  | T2 (vs C)<br>(2) | T1 (vs T2)<br>(3) | Control<br>Mean | Consumers |
|------------------------------|-------------------|------------------|-------------------|-----------------|-----------|
| Liquid Cash < \$1            | -2.54<br>(4.48)   | 4.58<br>(5.38)   | -7.13<br>(6.58)   | \$43.44         | 1,025     |
| Liquid Cash \$1+             | -5.17*<br>(2.42)  | -4.92*<br>(2.29) | -0.24<br>(2.99)   | \$66.85         | 5,689     |
| Liquid Cash < \$501          | -3.44<br>(2.69)   | -0.67<br>(2.76)  | -2.77<br>(3.47)   | \$58.73         | 3,796     |
| Liquid Cash \$501+           | -3.28<br>(3.51)   | -5.15<br>(3.30)  | 1.86<br>(4.30)    | \$69.07         | 2,918     |
| Liquid Cash < \$1,001        | -3.55<br>(2.49)   | -1.94<br>(2.49)  | -1.61<br>(3.18)   | \$60.45         | 4,726     |
| Liquid Cash \$1,001+         | -1.86<br>(4.24)   | -4.53<br>(3.90)  | 2.67<br>(5.10)    | \$69.94         | 1,988     |
| Minimum Liquid Cash < \$1    | -4.37<br>(2.54)   | -1.92<br>(2.67)  | -2.45<br>(3.32)   | \$59.70         | 4,230     |
| Minimum Liquid Cash \$1+     | -3.69<br>(4.03)   | -6.22<br>(3.49)  | 2.53<br>(4.78)    | \$69.44         | 2,484     |
| Minimum Liquid Cash < \$501  | -2.66<br>(2.26)   | -2.59<br>(2.18)  | -0.06<br>(2.84)   | \$62.05         | 6,252     |
| Minimum Liquid Cash \$501+   | -15.50<br>(8.53)  | 1.52<br>(8.83)   | -17.02<br>(9.97)  | \$79.79         | 462       |
| Minimum Liquid Cash <\$1,001 | -3.49<br>(2.22)   | -3.34<br>(2.16)  | -0.16<br>(2.79)   | \$62.41         | 6,422     |
| Minimum Liquid Cash \$1,001+ | -18.52<br>(10.03) | 3.25<br>(10.85)  | -21.77<br>(12.49) | \$81.69         | 292       |
| Time F.E.                    | X                 | X                | X                 |                 |           |
| Consumer Controls            | X                 | X                | X                 |                 |           |

*Notes:* The outcome in all regressions is the outcome: payments (\$). This table shows the estimates from treatment indicators, showing the heterogeneous effects, in percentage points, pooled across statement cycles 0 to 6. Each row of this table shows estimates for a different subsample, with the fourth column showing the number of consumers in each subsample. The rows starting with the prefix ‘Liquid Cash’ segment consumers by their liquid cash balance the day before the experiment went into the field, for example ‘Liquid Cash <\$1’ denotes consumers who had a liquid cash balance of less than \$1 at this point-in-time, and ‘Liquid Cash \$1+’ is the remaining consumers who had a balance of \$1 or more. The rows starting with the prefix ‘Minimum Liquid Cash’ segment consumers by their *minimum* liquid cash balance in the 90 days up to the day before the experiment went into the field, for example ‘Minimum Liquid Cash <\$1’ denotes consumers who had a minimum liquid cash balance of less than \$1 at any point in those 90 days, and ‘Minimum Liquid Cash \$1+’ is the remaining consumers who had a minimum balance of \$1 or more. For consumers who opened credit cards less than 90 days before the start of the experiment, we calculate this using 60 or 30 days, as available. All regressions include time fixed effects for each statement end date and a vector of pre-experiment consumer controls listed in Section 2.3. In all regressions, standard errors are clustered at the consumer-level and are shown in parenthesis. Statistical significance denoted at \* 5%, \*\* 1%, and \*\*\* 0.5% levels.

## C. Theory Appendix

### C.1 The Static Problem

**Setup.** A cardholder with statement balance  $B > 0$ , minimum  $m \in (0, B)$ , and cash  $a \geq 0$  chooses  $p \in [0, B]$  to maximize Equation (4), with  $C(p; a) = \int_0^p v(a - q) dq$ .

**Assumption C1.**  $v : (0, \infty) \rightarrow (0, \infty)$  is continuous and strictly decreasing, with  $\lim_{z \downarrow 0} v(z) = \infty$ ,  $\lim_{z \rightarrow \infty} v(z) = 0$ , and  $\int_0^\epsilon v(z) dz = \infty$  for every  $\epsilon > 0$ .

Under Assumption C1,  $C$  is strictly convex and  $C(p; a) \rightarrow \infty$  as  $p \uparrow a$ , so  $p < a$  always. Define the marginal pull toward paying below and above the minimum,

$$\pi_L \equiv \rho + \lambda(\eta_m + \eta_B), \quad \pi_H \equiv \rho + \eta_m + \lambda\eta_B, \quad \pi_L - \pi_H = \eta_m(\lambda - 1) \geq 0, \quad (6)$$

so that  $U'(p) = \pi_L - v(a - p)$  on  $(0, m)$  and  $U'(p) = \pi_H - v(a - p)$  on  $(m, B)$ :  $U$  is strictly concave on each interval, jumps up by  $F$  at  $m$ , and its slope drops by  $\eta_m(\lambda - 1)$  at  $m$ . Write  $x(\pi) \equiv v^{-1}(\pi)$  for the cash buffer associated with pull  $\pi$ , decreasing in  $\pi$ , and  $x_L \equiv x(\pi_L) \leq x_H \equiv x(\pi_H)$ . For  $a \in (m, m + x_L)$  define the cost of reaching the minimum from the unconstrained optimum below it,

$$\Delta_m(a) \equiv \int_{a-m}^{x_L} (v(z) - \pi_L) dz, \quad (7)$$

strictly decreasing in  $a$ , infinite as  $a \downarrow m$ , and zero at  $a = m + x_L$ ; let  $x_F \in (0, x_L)$  solve  $\Delta_m(m + x_F) = F$ .

**Payments as a step function of cash.** Maximizing Equation (4) piece by piece, below the minimum, at it, and above it, gives the payment in Equation (5) and the shape of its distribution. The proposition records both and separates the part of the bunching at  $m$  that the fee produces from the part that loss aversion adds.

**Proposition C1** (Step function). *Under Assumption C1, the optimal payment is Equation (5). Hence (i) there are mass points at  $m$  and  $B$ ; (ii) no payment is observed in the hole  $(m - (x_L - x_F), m)$ ; (iii) the density of payments is continuous immediately above  $m$ ; and (iv) the bunching interval  $[m + x_F, m + x_H]$  is the union of a notch component  $[m + x_F, m + x_L]$ , present when  $\eta_m = 0$ , and a loss-aversion component  $[m + x_L, m + x_H]$  whose width increases in  $\eta_m(\lambda - 1)$ .*

*Proof.* On  $[0, m)$ ,  $U'$  is positive iff  $a - p > x_L$ ; on  $[m, B]$ , positive iff  $a - p > x_H$ . If  $a > m + x_H$  then  $v(a - m) < \pi_H \leq \pi_L$ , so  $U$  is increasing on  $[0, m)$  and, with the upward jump  $F$  at

$m$ ,  $m$  dominates every  $p < m$ ; on  $[m, B]$  the maximizer is  $\min\{a - x_H, B\}$ , the first two lines of Equation (5). If  $m + x_L \leq a \leq m + x_H$  then  $\pi_H \leq v(a - m) \leq \pi_L$ , so  $U' \geq 0$  on  $[0, m)$  and  $U' \leq 0$  on  $(m, B]$ , and  $p^* = m$ . If  $m < a < m + x_L$  then  $v(a - m) > \pi_L$ ,  $U$  is decreasing on  $[m, B]$ , and the best point below  $m$  is  $p_L = \max\{a - x_L, 0\}$ ; substituting  $z = a - q$ ,  $U(p_L) - U(m) = \Delta_m(a) - F$ , so  $p^* = m$  iff  $a \geq m + x_F$ . If  $a \leq m$  the consumer cannot reach  $m$ . For  $a < m + x_F$ ,  $p_L < m + x_F - x_L$ , which is the hole. Continuity above  $m$  follows from  $p^* = a - x_H$  there. (iv) restates the two cases that give  $p^* = m$ .  $\square$

**Increasing the minimum.** Every threshold in Equation (5) sits a fixed distance above  $m$ , so a higher minimum moves the whole step function with it. The proposition records who moves and by how much, including consumers the minimum does not bind.

**Proposition C2** (A higher minimum). *Let the minimum rise from  $m$  to  $m' > m$ .*

- (i) *Consumers with  $a \in [m' + x_F, m' + x_H]$  pay  $m'$ ; consumers with  $a \in [m + x_F, m' + x_F)$ , who paid  $m$ , now pay  $\max\{a - x_L, 0\}$ .*
- (ii) *Consumers with  $a - x_H \in (m, m']$  paid above the old minimum and were not constrained by it; they now pay exactly  $m'$ .*
- (iii) *If  $\mu$  is concave on gains with  $\mu'(0^+)$  finite, every interior payer increases her payment,  $\partial p^* / \partial m > 0$ , most strongly close to  $m$ .*
- (iv) *With a grace-period notch  $G > 0$  at  $B$ , a target payer pays  $B$  rather than  $m$  iff  $G \geq \int_{a-B}^{a-m} (v(z) - \pi_H) dz$ ; the right-hand side is decreasing in  $m$ , so a higher minimum raises the share paying in full.*

*Proof.* (i)–(ii) follow from Equation (5) with  $m'$  in place of  $m$ , since every threshold has the form  $m + x$  with  $x$  independent of  $m$  ( $\Delta_m$  depends on  $m$  only through  $a - m$ ). (iii) The interior first-order condition is  $\Phi(p, m) \equiv v(a - p) - \rho - \eta_m \mu'(p - m) - \lambda \eta_B = 0$ , with  $\Phi_p = -v'(a - p) - \eta_m \mu''(p - m) > 0$  and  $\Phi_m = \eta_m \mu''(p - m) < 0$ , so  $dp/dm = -\Phi_m / \Phi_p > 0$ . (iv) The condition for  $B$  is  $G \geq \int_m^B (v(a - q) - \pi_H) dq = \int_{a-B}^{a-m} (v(z) - \pi_H) dz$ ; raising  $m$  removes a segment on which the integrand is nonnegative, because  $v(a - m) \geq \pi_H$  for a target payer.  $\square$

**Displayed options.** The app displays a menu  $M$  of one-click amounts; any other amount costs  $\chi \geq 0$  to enter. With  $p^*$  from Equation (5) and  $L(z) \equiv U(p^*) - U(z) \geq 0$ , the consumer pays the best menu option  $p^M$  if  $L(p^M) \leq \chi$  and  $p^*$  otherwise. Let  $h = B/2$ ; the experimental menus are  $M_C = \{m, h, B\}$ ,  $M_1 = \{h, B\}$ , and  $M_2 = \{B\}$ .

**Proposition C3** (An intermediate option). *Compare  $M_1$  with  $M_2$ . Consumers with  $p^* = B$  or  $p^* < m$  are unaffected. Among consumers with  $p^* \in [m, B]$  there is a cutoff  $\hat{p} \in (h, B)$*

such that: (i) a mass point appears at  $h$ , made of consumers with  $L(h) \leq \chi$  and  $p^* \leq \hat{p}$ ; (ii) payments rise for consumers just below  $h$  who previously entered  $p^*$ , and fall for consumers who previously rounded up to  $B$  and have  $p^* \leq \hat{p}$ , and for consumers just above  $h$ ; (iii) nothing changes elsewhere; (iv) the mean effect has ambiguous sign and vanishes as  $\chi \rightarrow 0$ .

*Proof.*  $U$  is strictly concave on  $[m, B]$  with maximum at  $p^*$ , so  $L$  increases with  $|z - p^*|$  on each side. For  $p^* \leq h$ ,  $U$  decreases on  $[p^*, B]$  and  $L(h) < L(B)$ . For  $p^* \in (h, B)$ ,  $U(B) - U(h) = \int_h^B (\pi_H - v(a - q)) dq$  is increasing in  $a$ , negative at  $a = h + x_H$  and positive at  $a = B + x_H$ ; let  $\hat{a}$  be its zero and  $\hat{p} = \hat{a} - x_H$ . Under  $M_2$  the consumer pays  $B$  iff  $L(B) \leq \chi$ ; under  $M_1$  she pays  $h$  iff  $L(h) \leq \chi$  and  $L(h) \leq L(B)$ , and  $B$  iff  $L(B) \leq \chi$  and  $L(B) < L(h)$ . Comparing the rules case by case gives (i)–(iii); (iv) follows from (ii).  $\square$

**Remark C1.** If  $\chi$  falls with experience, the switch from  $B$  to  $h$  is transitory; if a displayed amount also carries a small gain-loss weight  $\eta_h > 0$  while displayed, a mass point at  $h$  persists, consisting of consumers with  $v(a - h) \in [\pi_H + \eta_h, \pi_H + \lambda\eta_h]$ .

**Shrouding the minimum.** The treatments change what is displayed and nothing about payoffs, so under targeting they leave the optimum in Equation (5) unchanged; what they change is how a consumer who does not know the minimum to the cent gets to it. The proposition records the two routes and the predictions Section 3 states.

**Proposition C4** (Shrouding under targeting). *Suppose a target payer knows  $m$  only up to  $[m - \varepsilon, m + \varepsilon]$  when it is not displayed, that submitting an amount below  $m$  triggers a prompt revealing  $m$  at click cost  $\kappa > 0$ , and that the smallest round number weakly above  $m + \varepsilon$  is  $\tilde{p}$ . She pays  $\tilde{p}$  at overpayment cost  $\omega(a) = \int_m^{\tilde{p}} (v(a - q) - \pi_H) dq \geq 0$ , or probes and then pays  $m$ ; she probes iff  $\kappa < \omega(a)$ . Relative to a displayed minimum: final payments are unchanged up to a round-up of at most  $\tilde{p} - m$ ; the share paying exactly  $m$  falls and the share within a dollar above it rises; initial submissions below  $m$  and submissions per cycle rise; nothing else changes; and final payments do not vary with liquid cash.*

*Proof.*  $U$  and  $p^*$  do not depend on the menu, so both routes end at  $m$  or a round-up above it, and the consumer takes the cheaper.  $\omega \geq 0$  because  $v(a - q) \geq v(a - m) \geq \pi_H$  for  $q \geq m$  and a target payer.  $\square$

**Anchoring.** Model anchoring as a psychological cost  $\frac{\kappa_A}{2}(p - d)^2$  of deviating from a displayed amount  $d$ , with  $\kappa_A = 0$  when nothing is displayed.

**Proposition C5** (Anchor weight). *For an interior payer, to first order,*

$$p^A - p^* = \frac{\kappa_A}{\kappa_A + |U''(p^*)|} (d - p^*), \quad |U''(p^*)| = |v'(a - p^*)|. \quad (8)$$

A consumer at the kink or notch at  $m$  is unmoved for  $\kappa_A|d - m|$  below the kink size  $\eta_m(\lambda - 1)$  or the notch  $F$ .

*Proof.* Linearize the first-order condition  $U'(p) - \kappa_A(p - d) = 0$  around  $p^*$ . At the kink,  $U'$  jumps down by  $\eta_m(\lambda - 1)$ , so the anchored first-order condition has no solution away from  $m$  unless  $\kappa_A|d - m|$  exceeds the jump; the notch case is the same with  $F$ .  $\square$

Under anchoring with  $d = m$  displayed, shrouding increases payments by  $\frac{\kappa_A}{\kappa_A + |U''(p^*)|} (p^* - m)$  for every consumer with  $p^* > m$ , most for the cash-rich, and increases initial submissions. In a hypothetical scenario  $v$  is flat, so  $|U''| \approx 0$  and the anchor's weight is near one.

## C.2 The Dynamic Problem

**Dynamic environment.** Cycles are indexed by  $t$ ; per-cycle utility is  $u(\cdot)$ , strictly concave with Inada conditions; discount weights are  $(1, \beta\delta, \beta\delta^2, \dots)$ . Each cycle the consumer receives cash  $a_t$ , spends  $S_t$  on the card, pays  $p_t$ , and consumes  $c_t = a_t - p_t$ ; debt evolves as  $B_{t+1} = (1 + r)(B_t - p_t) + S_t$ .

**Spending.** Card spending  $S_t$  is taken as given. Present bias increases the level of spending, which is why a present-biased consumer carries a balance, but the model does not let spending respond to the payment. The commitment results require that a higher payment is not charged back, the re-borrowing margin Remark C3 discusses. This is conservative: my nudges do not change spending (Section 4), and when Quebec increased minimum payment requirements, spending did not rise to offset them (Allen et al., 2026).

**Reduction to the static problem.** Let  $\bar{c}$  be expected next-cycle consumption. A marginal dollar paid at  $t$  raises  $c_{t+1}$  by  $1 + r$  for a consumer who would otherwise have paid it next cycle; self  $t$  values this at  $\beta\delta(1 + r)u'(\bar{c})$  against a cost  $u'(c_t)$ . Dividing by  $u'(\bar{c})$  maps the problem into Equation (4) with

$$v(z) = u'(z)/u'(\bar{c}), \quad \rho = \beta\delta(1 + r), \quad (9)$$

so that  $v(\bar{c}) = 1$ , and every buffer in Equation (5) rises as  $\beta$  falls. More generally,  $v$  can be interpret as the marginal value of cash on hand in a classic buffer-stock problem, which includes the precautionary motive and satisfies Assumption C1.

**Proposition C6** (Present bias and the minimum). *Consider a consumer in a stationary hand-to-mouth state,  $a - m = \bar{c}$ .*

(i) *She is a target payer iff  $\beta\delta(1 + r) + \eta_m + \lambda\eta_B \leq 1$ .*

- (ii) If  $\beta = 1$  and  $\delta(1+r) > 1$  no such state exists: she pays above the minimum and amortizes the debt.
- (iii) For  $\beta < [1 - \eta_m - \lambda\eta_B]/[\delta(1+r)]$  the state exists: she pays the minimum every cycle, carries no cash, and revolves debt although she is neither at her limit nor unable to pay more.
- (iv) A naive consumer's plan for next cycle weights  $c_{t+1}$  and  $B_{t+2}$  by  $\beta\delta$  and  $\beta\delta^2$ , so the planned payment solves the  $\beta = 1$  problem with  $\rho^{\text{plan}} = \delta(1+r) > 1$ .

*Proof.* At  $a - m = \bar{c}$ ,  $v(a - m) = 1$ , so by Equation (5) the consumer is at  $m$  iff  $\pi_H \leq 1$  (or, when  $\pi_L < 1$ , iff  $F \geq \Delta_m(a)$ , which yields the same payment). If  $\beta = 1$  and  $\delta(1+r) > 1$  then  $\pi_H > 1$ , so  $p^* > m$  and  $c_t < \bar{c}$ , contradicting stationarity. (iii) restates (i) with  $c_t = \bar{c}$  self-confirming. (iv)  $\beta$  cancels from the ratio of the two weights.  $\square$

**What moves payments.** Because a target payer sits at a kink, her payment is locally invariant to anything that enters through  $\rho$  and responds fully to the location of the kink; the proposition records this, and the welfare slope that Section 6 uses.

**Proposition C7** (Disclosures versus minimums). *For target payers  $\partial p^*/\partial \rho = 0$  and  $\partial p^*/\partial m = 1$ ; for interior payers  $\partial p^*/\partial \rho = 1/|v'(a - p^*)|$ , small when cash is scarce, and  $\partial p^*/\partial m > 0$  under diminishing sensitivity. In the planner's units the value of paying  $m$  has slope  $-v(a - m) + \delta(1+r) + \eta_m + \lambda\eta_B$  in  $m$ , positive for a present-biased target payer and non-positive for a time-consistent one.*

*Proof.* At the kink or notch  $p^* = m$  for an open set of  $\rho$ ; differentiate  $v(a - p^*) = \pi_H$  for the interior; the welfare slope is the planner's marginal value, derived in the proof of Proposition C8(ii), evaluated at  $T = m$ .  $\square$

**Remark C2** (Relation to the sufficient-statistics evaluation in Allen et al. (2026)). *Allen et al. (2026) evaluate a minimum-payment mandate with  $\Delta W = \sum_{\theta} \gamma (-\Delta D^{\text{bias}}(\theta)) \phi(\theta) - \Lambda$ , where  $\gamma = (1 - \beta)\delta(1+r) u'(c_2)/u'(c_1) \approx 1 - \beta$  is the long-run self's loss per dollar of revolving debt,  $\Delta D$  the change in revolving debt net of the credit-limit response,  $\phi$  type shares, and  $\Lambda$  the access cost. For an interior payer,  $v(a - p^*) = \pi_H$ , so the planner's marginal value of a forced dollar is  $(1 - \beta)\delta(1+r)$  in the units of Equation (9), which converts to  $(1 - \beta)/\beta$  in date-1 dollars absent reference pulls and to  $1 - \beta$  under their market-price approximation: the model delivers  $\gamma$ .*

*For a consumer in the bunching interval,  $v(a - m) > \pi_H$  and the marginal value is  $(1 - \beta)\delta(1+r) - [v(a - m) - \pi_H]$ , with the gap in  $[0, \eta_m(\lambda - 1)]$  on the loss-aversion component and larger on the notch component, where the sign can flip. So  $\gamma$  is exact on the Euler margin,*

an upper bound at the kink, and wrong-signed for consumers who reach the minimum only by accepting a smaller cash buffer than they would otherwise keep, which is why excluding exact-minimum payers from the benefit is conservative.

**Remark C3** (Relation to Maxted (2025)). *Because revolving debt is net of payments (Gibbs et al., 2025), the re-borrowing margin of the present-bias dilemma of Maxted (2025) (his Proposition 10: where limits do not bind, a forced payment is charged back and a self-imposed commitment cannot raise welfare) is inside the sufficient statistic of Remark C2; the Quebec finding that revolvers with slack credit cut debt beyond the required-payment ceiling is the bracketing of payments from spending that the commitment results here assume. Corollary 5 of Maxted (2025), that sophisticates and naifs are observationally equivalent in consumption, is Proposition C8(iii) for payments.*

**Self-set targets.** Let the planning self, at the statement date, form an expectation  $T \in (m, B)$  of the payment, which is an additional reference point with weight  $\eta_T$  for the payment-date self. Both dates are in the planner's future, so her relative weights are  $\beta\delta : \beta\delta^2$  and  $\beta$  cancels.

**Proposition C8** (Self-enforcing targets). *(i)  $T$  is self-enforcing iff  $v(a - T) \in [\pi_H + \eta_T, \pi_H + \lambda\eta_T]$ . The highest self-enforcing target is  $T^* = a - x(\pi_H + \lambda\eta_T)$ , which exceeds  $a - x_H$  and rises with  $\beta$  and with  $\lambda\eta_T$ .  
(ii) The planner's optimum is  $T^{LR} = a - x(\delta(1 + r) + \eta_m + \lambda\eta_B)$ . She is constrained, setting  $T = T^* < T^{LR}$ , iff  $\lambda\eta_T < (1 - \beta)\delta(1 + r)$ .  
(iii) A naive consumer expects  $T^{LR}$ , pays  $T^*$ , and realizes a loss  $\lambda\eta_T(T^{LR} - T^*)$  each cycle.  
(iv) With cash uncertainty the expectation is a number  $\bar{T}$ , self-enforcing for  $a \in [\bar{T} + x(\pi_H + \lambda\eta_T), \bar{T} + x(\pi_H + \eta_T)]$ : payments bunch at  $\bar{T}$  with few payments just below it, and a minimum increase to  $m' \leq \bar{T}$  leaves these consumers unchanged.*

*Proof.* (i) With reference  $T$ , marginal utility is  $\pi_H + \lambda\eta_T - v(a - p)$  for  $p < T$  and  $\pi_H + \eta_T - v(a - p)$  for  $p > T$ ;  $p = T$  is optimal iff the first is nonnegative and the second nonpositive at  $T$ . (ii) At a deterministic personal equilibrium no gains or losses are realized, so the planner evaluates  $T$  by  $-C(T; a) + \delta(1 + r)T + \eta_m(T - m) + \lambda\eta_B(T - B)$ , increasing iff  $v(a - T) < \delta(1 + r) + \eta_m + \lambda\eta_B$ ; she chooses  $\min\{T^*, T^{LR}\}$ , and  $T^* < T^{LR}$  iff  $\pi_H + \lambda\eta_T < \delta(1 + r) + \eta_m + \lambda\eta_B$ . (iii) With reference  $T^{LR} > T^*$  the payment-date self is in the loss domain, where her optimum is  $T^*$ . (iv) Restate (i) with  $T$  fixed and  $a$  varying.  $\square$

Proposition C8 takes the target as a fixed dollar amount. When the minimum is displayed, above-minimum targets are often expressed in terms of it (Medina and Negrin, 2022), which changes how consumers respond to a change in the minimum.

**Proposition C9** (Targets indexed to the minimum). *Let a consumer’s self-set target be  $T(m) = km$  or  $T(m) = m + \ell$  with  $k > 1$ ,  $\ell > 0$ , carrying weight  $\eta_T$ , and suppose  $T(m)$  is self-enforcing at her cash level,  $v(a - T(m)) \in [\pi_H + \eta_T, \pi_H + \lambda\eta_T]$ . Then:*

- (i) *She pays exactly  $T(m)$ , above the minimum, so the minimum does not bind for her.*
- (ii) *If the minimum rises to  $m'$  and  $T(m')$  remains self-enforcing, she pays  $T(m')$ : her payment rises by  $k(m' - m)$  or by  $m' - m$  without the minimum binding. If  $T(m')$  is not self-enforcing because cash is too scarce, she pays  $a - x(\pi_H + \lambda\eta_T)$ , still above  $m'$  whenever that exceeds  $m'$ .*
- (iii) *If the minimum is shrouded and known only up to  $[m - \varepsilon, m + \varepsilon]$ ,  $T(m)$  cannot be entered exactly; the nearest self-enforcing target she can compute is a round number  $R$  not indexed to  $m$ . Her payment then bunches at  $R$ , and a subsequent change in  $m$  leaves it unchanged unless  $m'$  reaches  $R$ .*

*Proof.* (i) and (ii) follow from Proposition C8(i) with  $T$  replaced by  $T(m)$  and  $T(m')$ : the self-enforcement band is a condition on  $v(a - T)$  and moves with  $T$ . (iii) A target must be a number the consumer can submit; with  $m$  unknown to within  $\varepsilon$ ,  $km$  and  $m + \ell$  are not, and the self-enforcement condition holds for an interval of targets of width  $x(\pi_H + \eta_T) - x(\pi_H + \lambda\eta_T)$  in cash, which contains round numbers.  $\square$

Part (ii) is the response of unconstrained near-minimum payers to minimum increases in Keys and Wang (2019), Medina and Negrin (2022), Katz et al. (2026), and Allen et al. (2026), including paydown above the required-payment ceiling; part (iii) is the fall in payments at multiples of the minimum and the rise in round numbers under T1 and T2 (Appendix Table A6).

## D. Value Proximity Analysis (VPA)

### D.1 VPA Methodology

The recent marketing literature of Bartels et al. (2024) and Herzog et al. (2025) distinguishes between anchors and targets and also provides an empirical test—Value Proximity Analysis (VPA)—for whether behavior around a focal value is best characterized as psychological anchoring or instead as targeting i.e., reference-dependent preferences. The idea behind this test is to examine the distribution of observations around a focal value,  $V$ . VPA plots the proportion of observations that are at or above a focal value, starting with the 1% of observations with absolute values in the closest proximity to the focal value and expanding the proximity window by one percentage point until 100% of observations are included.

Anchors should have approximately 50% of observations above the focal value and increase under a low anchor as more observations are included (or decrease under a high anchor). In contrast, a target should have close to 100% of observations above the focal value, as consumers exert effort to achieve at least this value, and slope down as more observations are included. Anchors have a symmetric distribution around a focal value, whereas targets have an asymmetric distribution.

I estimate the relationship in an OLS regression with one observation for each proximity window  $w$  ( $Proximity_w \in \{1, 2, \dots, 99, 100\}$ ) of each treatment group  $g$  (C, T1, and T2). Let  $p_{g,w}$  denote a payment in window  $w$  of group  $g$ , so that  $\Pr(p_{g,w} \geq V)$  is the share of payments in that window at or above the focal value. The specification is:

$$\begin{aligned} \Pr(p_{g,w} \geq V) = & \alpha + \beta^{T1} T1_g + \beta^{T2} T2_g + \gamma Proximity_w + \theta^{T1} (Proximity_w \times T1_g) \\ & + \theta^{T2} (Proximity_w \times T2_g) + \varepsilon_{g,w} \end{aligned}$$

In this equation, the intercept,  $\alpha$ , reflects the value for the omitted category, the control group (C), and the regression includes two binary indicators for each of the treatments, T1 and T2. The regression includes a term,  $Proximity_w$ , to estimate how the relationship linearly varies with the proximity window, and it also includes interactions with T1 and T2 to allow for the slope to differ between C, T1, and T2. Because each window contains the narrower windows before it, the 300 rows are not independent, so I obtain standard errors by bootstrapping over consumers (10,000 draws, resampling consumers with all their cycles). The last two rows of Table A7 report the T2 – T1 contrasts, the effect of displaying the 50% option. I run this regression separately for each outcome,  $\Pr(p_{g,w} \geq V)$ , where the outcome is the fraction of payments in a proximity window that are at or above a threshold,  $V$ . The three thresholds that I test to determine whether they act as an anchor or a target for consumers are: the minimum payment amount, 50% statement balance amount, and statement balance amount in columns 1, 2, and 3, respectively, of Table A7.

The VPA approach has three clear predictions for distinguishing anchors and targets. First, under anchoring, the intercept would be near 50%, whereas under targeting it would be near 100%. Second, there would be a negative slope for targets or high anchors, and a positive slope for low anchors. These first two predictions can only be established when there are no changes in economic payoffs that coincide with the focal value being tested, which does not apply at the minimum or the statement balance. Proposition C1 shows that the fee and the end of interest produce the asymmetry on their own, whether or not consumers are reference dependent.

The third prediction can be tested because the experiment varies salience and not payoffs.

Under anchoring, both the intercept and the slope would significantly vary with the treatments that make particular amounts less salient, whereas if these are targets, the intercept and slope would not change significantly.

## D.2 VPA Results

Table A7 shows the VPA regression results, and these are consistent with the underlying VPA data presented in Appendix Figure A7. My VPA results are consistent with the minimum payment amount acting as a strong target rather than as an anchor. The intercept in column (1) of Table A7 is 98.5%, very close to 100%, and there is a significant negative slope. T1 changes neither. T2 lowers the intercept by 2.8 (s.e. 1.0) points, to 95.7%, with the slope unchanged, a parallel shift consistent with the small, insignificant increases in missed payments in Panel G of Figure 3. These VPA results are consistent with the rest of the analysis of my experiment, which shows that credit card payment choices are not due to anchoring to the minimum payment.

Column (2) of Table A7 examines the VPA for the 50% statement balance amount. The intercept, 76.8%, lies between the anchor and target benchmarks, which is what one expects at an amount with no discontinuity in economic payoffs, only a change in psychological payoffs. Comparing T1 to T2 isolates the effect of the 50% option itself: displaying it raises the intercept from 69.3% to 82.1%, a difference of 12.9 (s.e. 2.9) points, and steepens the slope from  $-0.51$  to  $-0.63$  per percentage point of proximity, a difference of 0.12 (s.e. 0.03). Displaying the 50% option therefore adds mass at or just above it in the narrowest windows, the spike documented in Section 4.4. The slope remains significantly negative in all three groups, inconsistent with a low anchor, but the shape of the distribution around the 50% amount depends on whether it is displayed, which a target's does not: it is a reference point only while displayed.

There are strong economic incentives for consumers to pay the statement balance amount, as doing so avoids incurring interest. Column (3) of Table A7 shows the VPA for the statement balance amount. The intercept is near 90%, and there is a significant negative slope with proximity. The intercept and the slope are not significantly affected by either treatment. Although, of course, the statement balance amount is not shrouded in any of the groups C, T1, or T2, it does become more prominent with the shrouding of alternative payment options in T1 and T2.